MEHLMANMEDICAL
HY GASTRO

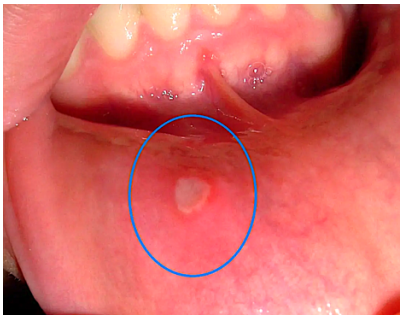
YouTube

@mehlmanmedical

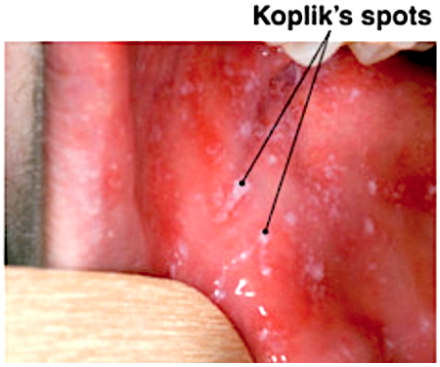
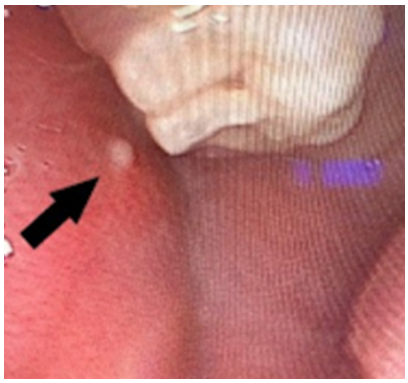
Instagram

@mehlman_medical

Disorders affecting lips/oral cavity for USMLE	
Peutz-Jeghers syndrome	- Combination of perioral melanosis (fancy word for hyperpigmentation around the lips/mouth) + **hamartomatous** colonic polyps. - USMLE will show image of lips pretty much always, and then they'll ask for what kind of polyps are seen in the colon (i.e., hyperplastic, tubulovillous, etc.), and the answer is just "hamartomatous."
Osler-Weber-Rendu	- Aka hereditary hemorrhagic telangiectasia; autosomal dominant. - NBME loves showing a mouth or fingernail picture of telangiectasias. - Q will give nosebleeds + show you the above pic. There can be high-output cardiac failure due to pulmonary AV fistulae. - GI bleeding can occur leading to anemia.
Plummer-Vinson syndrome	- Triad of iron deficiency anemia + esophageal webs (dysphagia) + angular cheilitis (cracked corners of mouth). - NBME Q can also mention pica (iron deficiency sign where patient eats clay, starch, or ice), or they can show spoon-shaped nails (koilonychia), which are a sign of severe iron deficiency.

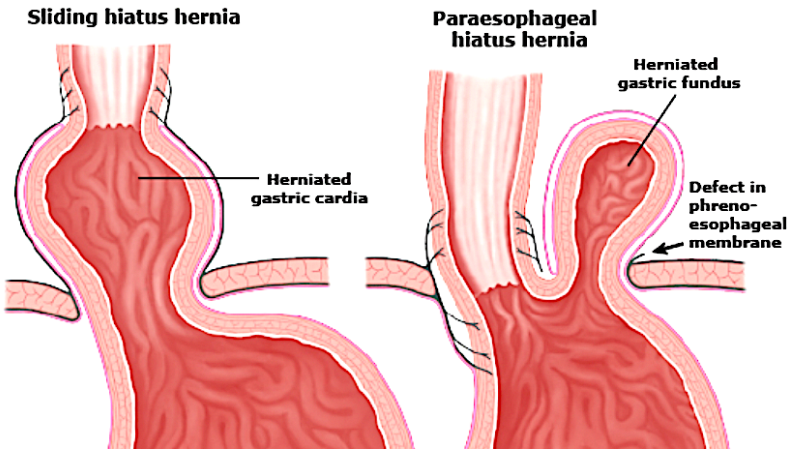
Lip psoriasis	- Just know it's possible. USMLE can mention it on upper lip or forehead, and this somehow confuses students, where they think it has to be on extensors only.
Aphthous ulcer	- Aka canker sore. - Usually appears as very painful lesion, self-resolving lesion on labial mucosa. - Not infective; idiopathic; thought to have very mild autoimmune association; sometimes associated with certain triggers like spicy food. 
Kawasaki disease	- Vasculitis that causes **5+ days of fever** + injected (red) eyes and/or lips/tongue + cervical lymphadenopathy + edema of dorsa of the hands + **desquamation of palms/soles** (often mentioned as palms/soles "rash," but not true rash). - Students obsess over coronary artery aneurysms as though they're so HY. They're not. USMLE basically never mentions them.
Perioral impetigo	- Often confused with herpes.

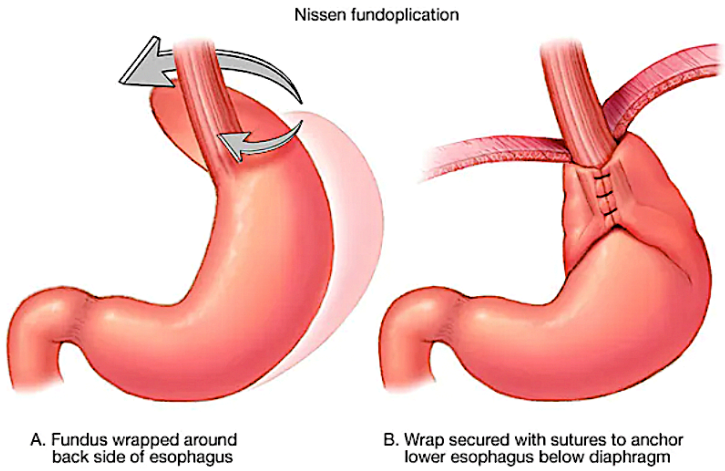
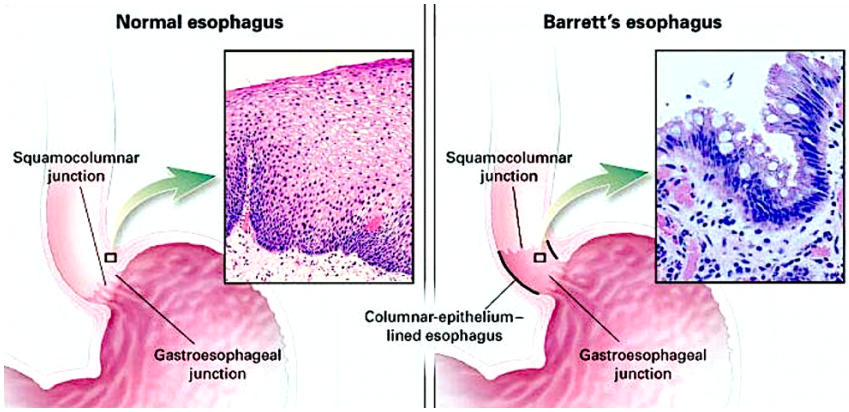
	- If they show you image of young kid in particular with lip lesions, it's usually impetigo caused by *S. aureus* or Group A Strep (*S. pyogenes*). - Can lead to PSGN (if caused by Strep), as discussed in the HY Renal PDF.
Herpes labialis	- Caused by HSV1 or 2. USMLE doesn't give a fuck about HSV1 being the lips and HSV2 being the genitals. Bunch of nonsense perpetuated by other resources. - Primary infection presents with fever and regional lymphadenopathy. Recurrences aren't as severe. - HSV goes latent in sensory nerves. - USMLE will show you above pic and then the answer is "DNA, enveloped, linear." Sounds nitpicky and low-yield, but it's one of the viral structures they like asking. Don't confuse with HepB, which is "DNA, enveloped, circular." I've made lots of YouTube clips on that stuff. - Treat with acyclovir → DNA polymerase inhibitor causing chain termination. HSV resistance to it will be due to altered thymidine kinase.
Scarlet fever	- Caused by Group A Strep. - Causes strawberry tongue / red lips + salmon-pink maculopapular body rash. - Treatment is penicillin to prevent rheumatic fever (type II HS); can also lead to PSGN (type III HS). I discussed this stuff in the HY Cardio and Renal PDFs.
Hand-foot-mouth	- Caused by Coxsackie A (an RNA virus under picornaviridae). - Causes benign, but contagious, lesions on, you guessed it LOL! – the hands, feet, and mouth. - Usually pediatric, but can present in adults (i.e., daycare workers, parents). - Don't confuse with coxsackie B, which can cause dilated cardiomyopathy, diabetes type I, and pleurodynia (latter I discussed in HY Pulm PDF).

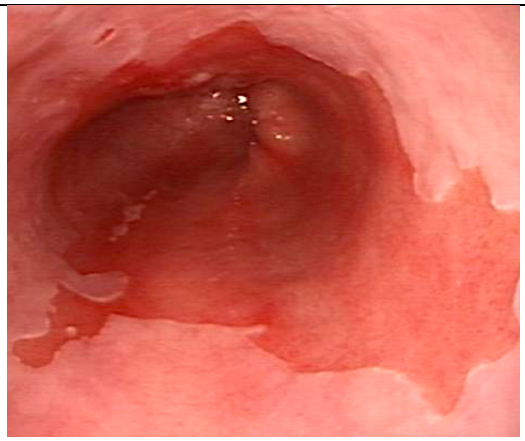
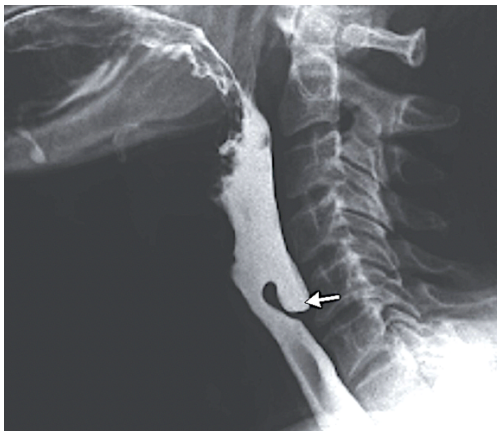
Herpangina	- Also caused by coxsackie A. - Presents as oropharyngeal vesicles or sores. - Can occur with or without hand-foot-mouth.
Koplik spots	- Caused by measles. - White/blue-ish spots on buccal mucosa.  - One of the ways to distinguish measles (rubeola) from rubella (German measles). - The same way you can memorize Koplik spots as unique to measles, you should memorize sub-occipital and post-auricular lymphadenophaty as unique to rubella. - Both measles and rubella have a head-to-toe maculopapular body rash. - Cough, coryza, conjunctivitis triad **is not** unique to measles and is simply a non-specific combo that can be seen for many viral infections in general. I've seen students get hardcore simple NBME Qs wrong where they think Dx is measles simply because these findings are listed. Whoever/whichever resources have perpetuated this triad as = measles should be hanged for absolute stupidity.
Sialolithiaisis	- Stones within the ducts of the salivary glands. - Stensen duct is the opening of the parotid duct (from parotid gland) into the oral cavity, which is located near the 2nd upper molar bilaterally.  - Sometimes a Q can say a patient has pain or inflammation on the buccal mucosa across from the second upper molar, and sialadenitis (inflammation) or sialolithiasis can be an answer.

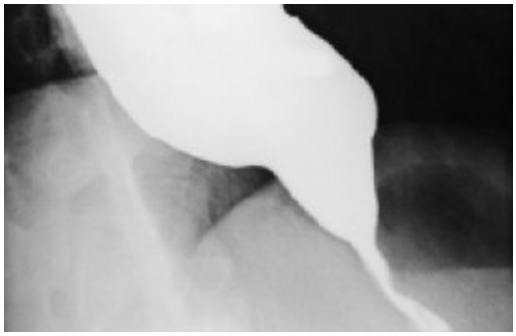
Leukoplakia	- White-ish, painless, rough patch on lateral tongue. - Precursor to squamous cell carcinoma of tongue. - Biggest risk factor is smoking / chewing tobacco.
Oral hairy leukoplakia	- Looks like leukoplakia, but caused by EBV. - Not considered premalignant (no dysplasia on biopsy).
Oropharyngeal candidiasis	- Painless white plaque on palate or tongue that bleeds when scraped. - Seen in immunocompromised patients, such as AIDS or organ transplant recipients on immunosuppressant agents. - Can be seen in patients with asthma who use inhaled corticosteroids. Patients need to rinse their mouths with water after use to ↓ candida risk. - Tx with nystatin mouthwash. - Odynophagia (painful swallowing) in an immunocompromised patient is candidal esophagitis until proven otherwise (Tx = oral fluconazole; not same as oral thrush where we use nystatin mouthwash).

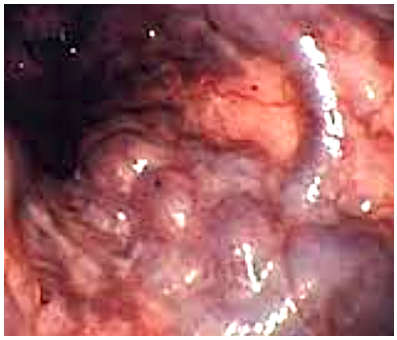
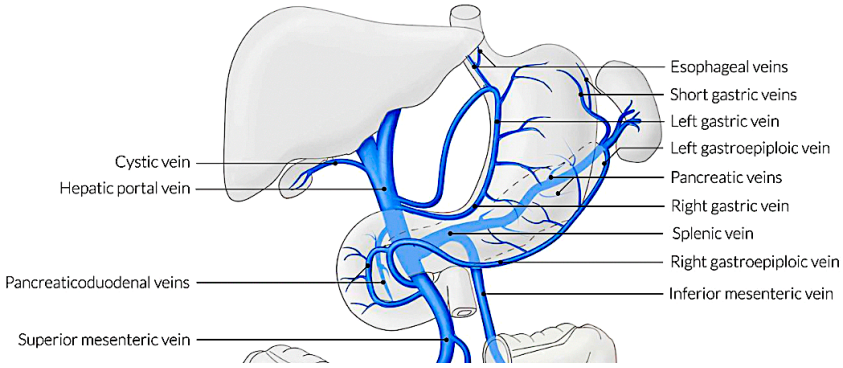
Salivary gland neoplasia	
Pleomorphic adenoma	- The answer on USMLE for benign salivary gland tumor that has variability in cell size and shape. - Most common salivary gland tumor; benign. - Pleomorphic = cells and nuclei have variability in size and shape.
Mucoepidermoid carcinoma	- The answer on USMLE for a salivary gland tumor that invades / has spread / if the patient has B-symptoms (fever, night sweats, weight loss). - Most common *malignant* salivary gland tumor. - Malignant means capable of invading, whereas pleomorphic adenoma will not invade.
Warthin tumor	- The answer on USMLE for salivary gland tumor where they say anything about it resembling lymphatic tissue. - Salivary gland tumor that has lymphoid stroma (i.e., looks like lymphatic tissue).

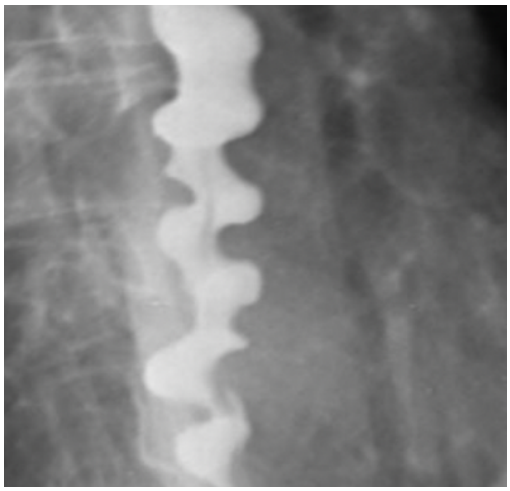
HY Esophageal conditions for USMLE	
Hiatal hernia	- Herniation of the stomach upward through the diaphragm. - 95% of the time, hiatal hernia is known as "sliding," where the cardia of the stomach rises upward through the esophageal hiatus (opening in diaphragm where esophagus passes). - "Abnormal relation of the cardia to the lower end of the diaphragm" is answer on NBME. - 5% of the time, it is paraesophageal, where the fundus of the stomach herniates upward through a hole in the diaphragm adjacent the esophagus. - "Protrusion of the fundus into the chest above the level of T10" = answer for paraesophageal hernia on NBME.  - Often associated with gastroesophageal reflux disease (GERD). - Diagnose with upper endoscopy (asked on Surg form). - Operative repair is the Tx (also asked on Surg form).
Gastroesophageal reflux disease (GERD)	- Irritation of esophageal mucosa by gastric acid; usually caused by ↓ **lower esophageal sphincter (LES) tone.** - USMLE wants you to know **obesity** is a risk factor for GERD, likely due to ↑ stretching and pressure applied to the LES. - As mentioned above, hiatal hernia is also a risk factor. - Classic presentation is burning in the throat/chest after eating a meal. - Can present with **nocturnal cough** or **recurrent pneumonitis.** These are HY findings in patients who don't have classic esophageal irritation symptoms. - Can lead to Barrett esophagus (discussed below), which leads to esophageal adenocarcinoma. - First step in diagnosis is 2-week trial of proton pump inhibitor (PPI), such as omeprazole, which will ↓ symptoms. - PPIs (e.g., esomeprazole) are more efficacious than H2-blockers (e.g., cimetidine). If you see both a PPI and H2-blocker as answers for a GERD Q, choose the PPI. - One 2CK Surg form has "2-week trial of H2-blocker" as an answer, but PPI isn't listed, so just know the Q is out there. - If trial of PPI doesn't work, 24-hour esophageal pH monitoring is the answer. - Nissen fundoplication is used last resort, but is asked on NBMEs.

	 - Answer can be written as "fundoplication" alone in patient with severe GERD. This also shows up on Peds, where they say infant with severe GERD and fundoplication is answer. - Pediatric GERD will present as coughing up milk 2-3x daily. It will not be described as high-energy or forceful (i.e., pyloric stenosis); it will not be bilious (i.e., duodenal atresia; annular pancreas); it will not occur as choking or spitting up during the first feed (i.e., tracheoesophageal fistula). - Answer to pediatric GERD Q is "immature lower esophageal sphincter". It's not complicated, but I've seen students say, "Wait, you can get GERD in kids?" - Tx for Peds GERD is thickened feeds (addition of dry rice to formula) and positional change prior to any use of PPIs.
Barrett esophagus	- Metaplasia of distal esophagus characterized by conversion of non-keratinized stratified squamous epithelium into columnar epithelium with goblet cells (which secrete mucous). - You'll sometimes here the latter described as "intestinal columnar epithelium." This is because the stomach doesn't have goblet cells; it has mucous neck (foveolar) cells. - Major risk factor for esophageal adenocarcinoma. 

	 - Can appear grossly red on upper endoscopy. Once visualized, the next best step is biopsy to confirm the presence of Barrett metaplasia.
Adenocarcinoma	- Affects lower 1/3 of esophagus. - HY pathogenesis is: GERD → Barrett esophagus → adenocarcinoma. - Will present in patient over 50 who has Hx of GERD with either 1) new-onset dysphagia to solids, or 2) dysphagia to solids *that progresses* to solids and liquids. - The "new-onset dysphagia" can refer to 3-6-month Hx in patient with, e.g., 10-20-year Hx of GERD. - 2CK wants immediate endoscopy in either of the above situations (i.e., don't choose barium first). - Sometimes rather than making you choose endoscopy straight up, they'll tell you in the last line an endoscopy was performed and shows a stricture, then they'll ask for next best step → answer = biopsy of the stricture.
Squamous cell carcinoma	- Affects upper 2/3 of esophagus. - Biggest risk factors for USMLE are **heavy smoking/alcohol use.** - Other risk factors like burns, chemicals, achalasia, etc., are mostly nonsense. - Will present in patient over 50 who is heavy smoker/drinker who has 1) new-onset dysphagia to solids, or 2) dysphagia to solids that *progresses* to solids and liquids. - Same as with adenocarcinoma, USMLE wants immediate endoscopy as the answer, followed by biopsy of a stricture or lesion if present.
Zenker diverticulum	- Outpouching of esophagus above the cricopharyngeus muscle (just above the upper esophageal sphincter).  - Mechanisms on USMLE are numerous and include, "cricopharyngeal muscle spasm," "uncoordinated swallowing," "dysphagia," and "increased oropharyngeal pressure."

	- False diverticulum (contains only mucosa + submucosa; true diverticula, in contrast, such as Meckel, include the muscular layer as well). - Often presents as overweight male over 40-50 with gurgling sounds when swallowing, or regurgitation of undigested food. - Halitosis (bad breath) is buzzy but only mentioned in maybe 1/3 of Qs. - Diagnosis is made with barium swallow. - Treatment on 2CK NBME is "diverticulectomy."
Achalasia	- Tightening (i.e., ↑ tone) of the LES due to loss of NO-secreting neurons in the Auerbach (myenteric) plexus. - New NBME wants **"enteric ganglia"** as the answer for site of problem. - Can be described on NBME as "dilated esophagus with constriction of the gastroesophageal junction." - Presents usually as dysphagia **to both solids and liquids from the start.** This is in contrast to cancer, which will be new-onset dysphagia to just solids, or *progression* from solids only to solids + liquids. The HY point is that neurogenic causes of dysphagia (i.e., achalasia) lead to solids + liquids dysphagia. - Almost always idiopathic; can rarely be caused by Chagas disease. - New 2CK NBME has "South American trypanosomiasis" as the answer for achalasia, which is aka Chagas disease. - First step in diagnosis is barium swallow, which shows a classic "bird's beak" appearance.  - After the barium is performed, esophageal manometry (a pressure study) is confirmatory. Biopsy is not traditionally performed. - Treatment is pneumatic (balloon) dilation, followed by myotomy (cutting of muscle fibers of the LES).
Systemic Sclerosis (aka scleroderma)	- As discussed in detail in the HY Pulm PDF, this is an idiopathic autoimmune disease characterized by multi-organ system fibrosis and hardening of tissues (i.e., sclerosis). - Can lead to GERD. An offline Step 1 NBME wants ↓ LES sphincter tone and ↓ esophageal peristalsis as an answer. This refers to the esophageal dysmotility component of **CREST** (**C**alcinosis, **R**aynaud's, **E**sophageal dysmotility, **S**clerodactyly, **T**elangiectasias).
Esophageal perforation	- Can either be iatrogenic (i.e., from endoscopy) or Boerhaave syndrome (from increased pressure due to retching, vomiting, or straining). - USMLE likes to say water-soluble contrast swallow shows accumulation in the mediastinum. If they don't mention this, they'll give either recent endoscopy or straining + acutely low BP and high HR, where you have to eliminate the other answers out of reasoning. - Water-soluble contrast (Gastrografin) is used in suspected perforation because the contrast will leak into the mediastinum and barium can cause mediastinitis. - In contrast (no pun intended), if patient has aspiration risk, we use barium and not Gastrografin because the latter can cause pneumonitis. - "Esophageal repair" is the Tx on NBME. Pretty straightforward.

Mallory-Weiss tear	- Classically retching/vomiting in an alcoholic that causes stretching and tearing of the gastroesophageal junction leading to **small amounts** of blood in vomitus. - Can also occur in patients with eating disorders or hyperemesis gravidarum. - It's to my observation on NBME material that the highest yield point about MW tear is that it does **not** cause high-volume hematemesis. If the vignette mentions high-volume blood, that is ruptured varix instead (discussed below).
Esophageal varices	- Enlarged and tortuous superficial esophageal veins due to portal hypertension in patient with cirrhosis or splenic vein thrombosis.   - The mechanism in the setting of cirrhosis: the **left gastric vein** connects the esophageal veins to the portal vein. In the setting of ↑ portal pressure, this pressure backs up to the left gastric vein, which backs up to the esophageal veins. Many USMLE Qs want "left gastric vein" as the answer for the vessel responsible for the varices. - In splenic vein thrombosis, the ↑ splenic venous pressure causes formation of collaterals to circumvent the thrombosis. This means nearby veins will form small tributaries/branches from the splenic vein. The left gastric vein is one of them → ↑ esophageal venous pressure. - It is to my observation on NBME Qs that 4/5 varices Qs will give a patient with **high-volume hematemesis.** In contrast, MW tear is 'just a little blood." I say 4/5, because 1/5 times you can get low-volume blood. The association is more or less: high-volume blood for esophageal condition always = varix, but small-volume can be either MW tear or varix. - 1/5 Qs might say "just a little blood" if the varix is friable but not overtly ruptured, but the vignette will be obvious (i.e., cirrhosis) + they might just ask for "left gastric vein" as the answer. - Ruptured varices are lethal ~50% of the time. The patient will vomit high volumes of blood. - Propranolol is prophylaxis, not for acute treatment. - Treatment is endoscopy + banding. - Octreotide can also be used for acute bleeding, but endoscopy + banding is best answer on USMLE if you are forced to choose.

	- There is a 2CK Q where patient has ↓↓ BP due to ruptured varix and the answer is "IV fluids," where endoscopy is the wrong answer. I've had a student say, "Wait but I thought you said endoscopy and banding is what we do first." And my response is, yeah, but you always have to address ABCs first on 2CK. They could by all means say patient has O2 sats of 50% and the answer would be give oxygen before fluids.
Diffuse esophageal spasm	- Idiopathic spasm of the esophagus. - All you need to know is that this that causes pain that can mimic angina pectoris, but patient will be negative for cardiac findings/disease.  - Barium shows a corkscrew appearance. - "Nutcracker esophagus" is a similar condition that is due to idiopathic hyperperistalsis and causes pain similar to DES.
Eosinophilic esophagitis	- Idiopathic allergic condition of the esophagus that presents with difficulty swallowing. - Biopsy shows eosinophils in epithelium (it's like no shit). - Endoscopy shows "trachealization" of the esophagus with concentric rings. - Patient might have history of asthma or atopy.
Schatzki ring	- Garbage diagnosis for USMLE. Doesn't exist. Mentioning it here as a negative for USMLE giving a fuck cuz otherwise someone will be like, "Mike u didn't mention Schatzki ring." - Technically a narrowing of the distal esophagus that could be considered as a differential alongside achalasia. Once again, never seen this assessed as a correct answer on any NBME material. Maybe just once as a distractor. Garbage diagnosis.

HY GI exocrine hormones quick review	
- Exocrine means secreted across epithelium into the GI tract. In contrast, endocrine means into the blood.	
Cholecystokinin (CCK)	- Produced by "neuroendocrine cells of the small bowel." - Does 3 things: 1) Causes gall bladder contraction; 2) Causes relaxation of sphincter of Oddi; 3) ↑ exocrine pancreas secretion of proteases and lipases into the duodenum. - Impetus for secretion is macronutrients (i.e., fat, protein, carbs) entering the duodenum from the pylorus, although fats are classically the major stimulator. - Even though in theory acid can lead to ↑ secretion of CCK, this is **wrong** on USMLE. It is **macronutrients** that cause ↑ CCK release on USMLE. - Answer on USMLE for the hormone that is responsible for pain in biliary colic (gallbladder pain from cholelithiasis, where a stone transiently obstructs entrance into the cystic duct during gallbladder contraction). - Step 1 NBME Q will give you pain from biliary colic after patient eats a fatty meal, and then they'll ask for where the hormone responsible is secreted → answer = "neuroendocrine cells of small bowel."
Secretin	- Produced by S cells of the duodenum, but USMLE doesn't give a fuck. - What they care about is that secretin causes ↑ **bicarb release** from the exocrine pancreas into the duodenum, and that **stomach acid** entering the duodenum from the pylorus is the impetus for secretion. - What USMLE will do is show you a line graph representing pancreatic secretions, where you can see there's ↑ bicarb release but no ↑ in proteases and lipases, and the answer is just "secretin." - Secretin should normally suppress gastrin levels. In patients with gastrinoma (Zollinger-Ellison syndrome), gastrin does not suppress in response to secretin. Some students think gastrin levels go up in response to secretin, which is not true. It is merely the case that gastrin levels remain high / do not suppress.
Histamine	- Produced by **enterochromaffin-like cells** of the GI tract. Weird name, but just deal with it. "Regular enterochromaffin" cells secrete serotonin in the GI tract. - Binds to H2 receptors on gastric parietal cells, stimulating acid secretion. - Enterochromaffin-like cell hyperplasia occurs in chronic gastritis (on NBME); this makes sense, since we have atrophy or destruction of parietal cells in chronic gastritis, meaning histamine production goes ↑ to compensate. - Cimetidine is HY H2-blocker that ↓ stomach acid secretion; can cause gynecomastia. It also inhibits P-450 (↑ serum levels of other drugs). Ranitidine does not inhibit P-450.
Gastrin	- Secreted by G cells. - ↑ acid via two main mechanisms: 1) directly agonizes gastrin receptors on parietal cells; 2) stimulates enterochromaffin-like cells to secrete histamine, which in turn ↑ stomach acid. - Gastrin levels rise in chronic gastritis and *H. pylori* infections. - Gastrinoma (aka Zollinger-Ellison syndrome) causes recurrent duodenal ulcers and sometimes jejunal or ileal ulcers; can be part of MEN 1.
Vasoactive intestinal peptide (VIP)	- Causes ↑ water content / volume of pancreatic secretions. - I'd say VIP is weirdly the highest yield GI hormone on 2CK Surg forms In particular. VIPoma shows up everywhere. - ~2/3 of Qs will be WDHA syndrome → Watery Diarrhea, Hypokalemia, Achlorhydria (↑ serum pH + low serum Cl^-). - ~1/3 of Qs will be nothing about diarrhea, but will mention ↑ serum pH + low K^+ + facial flushing. A 2CK NBME Q gives pH of 7.56 and potassium in the 2s + facial flushing. - USMLE will show graph of a hormone that causes ↑ volume of pancreatic secretions but pH and enzyme concentrations don't ↑ → answer = VIP.
Motilin	- Stimulates peristalsis in the absence of food. - Responsible for borborygmi (GI sounds).

	- Erythromycin (yes, the antibiotic) can agonize motilin receptors in patients with gastroparesis.
Glucose-dependent insulinotropic peptide	- Aka gastric inhibitory peptide (GIP). USMLE loves this hormone. - Causes ↑ insulin release in response to oral macronutrients (not just carbs). - What you need to know is that it is GIP that is responsible for oral glucose being utilized faster than IV glucose, since we get ↑ insulin spike if oral. USMLE will ask this two ways: 1) Two people both get 50g of glucose. One gets it orally. One gets it IV. The one who gets it orally utilizes it faster. Why? → answer = "glucose-dependent insulinotropic peptide." 2) Two people get 50g IV glucose. One also gets a small amount of oral fatty acids at the same time. The latter person utilizes the IV glucose faster. Why? → answer = "glucose-dependent insulinotropic peptide."

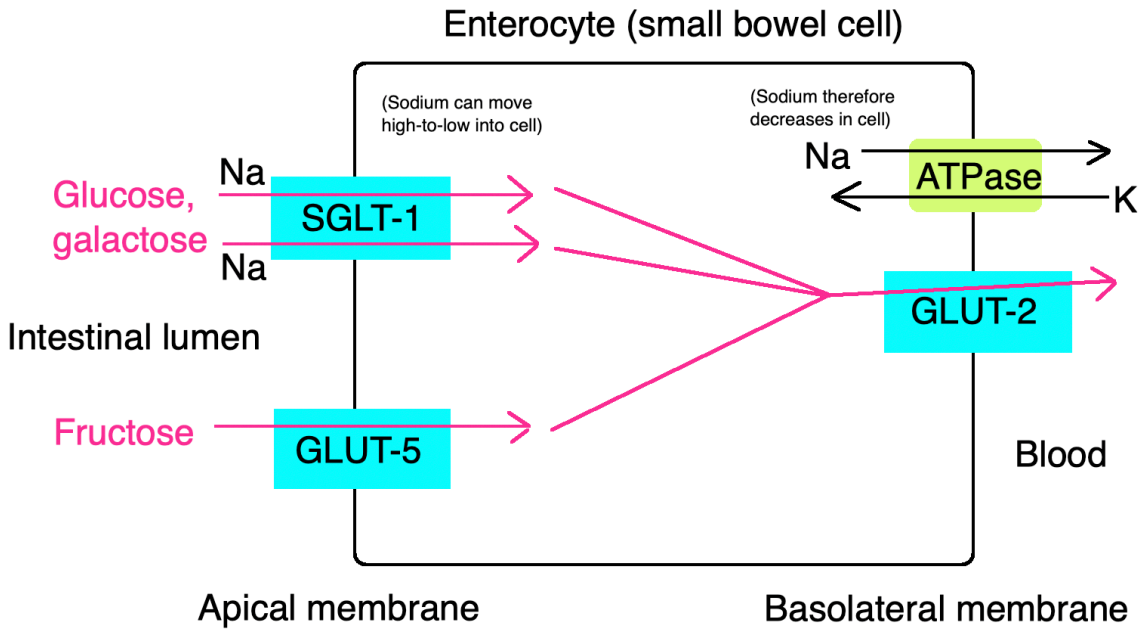
HY GI enzymes quick review	
Amylase	- Secreted by both salivary glands and the pancreas to digest starch into monosaccharides. - Pancreatic amylase is ↑ in acute pancreatitis. The weird detail I can communicate is that even though it's not as reliable or specific as lipase for pancreatitis, various 2CK NBME Qs will give only isolated ↑ amylase in some Qs, without even listing lipase. So if you get a Q where amylase is high and lipase isn't listed, don't be confused. It's still pancreatitis.
Lipase	- Secreted by both salivary glands (lingual lipase) and pancreas to break down fats. - ↑ in pancreatitis.
Pepsin	- Secreted by chief cells of gastric mucosa as pepsinogen, which is a zymogen (means pro-enzyme that is inactive). - Converted into active pepsin by stomach acid, which then breaks down protein.
Intrinsic factor	- Secreted by parietal cells (in addition to acid). - Binds vitamin B12 and enables its absorption at the terminal ileum. - ↓ in gastrectomy and pernicious anemia (discussed later).
Enterokinase	- Brush border enzyme that converts trypsinogen from the pancreas into trypsin.
Trypsin	- Secreted by exocrine pancreatic acinar cells as the zymogen trypsinogen. - Converted by **enterokinase** at the brush border into trypsin. - Trypsin is a protease that breaks down protein. - Once active trypsin is formed, it activates other zymogens (e.g., chymotrypsinogen) into active proteases. - USMLE will ask about enterokinase, followed by wanting you to know that once we've got active trypsin, it is the latter that now activates other things. Enterokinase merely gets things started.
Serotonin	- 90% of serotonin is secreted by enterochromaffin cells of the GI tract. - Metoclopramide is an anti-emetic + prokinetic agent (means ↑ peristalsis). It is a D2 antagonist but also an antagonist of serotonin 5HT3 and agonist of 5HT4 receptors. The effects on serotonin receptors ↑ gut peristalsis. - Serotonin is also produced by carcinoid tumors, which are neuroendocrine tumors of small bowel, appendix, or lungs that are S-100 (+) and consist of small blue cells. - Carcinoid syndrome will present as flushing, tachycardia, diaphoresis, and diarrhea. Tricuspid regurg can also be seen. - Diagnose with urinary 5-hydroxyindole acetic acid (5-HIAA). - Don't confuse with serotonin syndrome, which is a drug interaction (e.g., starting a monoamine oxidase inhibitor too soon after discontinuing an SSRI, or if a patient takes St John wort with an SSRI).
GLP-1	- Glucagon-like peptide 1. - Stimulates insulin secretion. - Exenatide + liraglutide are GLP-1 receptor agonists for type II diabetes.

DPP-4	- Dipeptidyl-peptidase 4. - Breaks down GLP-1. - Therefore, DPP-4 inhibitors ↑ GLP-1 and ↑ insulin release. - NBME asks which drug listed ↑ insulin release (i.e., is an insulin secretagogue); answer = linagliptin (DPP-4 inhibitor).

HY GI endocrine hormones quick review	
Insulin	- Produced by β-islet cells in the tail of pancreas. - ↓ blood glucose by two main mechanisms: 1) ↑ GLUT-4 on skeletal muscle and adipose tissue; 2) ↑ glucokinase activity in the liver (hexokinase equivalent at liver), which pulls glucose out of the blood to be stored as glycogen. - ↑ Fatty acid and protein synthesis. Also ↑ lipoprotein lipase activity (fat storage). - Causes dephosphorylation of enzymes (I talk about this in detail in my HY Biochem pdf). - Insulin secretion is HY: glucose enters β-islet cells via GLUT-2 → ↑ ATP production within β-islet cell → closure of ATP-gated K^+ channel on membrane of β-islet cell → K^+ builds up inside the cell → more positive charge in cell → depolarization of cell → causes Ca^{2+} to move into cell → triggers insulin vesicle efflux from cell. - Insulin is normally produced as pro-peptide that must have C-peptide cleaved off as part of the process. C-peptide and insulin are co-secreted, meaning their serum levels should match one another. If patient has ↑ insulin but ↓ C-peptide, answer = exogenous injection. If C-peptide is normal/high, insulin production is endogenous. First step is checking serum hypoglycemic levels ("serum hypoglycemic" = type II diabetes med that ↓ glucose, such as sulfonylureas and meglitinides). If serum hypoglycemics are negative, then do CT of abdomen to check for insulinoma. - Insulinoma (and ↑ insulin in general) cause **Whipple triad:** 1) hypoglycemia; 2) symptoms of hypoglycemia (tachycardia, tremulousness); 3) improves with a meal / gets worse between meals. - Insulin is absent in type I diabetes and ↑↑ in early type-II diabetes (i.e., hyperinsulinemia) with beta-islet cell hypertrophy). - Insulin inhibits ketone formation, so we have ↓ ketones in type II, but ↑ ketones in type I. - Q might give you easy vignette of type II diabetes + ask what you'd see in the patient → answer = "hyperinsulinemia," whereas "ketonuria" is wrong, for instance. HY stuff that's easy to get wrong. - Hyperinsulinemia causes anovulation / polycystic ovarian syndrome (see my Repro PDF). - USMLE can show you pic of acanthosis nigricans, which is almost always due to insulin resistance. - Patients with chronic pancreatitis and pancreatectomy can have diabetes (loss of pancreatic tail).
Glucagon	- Secreted by α-cells of the pancreatic tail. - Causes ↑ serum glucose and phosphorylation of enzymes. - USMLE wants you to know glucagon ↑ cAMP via G-alpha-S G-proteins. - Glucagonoma will present as ↑ serum glucose and a body rash called necrolytic migratory erythema. Don't confuse with the facial flushing seen with VIPoma; in addition, VIPoma doesn't ↑ glucose levels. - Patients who receive exogenous insulin can sometimes have prolonged or exaggerated hypoglycemic effects. If this occurs, an answer on USMLE for why this happens is "lack of counter-regulatory glucagon." Sounds weird, but it shows up more than once on NBMEs. Essentially, patients with diabetes, chronic pancreatitis, or pancreatectomy are not just prone to losing the β-islet cells, but they can also lose the α-cells. When glucose goes ↓, glucagon should go ↑ to compensate, but if this can't happen, glucose stays low. - Question on NBME with chronic pancreatitis (chronic burnout from repeated acute pancreatitis; discussed more later) wants you to know that the arrows are: ↓ insulin production, ↓ glucagon production, no-change peripheral response to insulin.
Somatostatin	- Secreted by delta-cells of pancreatic tail.

	- ↓ secretion of most GI hormones, as well as growth hormone. - Somatostatinoma presents as steatorrhea (probably due to ↓ pancreatic lipase secretion). - Octreotide is a somatostatin analogue that can be used in addition to endoscopic banding for esophageal varices Tx → leads to ↓ portal blood flow/pressure.
Ghrelin	- Produced by enteroendocrine cells of GI tract. - All you need to know is that this hormone makes you feel hungry. - Blood levels are highest **just at the start of the meal.** USMLE will show you a graph of ghrelin levels, and they ask where on the graph corresponds to the start of a meal, and the answer is at the peak. Not complicated. But I've seen innumerable students get this wrong.
Leptin	- Produced mostly by adipocytes. - Makes you feel full (i.e., opposite of ghrelin). - Also plays important role in hypothalamic/anterior pituitary secretion of gonadotropins. - Leptin is ↓ in those with low BMI/anorexia → ↓ GnRH → ↓ LH + ↓ FSH → amenorrhea.

Intestinal monosaccharide transport



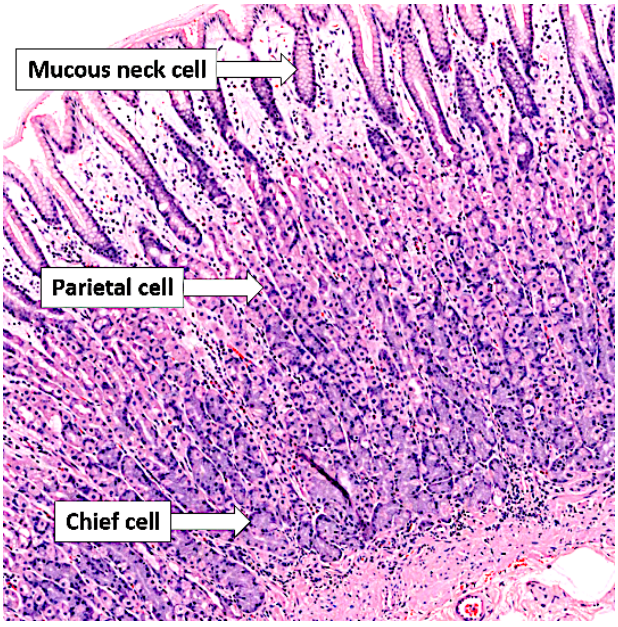
- GLUT transporters allow for movement via facilitated diffusion, where the sugars move down their concentration gradients (i.e., from high to low).
- SGLT-1 functions via secondary active transport, where a Na/K ATP-ase on the basolateral membrane pumps Na out of the enterocyte into the blood. This lowers Na within the enterocyte, creating a favorable high-to-low gradient for Na from the intestinal lumen into the cell. This gradient then drives the movement of glucose and galactose against their concentration gradients into the cell, allowing for more efficient absorption.

Peptic ulcers HY causes

- "Peptic ulcer" is an umbrella term that refers to both duodenal and gastric ulcers.
- Gastric ulcers cause pain immediately with meals (due to ↑ acid secretion). Patients can sometimes lose weight due to aversion to pain from eating.
- Duodenal ulcers cause pain 1-2 hours after meals (due to ↑ acid entering duodenum). With meals, the pylorus tightens, thereby relieving any residual discomfort. Patients may gain weight since eating ↓ pain.

Helicobacter pylori	- Responsible for almost all duodenal ulcers (>95%), whereas it causes a lower % (only >60%) of gastric ulcers. This is merely because the latter are caused by many other things as well, so we simply have ↓ fraction caused by *H. pylori.* In other words, there's no special tropism of *H. pylori* toward duodenal over gastric mucosa.

	- Mechanism for ulcers that shows up on USMLE is: "secretes proteinaceous substrates that damage mucosal lining." This is correct over "↑ gastric acid secretion" if both are listed side-by-side, even though *H. pylori* does ↑ gastrin levels, which ↑ acid secretion. - Produces urease, which causes ↑ ammonia production around the organism, allowing it to survive in the ↓ pH of the stomach. - Antral/pyloric ulcers can lead to gastric outlet obstruction. They will mention this in a Surg Q as a "succussion splash." - ↑ risk of MALT lymphoma, a type of B-cell lymphoma. - Diagnose *H. pylori* with urease breath test or stool antigen. - Treat *H. pylori* with CAP → clarithromycin, amoxicillin, PPI (e.g., omeprazole). - USMLE really doesn't give a fuck about the treatment, but CAP is safe to know. Metronidazole, tetracycline, bismuth, and PPI tetrad is used if CAP fails (students ask about those other drugs). - Perforated duodenal ulcer will present as sudden-onset rigid abdomen (involuntary guarding) and abnormal vitals in patient with history of abdo pain after meals (means Hx of duodenal ulcer). USMLE wants "X-rays of the chest and abdomen" to look for air under the diaphragm (HY finding that indicates ruptured viscus).
Gastrinoma	- Zollinger-Ellison syndrome causes recurrent duodenal ulcers and sometimes jejunal or ileal ulcers. - Can be part of MEN1 or idiopathic. - *H. pylori* is more common than gastrinoma. As mentioned above, >95% of duodenal ulcers are due to *H. pylori*. There is an NBME Q where they give older male with a duodenal ulcer + no other information + they ask for most likely cause → answer = testing for *H. pylori*; gastrinoma is wrong. - Gastrinoma can also present with diarrhea. I've seen this detail show up in innumerable NBME Qs. So they might say patient with history of recurrent abdominal pain and diarrhea. - Vignettes can be tricky with gastrinoma and tell you the patient has 8-10 watery stools daily, where you say, "That sounds like VIPoma." But they'll also tell you the patient has history of abdo pain after meals.
NSAIDs	- Cause gastric ulcers. I haven't seen these cause duodenal ulcers on USMLE. - Prostaglandins are necessary for stimulation of gastric alkaline mucous production and maintenance of the gastric lining. NSAIDs → ↓ prostaglandin production → disruption of gastric lining. This can lead to both ulcers and irritation (gastritis). - USMLE wants you to know that PPIs are first-line for ulcer treatment and prevention in general, but that **misoprostol** is a PGE1 analogue that is used after PPIs in NSAID-induced ulcers (i.e., we're replenishing the ↓ prostaglandin from NSAIDs).
Burns	- Called Curling ulcers (think: curling irons are hot). - Loss of fluid post-burns from evaporation → ↓ blood flow to stomach → ischemic ulcers. - Rare, but just know they exist.
↑ ICP	- Called Cushing ulcers. - Head trauma or brain tumor → ↑ intracranial pressure (ICP) → ↑ parasympathetic outflow to stomach → ↑ ACh binding to muscarinic receptors at parietal cells → ↑ acid production. - Don't forget that the 3 synergistic mechanism for acid production are 1) gastrin binding directly to gastrin receptors on parietal cells; 2) gastrin stimulating enterochromaffin-like cells to secrete histamine, which then binds to H2 receptors on parietal cells; and 3) direct parasympathetic activity, where ACh binds to muscarinic receptors on parietal cells.
Smoking/Alcohol	- Not tested as overt causes of ulcers on USMLE. But you should know that smoking and alcohol are believed to *decrease healing* of pre-existing ulcers.

<table>
<tr><th colspan="2">Gastritis causes + HY points</th></tr>
<tr><td colspan="2">- Acute gastritis will present as irritation leading to bleeding of gastric mucosa (e.g., from NSAIDs).
- Chronic gastritis will be an atrophy or autoimmune destruction of the mucosa associated with ↓ mucosal thickness, ↓ HCl production (achlorhydria), ↑ gastrin production, and enterchromaffin-like cell hyperplasia (↑ histamine production to compensate for ↓ acid). NBME will give you this constellation of findings and then just simply have “chronic gastritis” as the answer, with acute gastritis not even listed. It’s not hard, but I’ve seen students miss this a lot.</td></tr>
<tr><td>H. pylori</td><td>- Causes what is referred to as “Type B gastritis,” which is inflammation tending to affect the antrum of the stomach.
- Can lead to pyloric channel ulcers + gastric outlet obstruction. This will present on 2CK Surg forms as a “succussion splash.”</td></tr>
<tr><td>Pernicious anemia</td><td>- Causes what’s referred to as type A gastritis,” which affects mostly the fundus/body of the stomach.
- Autoimmune antibody-mediated destruction of parietal cells.
- Sometimes antibodies can be against intrinsic factor.
- Ultra-HY cause of B12 deficiency (↑ MCV + hypersegmented neutrophils +/- neuropathy) on USMLE.
- Associated with other autoimmune diseases, e.g., vitiligo. So if they give you ↑ MCV in patient with, e.g., type I diabetes, you should think, “Autoimmune diseases go together, so if the patient has one AA disease, he/she has ↑ propensity for others.” Don’t worry about strict HLA associations here.
 

- There is Q on an NBME exam where they ask you to identify the parietal cell. Histo overall is LY for Step 1, but they want you to know that parietal cells are the “midway” cells between the gastric surface superficially and the deeper, basophilic chief cells. Notice how the parietal cells are slightly lighter/more eosinophilic (pink) in comparison to the chief cells, which are darker/more basophilic (purple).</td></tr>
<tr><td>NSAIDs</td><td>- Answer on USMLE for GI bleeding in someone taking, e.g., indomethacin or naproxen.
- Causes type A gastritis, affecting the fundus/body. USMLE doesn’t specifically give a fuck, but you should basically be like, “H. pylori causes antral gastritis, whereas other causes like NSAIDs and pernicious anemia are fundus/body of stomach.”
- Prostaglandins are necessary for stimulation of gastric alkaline mucous production by foveolar cells and maintenance of the gastric lining. NSAIDs → ↓ prostaglandin production → disruption of gastric lining. This can lead to both gastritis (inflammation) and ulcers.
- As mentioned above for ulcers, misoprostol can be used in these patients following PPI.</td></tr>
</table>

Gastric cancer points
- Perpetuated over the years as "intestinal vs diffuse" types. USMLE doesn't care. What you need to know is that histo will show **signet-ring cells that contain mucin.** And you don't need to know the histo for this either. - Grossly, can cause a linitis plastica, which is a leather bottle appearance to the stomach. - Gastric metastases can spread hematogenously to the ovaries (Krukenberg tumors). These will show up as bilateral ovarian lesions that show, you guessed it, **signet-ring cells that contain mucin.** - ↑ incidence of gastric cancer in Japan due to ↑ nitrosamines/smoked foods. - MALT lymphoma is a B cell lymphoma that can be caused by *H. pylori*. This does not have signet ring cells. - Acanthosis nigricans can be associated with gastric cancer (even though most of the time, it just means insulin resistance). - Virchow node (Troisier sign) → palpable supraclavicular lymph node that can reflect visceral malignancy, especially gastric cancer.

Hyperbilirubinemia	
- Unconjugated hyperbilirubinemia is due to causes before the liver, such as high RBC turnover or inability to conjugate at the liver. - Conjugated hyperbilirubinemia is due to post-hepatic obstruction. - If the problem is at the liver (intra-hepatic), there can be either indirect or direct hyperbilirubinemia.	
Unconjugated (indirect)	- Bilirubin is produced from breakdown of heme from RBCs at the spleen. It will initially be "unconjugated" in this form, where it merely leaves the spleen non-covalently bound to albumin and is not water-soluble – i.e., it won't show up in the urine. Unconjugated bilirubin is aka indirect bilirubin. - In the setting of hemolysis or ↑ RBC turnover (i.e., sickle cell, **hereditary spherocytosis, blood given during surgery**), where we have ↑ RBC breakdown, we get ↑ indirect bilirubin. - Once it arrives at the liver, it is taken up by the liver and conjugated to glucuronide, making it water-soluble. Conjugated bilirubin is aka direct bilirubin. - If the there's a problem with uptake at the liver (i.e., acute hepatitis), or there is deficient conjugation enzyme (Gilbert syndrome, Crigler-Najjar), indirect bilirubin also goes ↑. - Hemolysis or ↑ RBC turnover can cause direct bilirubin to go ↑ sometimes as well, but the shift will be more toward indirect being ↑. I point this out because students often get confused by this, but if we have ↑ indirect bilirubin going into the liver, then this means ↑ direct bilirubin going out. A 2CK NBME Q gives 9 packs of RBCs given during surgery → days later, total bilirubin is 5.0 and direct bilirubin 2.3 → answer = "overproduction of bilirubin." - In regard to neonatal labs (i.e., first month of life), I should note that hematocrit % can be in the 50s (NR 45-61; in adults, NR is 40-50). 2CK, for instance, will often show Hct as 56% in a newborn and the student is like "Wow that's high!" where they think there's neonatal polycythemia or some other pathology, but it's actually normal. Unconjugated jaundice can occur in neonates because of ↑ RBC turnover where HbF is replaced with HbA.
Conjugated (direct)	- Secreted into bile from the liver. If we have a bile duct obstruction, we get ↑ direct bilirubin. - ALP also goes ↑ in bile duct obstruction because it is secreted by bile duct epithelium. This means the combo of "↑ ALP + ↑ direct bilirubin" is very buzzy for bile duct obstruction. - If ALP is ↑ but direct bilirubin not high, this can be due to things like bone fractures or Paget disease. - If direct bilirubin is ↑ but ALP is not high, this can be due to intra-hepatic pathology like viral hepatitis. - GGT will also go up with bile duct obstruction, since it is also secreted by bile duct epithelium, but USMLE actually rarely mentions this one. What they want you to know is GGT spikes with acute alcohol consumption / binge drinking. - In the event of bile duct obstruction, not only will ALP, GGT, and direct bilirubin go ↑, but the urine becomes darker from ↑ direct bilirubin in it (and urobilin, which comes from direct bilirubin, but USMLE doesn't assess this). Direct bilirubin, since it is water-soluble, shows up in the urine. In addition, stools become lighter/pale (aka "acholic stools"), since

	there is ↓ direct bilirubin making it to the intestines, which means ↓ stercobilin production (pigmentation in stool). In other words, "dark urine + pale stools" is buzzy for bile duct obstruction the same way "↑ ALP and ↑ direct bilirubin" is. - Acute hepatitis, as mentioned above, can also cause ↑ direct bilirubin due to ↓ secretion into bile, in addition to ↑ indirect. But this makes sense, since it is literally an intra-hepatic pathology. - Highest yield cause of ↑↑ direct bilirubin on USMLE is biliary atresia in neonates (discussed later). This is all over the place, whereas Crigler-Najjar (↑↑ indirect bilirubin in neonates) is nonexistent. - Dubin-Johnson and Rotor syndromes are virtually nonexistent on USMLE but cause ↑ direct bilirubin in adults due to ↓ bile excretory pumps at the liver. Students get hysterical about these because they sound weird, but they're LY. Dubin-Johnson is asked once on a new 2CK NBME form. But apart from that, LY. - Cholangitis (inflammation of bile ducts), choledocholithiasis (stone in biliary tree), choledochal cyst, head of pancreas cancer (impingement on common bile duct), and cholangiocarcinoma (bile duct cancer) are all HY causes of bile duct obstruction. I discuss all of these conditions in detail below.

Bilirubin uptake/secretion diseases	
Gilbert	- Pronounced "Jeel-BEAR, not "GILL-burt." Gastroenterologist I met once tripped out over med students pronouncing this wrong. - Partial deficiency of bilirubin uptake enzyme at the liver (UDP glucuronosyltransferase). - Presents as isolated ↑ indirect bilirubin with yellow eyes in young adult with stress factor, such as studying for exams, or recent surgery/trauma. The patient will otherwise be completely healthy. - No treatment necessary. - One NBME Q gives this post-appendectomy, which confuses students. Remember that surgery is a stressor/trauma. A wrong answer for this Q is bile duct trauma, since the latter would be direct, not indirect, in hyperbilirubinemia.
Crigler-Najjar	- Near-absence of UDP-glucuronosyltransferase causing ↑↑ indirect bilirubin in neonate. - Nonexistent yieldness but students ask about this.
Dubin-Johnson	- ↓ excretion of bilirubin into bile at the liver due to ↓ secretory pumps. - Causes ↑ direct bilirubin in otherwise healthy adult. - Can cause black liver in theory. - Asked once on new 2CK NBME. Apart from that, nonexistent on USMLE material.
Rotor	- Same as Dubin-Johnson but no black liver. - Nonexistent diagnosis on USMLE. Students get maniacal over this supposedly not having a black liver, whereas Dubin-Johnson has a black liver – "Rotor! That's the one that has no black liver but Dubin-Johnson does!" Relax. Try memorizing stuff that actually increases your score rather than dumb shit that just makes you feel better.
Biliary atresia	- One of the highest yield Peds diagnoses. Answer on USMLE for ↑↑ direct bilirubin in a kid under the age of 6 weeks. - Caused by lack of development of the intrahepatic bile ductules and biliary tree. - USS will be done first, but USMLE wants **liver biopsy** to confirm the diagnosis (shows up on NBME form; cholangiography not listed, but I've had students ask about the latter). - Kasai procedure is done to treat, followed by liver transplant if unsuccessful.

Autoimmune liver conditions	
Primary biliary cirrhosis	- Inflammation of bile ducts within the liver, leading to their destruction. - Answer in a woman 20s-50s who has generalized pruritis, ↑ serum cholesterol, ↑ ALP, ↑ direct bilirubin. - USMLE loves to mention **Hx of autoimmune disease in the patient or family member** (because autoimmune diseases go together). So they'll say she has type I diabetes mellitus, SLE, or vitiligo; or her brother has RA, etc. - They can mention a stone is present in the gallbladder on ultrasound, which gets some students real emotional / confused, but it makes sense since patient has ↑ cholesterol. It's just a distractor point and doesn't relate to the actual diagnosis at hand. - Diagnose with anti-mitochondrial antibodies as first step, followed by liver biopsy to confirm. - Initial Tx = ursodeoxycholic acid (ursodiol). - New NBME material for Step 1 asks a couple Qs on fat-soluble vitamin malabsorption for PBC (deficiency of A and D on NBME for each Q, respectively, with no info supporting those presentations; they just ask for hypothetical vitamin deficiency). In theory, would be due to biliary obstruction, where ↓ bile entering small bowel merely means ↓ fat absorption. This is nothing special to PBC, but I mention it because it's asked twice.
Autoimmune hepatitis	- Young adult with ↑ LFTs who has (+) anti-smooth muscle antibodies.

HY Hepatobiliary conditions	
Pancreatic cancer	- Head of pancreas cancer impinges on common bile duct, resulting in obstructive jaundice (↑ ALP + ↑ direct bilirubin) in smoker with weight loss, or in patient who had gallbladder taken out years ago (so obstruction clearly can't be due to a stone in common bile duct). - Patient will not be febrile; can have dull abdominal pain. - Courvoisier sign is a painless, palpable gallbladder in an afebrile patient who's jaundiced. This is pancreatic cancer until proven otherwise and is pass-level. - **Pancreatic enzymes are normal in pancreatic cancer.** - USMLE wants **CT of the abdomen** to diagnose. - Whipple procedure is done to remove head of pancreas. If the cancer is isolated to the tail, distal pancreatectomy is the answer. - Patients with pancreatectomy need pancreatic enzyme supplementation. The exam can write this as "pancrelipase."
Cholangiocarcinoma	- Bile duct cancer. Answer on USMLE if the vignette sounds like pancreatic cancer but they tell you in the last line CT is negative. - Answer is ERCP as next best step. - Smoking is risk factor, same as pancreatic cancer. - Can rarely be caused by *Clonorchis sinensis* (trematode).
Cholelithiasis	- Stones in the gallbladder. - Presents with biliary colic, which is acute-onset waxing/waning spasm-like pain in the epigastrium or RUQ. - Pain is due to **cholecystokinin** causing gall bladder contractions, where a stone within the gallbladder transiently obstructs flow of bile into cystic duct. They ask this on 2CK as well, where patient will biliary colic and answer is "obstruction of cystic duct opening by stone." Not dramatic. - Classic demographic is 4Fs = Fat, Forties, Female, Fertile for cholesterol stones. This is because estrogen upregulates HMG-CoA reductase, causing ↑ cholesterol synthesis and secretion into bile. NBME will give you standard vignette of 4Fs, and then the answer will just be "increased secretion of cholesterol into bile."

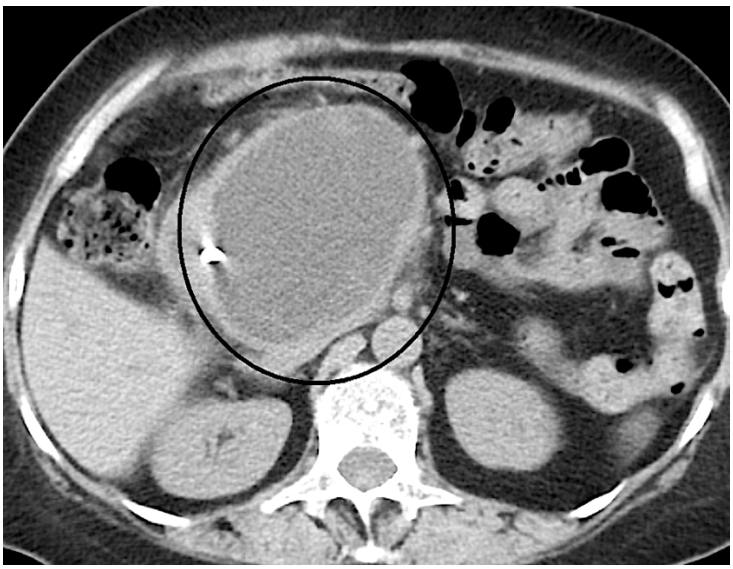
	- Another NBME Q gives vignette of cholelithiasis, and then rather than asking for the diagnosis, they ask what the patient most likely has → answer = "lithogenic bile." Slightly awkward, but means promoting the formation of stones. - ↑ cholesterol stones in pregnancy not just due to estrogen effect but also because progesterone slows biliary peristalsis (biliary sludging). Tangentially, progesterone also slows ureteral peristalsis, which is why there's ↑ risk of pyelonephritis (as discussed in the HY Renal PDF). - Cholesterol stones most common, but pigment (calcium bilirubinate) stones are exceedingly HY for hereditary spherocytosis and sickle cell, due to ↑ RBC turnover. - USMLE wants you to know that splenectomy is Tx for hereditary spherocytosis to ↓ incidence of cholelithiasis. Sometimes the vignette wants "cholecystectomy + splenectomy" as combined Tx. Or if they say this was in a parent's Hx in patient with low Hb, you know the Dx is hereditary spherocytosis (autosomal dominant). - Infections can also sometimes cause pigment stones, but LY. - Diagnose with abdominal ultrasound. - Ursodeoxycholic acid (ursodiol) ↓ **secretion of cholesterol into bile.** USMLE just wants you to know this MOA + that it can be used in patients with asymptomatic cholelithiasis, those declining cholecystectomy, and in pregnancy.
Cholecystitis	- Cholelithiasis + fever. - Vignette will sound exactly like cholelithiasis, but if we add a fever on top of it, we now just call it cholecystitis. - Virtually always due to obstruction by a stone + infection as a result. - Diagnose with abdominal ultrasound, showing the stones. - If ultrasound is negative, do a HIDA scan. - Tx is cholecystectomy. - Emphysematous cholecystitis is air visualized within the biliary wall. USMLE wants *Clostridium perfringens* as the organism (can cause gas gangrene anywhere) in this case. - Chronic cholecystitis is calcification of the gallbladder (aka chronic calculous cholecystitis, or "porcelain gallbladder"). It is due to repeated bouts of acute cholecystitis. Cholecystectomy is indicated due to ↑ risk of developing cancer of the gallbladder. - ALP and bilirubin will not be increased 19/20 questions. This is because inflammation of the gallbladder doesn't mean we have any form of common bile duct obstruction. However there is one nonsense Q on a 2CK NBME where 4 of the answers are wildly wrong, with correct answer being cholecystitis in setting of high ALP + direct bilirubin. Since cholecystitis is caused by stones virtually always, the implication is patients can occasionally have concurrent choledocholithiasis.
Choledocholithiasis	- Stones anywhere within the biliary tree. Don't confuse with cholelithiasis. - Obstructive jaundice (↑ ALP + ↑ direct bilirubin) in patient who has Hx of cholelithiasis or Hx of cholecystectomy performed within the past 2 weeks. - Regarding the latter, the Q can say patient had cholecystectomy performed a week ago + "intra-operative cholangiography was not performed." The implication is bile duct patency is supposed to be visualized during cholecystectomy to ensure there isn't a stone retained within the biliary tree. However, the Q need not say "intra-operative cholangiography was not performed." This explication merely makes the Q pass-level. - **USMLE wants abdominal ultrasound followed by ERCP** as the answer. ERCP tends to show up as what they want, but there is a Q floating around somewhere that asks for ultrasound first, so know that sequence.

	- As I talked about earlier, I've never seen MRCP as a correct answer. - Being able to discern choledocholithiasis from pancreatic cancer is vital for USMLE. If they give you obstructive jaundice but say the patient is a heavy smoker with weight loss, or hasn't had a gallbladder for 25 years, you know it's head of pancreas cancer, not choledocholithiasis. And then, once again, if it sounds like pancreatic cancer but the CT is negative, that's cholangiocarcinoma and we do ERCP.
Gallstone pancreatitis	- **Gallstone pancreatitis** is a type of choledocholithiasis in which the stone has descended distally enough in the common bile duct that it now obstructs the **hepatopancreatic ampulla.** This results in a backflow of pancreatic enzymes to the pancreas causing damage → acute pancreatitis. - USMLE wants "hepatopancreatic ampulla" as correct over "common bile duct" if they ask you to choose location of the obstruction. Slightly odd, since the stone is still in the common bile duct, but hepatopancreatic ampulla blockage is *why* we have the pancreatitis. - Vignette will give obstructive jaundice *but also high pancreatic enzymes*. - In other words, the Q will give ↑ ALP, ↑ direct bilirubin, and ↑ amylase / lipase. - I repeat that pancreatic cancer will never have elevated enzymes on USMLE, so this is a pass-level means to distinguish. - For gallstone pancreatitis, do **USS → CT of abdomen → ERCP** if they force you to choose based on the answer choices. Ultra-annoying, but USMLE will have different answers. A 2CK NBME Q has CT as answer prior to ERCP. But I believe it's on the Free 120 that USS is correct over CT. So this causes confusion with students. So go with USS → CT → ERCP.
Cholangitis	- Inflammation of the bile ducts. You must know **Charcot triad for cholangitis: 1) fever, 2) jaundice, 3) abdominal pain.** We classically learn this as "RUQ/epigastric pain," but I can tell you USMLE doesn't give a fuck and will just say "abdominal pain." I've had students get cholangitis Qs wrong because they're like, "I thought it had to be RUQ pain though." And I'm like, yeah, I agree, but USMLE doesn't give a fuck. - Will present 3 ways on USMLE: - 2/5 Qs will be ascending cholangitis, which is infection from ascension of bowel flora (*E. coli*, *Bacteroides*) up the common bile duct. - 2/5 Qs will be primary sclerosing cholangitis, which is inflammation of the bile ducts in the setting of IBD (always due to ulcerative colitis on USMLE, but can rarely be due to Crohn in real life). p-ANCA can be positive, and there can be a beaded appearance of the common bile duct on cholangiogram. On 2CK NBME 10, there is a Q on primary sclerosing cholangitis where the patient is afebrile (rare; i.e., we don't have full Charcot triad), but they say patient has UC + obstructive jaundice + ERCP shows "narrowing of the bile ducts," so the answer is evidently PSC. - 1/5 Qs will just give you a patient with history of cholelithiasis, where the implication is a stone has descended causing choledocholithiasis, which in turn led to obstruction + inflammation of the common bile duct. - USMLE wants antibiotics + **ERCP to diagnose and treat.** Biliary drainage by ERCP ↓ morbidity and mortality.
Acute pancreatitis	- ↑ amylase and/or lipase in patient who has abdominal pain. - Pain need not radiate to the back. - Alcoholism and stones are two biggest risk factors. - Hyperlipidemia and hypercalcemia can also occasionally be causes. - Can be caused by drugs (didanosine is HIV NRTI; exenatide is a GLP-1 analogue for diabetes). - Scorpion sting and mumps as causes show up on USMLE, but "show up" doesn't = HY. In other words, if someone asks you for risk factors for

pancreatitis and you immediately yell "scorpion sting!" before smoking and alcohol you'll come off as a low-IQ cuck.
- Can be caused by choledocholithiasis (gallstone pancreatitis; ↑ pancreatic enzymes in setting of ↑ ALP and ↑ direct bilirubin).
- Results in enzymatic fat necrosis.
- You don't need to memorize Ranson criteria (elaborate criteria for Surg to determine prognosis in pancreatitis). What you need to know are two main points:

> 1) ↓ Serum calcium and ↑ glucose are bad prognostic indicators. In fact, I'd say low calcium is quite possibly the highest yield variable on USMLE for pancreatitis. There are many long/vague pancreatitis Qs out there, where they'll mention ↓ calcium in the stem, and you're like "Boom pancreatitis."
> 2) The degree of lipase and/or amylase elevation doesn't correlate with prognosis. In other words, if amylase is 1200 versus 10,000, that's not important for patient outcome. The rest of the Ranson criteria is more Surg rotation masturbation info and otherwise nonsense for USMLE.

- It's to my observation NBME Qs will sometimes list high amylase without mentioning lipase for whatever reason. Lipase is more specific for pancreatitis, but it's what NBME sometimes does, so don't be confused.
- HY initial management for 2CK is a triad: 1) NPO (nil per os; or nothing by mouth; this can be written as simply **bowel rest** as NBME answer); 2) NG tube decompression; 3) IV fluids with normal saline. This triad is not specific for pancreatitis, and can be done for a number of GI-related conditions, including cholangitis and stones. But it just tends to get tested a lot for pancreatitis.
- After this initial triad, "CT scan of the abdomen" is answer on NBME as next best step to look for any degree of fluid collection. If a pancreatic pseudocyst is present, NBME answer is to drain this by ERCP.
- It's to my observation USMLE doesn't assess antibiotics as part of pancreatitis Tx, but apparently the carbapenems (like imipenem) have fantastic penetration of pancreatic tissue.



- A 2CK NBME Q gives pancreatitis + similar image as above, and the answer is just ERCP (for drainage of the pseudocyst within black circle).

	- A necrosectomy is removal of pancreatic tissue in the setting of high fraction of necrosis. Patients with pancreatectomy require enzyme replacement (pancrelipase). - Acute pancreatitis is HY cause of ARDS.
Chronic pancreatitis	- **Steatorrhea in alcoholic** or patient with recurrent acute pancreatitis episodes → results in pancreatic burnout with ↓ production of pancreatic enzymes → ↓ proteases/lipases → malabsorption. - In other words, we have **normal** pancreatic enzymes – i.e., they are **not** elevated. - There is no pain in chronic pancreatitis. Think of this condition as effectively "cirrhosis but of the pancreas." - Question can say alcoholic has ↑ fat and protein seen in stool. - In addition to burnout of the exocrine pancreas leading to malabsorption and steatorrhea, the endocrine pancreas can also burnout leading to diabetes mellitus. - D-xylose test is normal. This is a monosaccharide that is readily absorbed by the small bowel insofar as the intestinal lining is intact. So if we get steatorrhea + normal D-xylose, this can be due to pancreatic insufficiency or lactose intolerance. If D-xylose test is abnormal, we know the malabsorption is due to an intestinal lining issue, such as Celiac or Crohn. - If the USMLE gives you a CT for chronic pancreatitis, they will show you calcification within the pancreas. - Tx = pancreatic enzyme supplementation (pancrelipase).
Bile duct leak	- The answer on 2CK Surg if they give you patient with cholecystectomy within the past week who now has fever, abdominal pain, and ↑ direct bilirubin. - During the surgical anastomoses created during cholecystectomy, sometimes there can be a post-op bile leak, literally. Weird diagnosis but it shows up on occasion.
Sphincter of Oddi dysfunction	- The answer on USMLE if the Q sounds like gallstone pancreatitis (i.e., ↑ ALP, ↑ direct bilirubin, ↑ pancreatic enzymes) but the patient hasn't had a gallbladder for many years (so it can't be a stone causing the obstruction).
Gallstone ileus	- Gallstone enters small bowel and causes an intestinal obstruction. - The answer on USMLE if they say "air in the biliary tree" (pneumobilia). - Can be caused by cholecystoduodenal fistula, which is when a patient with cholelithiasis can develop a conduit between the gallbladder and small bowel. - What the USMLE will do is give you a long, rambling paragraph where they say "air in the biliary tree" in the last line. The answer will then just be "gallstone ileus" straight up, or it will be "cholecystoduodenal fistula."

Choledochal cyst	- Asked on 2CK Surg. What they're going to do is give you a long, nonsense paragraph with ↑ ALP and ↑ direct bilirubin + they say "CT shows a cystic structure within the biliary tree." Answer = "simple excision of the cyst." Students are huh, what's going on here? → choledochal cyst. You have to just excise it. Bullshit/dumb diagnosis. Take it up with NBME not me.
Primary biliary cirrhosis	- As mentioned earlier, this will be a woman 20s-50s who has generalized pruritis, ↑ cholesterol, ↑ ALP, ↑ direct bilirubin, and Hx of autoimmune disease in her or a relative. - Diagnose with anti-mitochondrial antibodies as first step, followed by liver biopsy to confirm. - Initial Tx = ursodeoxycholic acid (ursodiol).
Spontaneous bacterial peritonitis (SBP)	- One of the highest yield diagnoses on 2CK. - Bacterial infection of peritoneal fluid by mixed enteric flora. - The answer on USMLE when they give you diffuse abdominal pain and fever in one of the three following scenarios: 1) cirrhosis; 2) recent peritoneal dialysis; 3) nephrotic syndrome. - Shifting dullness" or a "fluid wave" are buzzy for ascites, but questions will often omit these descriptors from stems. What they care about is you identifying when a patient either has a major risk factor for ascites, or are aware of a peritoneal intervention as the cause for the presentation. - In other words, they can tell you a patient has cirrhosis + abdo pain + fever, where they don't have to mention a fluid wave, but you just have to infer, "Well he/she clearly has major risk factor for ascites, so this sounds like SBP." - There is a 2CK Peds Q where they say kid with minimal change disease (nephrotic syndrome) has abdo pain + fever → answer = spontaneous bacterial peritonitis. - Next best step in diagnosis is abdominal **paracentesis**. This refers to aspiration of fluid from the peritoneal cavity. Do not confuse this with pericardiocentesis. "Paracentesis" as an answer shows up everywhere, and I've seen students avoid it because they're like, "What? I thought that meant pericardiocentesis." - After the paracentesis is done, USMLE wants a very specific order for what to do next. **Choose "white cell count and differential" first if it's listed, followed by "gram stain and culture of the fluid."** - This order is important because "gram stain and culture of the fluid" is a correct answer in one NBME Q but wrong answer in another NBME Q, where "white cell count and differential" is correct to do first. - The reason white cell count and differential is done first is because SBP is diagnosed when paracentesis shows >250 WBCs/μL. - There is a new 2CK Q where they give ascites in patient with cirrhosis, but do not mention fever or abdominal pain. However, they say paracentesis shows 900 WBCs/μL → answer = "antibiotic therapy." It should be made clear though that ~6/7 SBP Qs will give abdo pain and fever. - Treatment is ceftriaxone.
Focal nodular hyperplasia	- Obscure diagnosis where they tell you adult has an abdominal CT for unrelated reason and has a 1-2-cm hepatic lesion **with a central scar.** There will be zero mention of trauma or infection. Every student says WTF. - This is a lesion of hepatocellular hyperplasia **that does not require treatment.** This is the answer on NBME, where they tell you CT shows hepatic lesion with central scar. Answer = "no further diagnostic studies indicated."

HY cirrhosis points
- Cirrhosis is a small, shrunken, burnt out liver due to chronic disease. - HY causes are alcoholism, HepB/C, Wilson disease, hemochromatosis, NASH, α1-antitrypsin deficiency etc. - "Burned out" means LFTs are normal or low – i.e., there is not transaminitis as with acute hepatitis. - **USMLE likes ↑ PT** and ↓ clotting factor synthesis in cirrhotic patients. PTT will go up also, but for some reason high PT is what often shows up in Qs. - Hyperammonemia occurs due to ↓ urea cycle activity (normally occurs in liver). This can cause hepatic encephalopathy (confusion) and asterixis ("hepatic flap" of the hands). - USMLE likes acute exacerbations of hyperammonemia caused by **GI bleeds** → ↑ ammonia absorption. - Spontaneous bacterial peritonitis (SBP) is ↑↑ HY on 2CK, as discussed above. - Esophageal varices from ↑ portal pressure that backs up to **left gastric vein** (discussed earlier). - Caput medusae are visible periumbilical veins (superior epigastric veins).

Systemic inflammatory response syndrome (SIRS)	
- Knowing this short table is vital for understanding many IM and Surg Qs on 2CK. If you're studying for Step 1, I'd say it's still worth knowing this. - In the setting of stress (i.e., due to trauma, surgery, autoimmune flare, infection), catecholamines and ↑ sympathetic activity might shift the patient's vitals out of the normal range. - The reason knowing SIRS is important is because the patient can have abnormal vitals **without** having an infection.	
SIRS	- 2 or more of the following: - Temperature <36C or >38 (<96. 8F or >100.4). - HR >90. - RR >20. - WBCs <4,000 or >12,000.
Sepsis	- SIRS + source of infection.
Septic shock	- Sepsis + low BP.
- Sometimes you will see a patient's vitals slightly out of the normal range in the setting of trauma, surgery, or autoimmune flares, and you have to be able to say, "There's no infection. That's just SIRS from sympathetic activation." - Knowing if a patient is septic is important for management of patients on 2CK, where sometimes antibiotic regimens are stepped up. For example, when treating PID, if the patient is septic, intravenous ceftriaxone and azithromycin is correct on one of the 2CK NBMEs; IM ceftriaxone and oral azithromycin is wrong. This is because the latter is for most patients who have PID but aren't septic. - Ceftriaxone is frequently an answer on 2CK for in-hospital patients who are septic from a variety of community-acquired conditions, e.g., pneumonia, pyelonephritis, prostatitis. For instance, community-acquired pneumonia is empirically treated with azithromycin (on 2CK NBME 8), but if patient is septic, we can go straight to ceftriaxone (have seen this more than once on 2CK NBMEs). - For hospital-acquired infections in which patients are septic, NBME goes hard-hitting with vancomycin PLUS ceftazidime or cefepime. This regimen covers MRSA and *Pseudomonas*.	

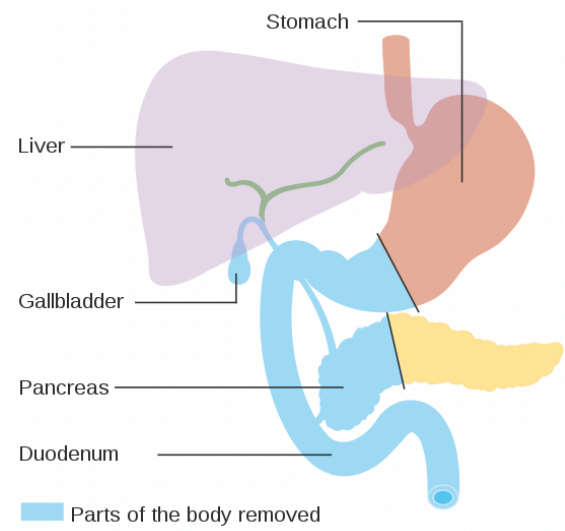
Miscellaneous 2CK Surgery intestinal diagnoses	
Adhesions	- One of the highest yield causes of small bowel obstruction (SBO) on USMLE. - Occur as a result of major abdominal surgery – i.e., hysterectomy, laparotomy, appendectomy. - Prior surgery results in inflammation that yields fibrin deposition within the peritoneal cavity. This causes "fibrous adhesions" over time that the small bowel can get caught in. - Vignette will say, e.g., 50-year-old who had laparotomy 10 years ago for gunshot wound now has obstipation (failure to pass flatus in addition to stool) for 2 days + SIRS. This is pass-level and answer is just "adhesions."

	- New 2CK NBME has “mechanical obstruction of small intestine” as the answer in patient with Hx of appendectomy. Pass-level, but that’s another way they can write the answer. - As mentioned earlier, for many abdo conditions – i.e., SBO, pancreatitis, cholangitis, etc., a general “triad” of NPO, IV fluids, and NG tube is done first. In other words, you might get a simple adhesions Q, and then the answer is just “0.9% saline” as the first step, where bowel rest and NG tube aren’t listed. Or the answer will be "nasogastric decompression," where fluids were already given and bowel rest isn't listed. - If the patient is stable, do abdominal X-ray (AXR) if SBO is suspected to look for gas / obstruction. A very buzzy AXR finding that means SBO (and not specific for adhesions for that matter) is “dilated loops of small bowel with air-fluid levels.” In general, you should remember: "AXR is done when we're looking for gas," whether that be SBO, sigmoid volvulus, toxic megacolon, etc. - If the patient is unstable (i.e., low BP), go straight to laparotomy. Never do a CT scan in an unstable patient on USMLE (this includes patients with normal BP *after* IV fluids given to restore low BP – i.e., the fluids are just acting as a temporizing measure and you need to find source of problem asap). - Crohn disease can cause *intraluminal fibrotic strictures*, **not** adhesions. There is a 2CK Q that lists both as answer choices. Strictures due to Crohn are internal / within the lumen; adhesions are external fibrotic bands.
Post-op ileus	- Surgery is a stressor / form of trauma. Especially post-abdominal surgery in which the bowel has been touched/manipulated, there can be acute diminution of peristalsis resulting in pseudo-obstruction (i.e., the patient is obstipated but there’s no physical obstruction). - Opioids can ↓ peristalsis / cause constipation. These should be considered as a contributing factor to ileus. An important rule for USMLE is that we **always treat pain fully,** even if patient has Hx of drug abuse or ileus. There is a 2CK Q where a patient has ileus post-major surgery; answer is “maintain dose of opioids + add stool softeners.” - If the Q gives you Bristol 7 stool (watery diarrhea; no solid pieces) with laxatives but Bristol 1 (severe constipation; pellet-like stools) when on nothing, answer on NBME is “add fiber,” with the aim being Bristol 3-4.
Ogilvie syndrome	- Large bowel pseudo-obstruction. - **Classically follows hip surgery** for whatever reason. - Think of this as ileus but of the large, not small, bowel. “Ileus” sounds like ileum, which is part of small bowel. So ileus isn’t used to refer to large bowel pseudo-obstruction. - Abdominal x-ray is what we do first in stable patients when we are looking for suspected obstruction. If patient is unstable, always go straight to laparotomy.
Duodenal hematoma	- Shows up on NBME. You just need to know this is the answer for someone who has ↓ hemoglobin + bilious vomiting. - The ↓ Hb is because the patient is bleeding internally. - Bilious vomiting implies obstruction (almost always duodenal).
Duodenal perforation	- Aka ruptured viscus, meaning “organ,” but almost always refers to duodenum. - Caused by Hx of duodenal ulcers. - Highest-yield point is that this is diagnosed with **chest and abdominal x-rays** showing **air under the diaphragm.** Presents two ways: 1) Q will give patient who has Hx of abdo pain after meals (implying Hx of duodenal ulcer) + now has SIRS + acutely worse abdo pain + rigid abdomen → answer = “x-rays of the chest and abdomen.” 2) Q will say acute-onset epigastric pain as though the patient was kicked in abdomen + board-like rigidity + x-rays show air under the diaphragm → answer = “immediate surgical exploration of upper abdomen.” - Not a hard presentation or management, but it shows up all over the place + students get this wrong all the time. Easy points.

Subphrenic abscess	- Complication of splenectomy. - Patient will have fever + abdo pain + leukocytosis post-splenectomy + they'll ask for diagnosis → answer = subphrenic abscess. Tx = drain.
Dumping syndrome	- Answer on 2CK in patient who has diarrhea + hypoglycemia post-major surgery of the stomach/small bowel. - Post-gastrectomy, for instance, "rapid transit of hyperosmolar chyme" is an answer on NBME for the mechanism. This triggers a ↑↑ spike in insulin due to glucose-dependent insulinotropic peptide.
Blind loop syndrome	- Answer on 2CK in a patient who has diarrhea post-major surgery of the stomach/small bowel **without** hypoglycemia. The vignettes can otherwise sound pretty similar. - This is when a part of the proximal small bowel forms a crevice or nook in which peristalsis bypasses this "blind" segment, thereby creating stasis within it. This can lead to small intestinal bacterial overgrowth (SIBO) and diarrhea.

HY Referred pain	
Spleen	- Splenic laceration → ULQ pain +/- can refer to left shoulder (Kehr sign).
Diaphragm	- Diaphragmatic irritation can cause pain going to left shoulder (asked on NBME); spleen is wrong answer. The key here is they ask "irritation."
Gallbladder	- RUQ or epigastric pain +/- can refer to right shoulder.
Appendix	- Epigastric pain initially (visceral peritoneal inflammation) that migrates to RLQ (parietal peritoneal inflammation). - USMLE Q will ask straight up why patient has RLQ pain after initially only have epigastric pain → answer = "inflammation of parietal peritoneum." Not hard.

GI Surgeries for 2CK	
- There are countless fancy GI surgeries that could be discussed just for the sake of it, but it's a vehement waste of time. I'm going to comment on just stuff I think is HY for USMLE, without the nonsense.	
Gastrectomy	- The Billroth II, or gastrojejunostomy, is a partial gastrectomy that shows up sometimes on Surg forms as causing dumping or blind loop syndromes. - In other words, they'll say a patient has diarrhea following a Billroth II gastrojejunostomy and then ask for diagnosis → answer = blind loop syndrome. - Or they'll give diarrhea + hypoglycemia (dumping syndrome), and the answer is just "rapid transit of hyperosmolar chyme." - Can cause B12 deficiency due to removal of parietal cells. Can also cause B1 deficiency (asked twice on new 2CK NBMEs) → confusion/nystagmus (i.e., Wernicke findings; B12 deficiency doesn't present with these); can have (+) Romberg sign in both B1 and B12 deficiency Qs (for B1, this is part of dry beriberi; for B12, this is part of subacute combined degeneration). - Done for gastric cancer and perforations.
Gastric bypass	- In contrast to gastrectomy, no part of the stomach is removed here. A pouch is made where part of the stomach is merely walled off from food passage. - Answer on Family Med forms weirdly enough for the most effective way to manage weight in someone who is morbidly obese, albeit last resort. Not sure why this is on FM forms but they ask it. - Sequelae same as for gastrectomy. B12 deficiency can occur due to disruption of parietal cells, even though stomach is not removed. - Another name for gastric bypass is roux-en-Y, which will show up tons on 2CK Surg forms as a wrong/distractor answer. I've only seen "gastric bypass surgery" as correct on FM forms, but never "roux-en-Y" on Surg forms.
Whipple procedure	- Aka pancreaticoduodenectomy. - Used for **head of pancreas** cancer.

	- Because of interconnected blood supply, removal of head of pancreas requires removal of the gall bladder, duodenum, and proximal jejunum.  - As I said, just know that this is done for head of pancreas cancer.
Distal pancreatectomy	- Answer on Surg form for tail of pancreas cancer. Not complicated, but it's asked.
Colectomy	- Even though segmental colectomies (many variants exist) can be used to treat localized cancers of the colon, it's to my observation that basically all colon cancer Qs on NBME have total colectomy as the answer if they force you to choose. - A 2CK NBME Q gives an 18-year-old with recently diagnosed FAP, and the answer is "total proctocolectomy." Serial colonoscopy is wrong answer, since chance of cancer is 100%. USMLE wants total removal once patient is 18. - A difficult newer 2CK NBME Q gives patient with uncontrolled colonic bleeding and severely low Hb despite transfusions + they say a radiouptake scan shows localization to one part of the colon → answer = total colectomy; hemicolectomy is wrong answer.
Laparoscopy	- Keyhole surgery used classically for appendectomy, cholecystectomy. - Highest yield indication on USMLE is diagnosis/treatment of endometriosis. - Patient must be hemodynamically stable (i.e., normal BP). If low BP, do laparotomy, not laparoscopy.
Laparotomy	- Used for unstable (low BP) patients after abdominal trauma (e.g., ruptured spleen from MVA); also for ruptured ectopic pregnancies where patient is unstable. - If patient is unstable, **never pick CT** on USMLE. Go straight to laparotomy. - If patient has abdominal trauma + low BP + was given fluids and now has normal BP, do not do CT. Go straight to laparotomy. The fluids are just a temporizing measure (buying time), but the underlying problem still needs to be fixed asap. - Penetrating trauma to the abdomen (usually gunshot wounds) requires immediate laparotomy, even if the patient is stable. Below the level of the nipples is considered "abdomen" in this scenario. - USMLE loves this as major cause of post-surgical adhesions (SBO months to years later). - Celiotomy is another name for laparotomy that shows up on some of the 2CK CMS Surg forms. It's a correct answer on one of the forms, where every student says wtf. Old-school surgeons might remark on differences between laparotomy and celiotomy, but USMLE uses the terms interchangeably.
Splenectomy	- Done in setting of trauma (splenic laceration), hereditary spherocytosis, or ITP. - "Autosplenectomy" refers to loss of spleen due to repeated microinfarcts in sickle cell.

	- "Asplenia" can refer to anyone who's had splenectomy, autosplenectomy, or who didn't develop a spleen from birth (Ivemark syndrome on 2CK NBME). - The white pulp of the spleen ordinarily contains 50% of the immune system's reservoir of macrophages. - Patients are susceptive to encapsulated organisms because the spleen has a high phagocytic capacity, where opsonization with C3b and IgG, followed by phagocytosis in the white pulp, is how they are cleared. - Patients need vaccines against *Strep pneumo, Neisseria meningitidis,* and *Haemophilus influenzae* type B. - Penicillin prophylaxis is given to patients with asplenia until age 5, or until 1 year post-surgical splenectomy. If the USMLE forces you to choose which organism this protects against the most, the answer is *Strep pneumo.* - However, if USMLE vignette tells you a patient with asplenia misses a penicillin prophylaxis dose + now has sepsis, the Tx is a 3rd generation cephalosporin (i.e., cefotaxime or ceftriaxone), not penicillin. The latter is only for prophylaxis. - Howell-Jolly bodies (nuclear remnants) are seen within RBCs on a smear in patients with asplenia. This is because senescent and abnormal RBCs are ordinarily cleared out by the red pulp of the spleen. - USMLE wants you to know the lifespan of a RBC is 4 months (118 days). They ask this on NBME as how long it will take for all carbon monoxide to be cleared out of the blood in the setting of exposure (i.e., not until all RBCs exposed are removed).

HY GI Diagnostic modalities for Surg/IM (2CK mostly)	
- Same as with the GI surgeries table above, we could discuss endless indications for the below modalities, but I'm staying concise here with just the highest yield points.	
Nasogastric tube	- "Insertion of NG tube" is the answer for tracheoesophageal fistula and choanal atresia (discussed in HY Pulm PDF). - The NG tube will hit the blind pouch in the proximal esophagus for TEF. - For choanal atresia (non-patency of the nasal passages), it simply can't be inserted through the nose.
Barium swallow	- For suspected Zenker; shows outpouching. - First step in achalasia (before manometry); shows bird's beak. - Do not use for esophageal perforation. Can cause mediastinitis if it leaks out through a hole in the esophagus. - Can be used for patients with aspiration risk, since it doesn't cause pneumonitis.
Gastrografin swallow	- Aka water-soluble contrast swallow. - Used for esophageal perforations in hemodynamically stable patients because it doesn't cause mediastinitis. - Do not use if patient has aspiration risk; causes pneumonitis. Use barium instead. - I've seen NBME Q where there is esophageal perforation and the patient has low BP (hemodynamic instability). Answer is emergent thoracotomy. Could be thought of as "laparotomy but of the chest instead," where we don't have time for imaging if the patient is unstable.
Esophageal manometry	- Answer for achalasia after barium swallow shows bird's beak. - A pressure study of the esophagus.
Endoscopy	- Aka esophagogastroduodenoscopy. - Immediate answer for any patient with new-onset dysphagia and Hx of GERD or heavy smoking/alcohol (for esophageal cancer). Then biopsy any lesion/stricture. - Endoscopy + banding for esophageal varices emergent management. - Diagnosis of hiatal hernia (asked on NBME). - Done in patients over 50 who have *H. pylori* positivity (on newer 2CK form).
Capsule endoscopy	- Always wrong fucking answer on USMLE.
Upper GI series	- Congenital midgut volvulus in pediatrics; will show a corkscrew. - This is a contrast swallow followed by X-rays to visualize the upper GI tract.

Abdominal ultrasound	- Cholelithiasis. - First step to diagnose cholecystitis; if negative, do HIDA scan. - First step for choledocholithiasis, then do ERCP. - Intussusception Dx, then do enema (definitively diagnostic and therapeutic). - Pyloric stenosis.
HIDA scan	- Answer for confirmatory diagnosis of cholecystitis (not cholelithiasis alone) if ultrasound is negative. - Radiocontrast is injected + secreted into bile. If gallbladder lights up, there is no obstruction of the cystic duct and it is negative; if gallbladder doesn't light up, we know there's an obstruction by a stone and it confirms cholecystitis. This is because almost all spontaneous cholecystitis cases are due to obstruction by a stone.
CT of abdomen	- USMLE won't force you to choose, but just assume contrast CT is always used. The only times non-contrast CT will be an answer is for urolithiasis and intracranial bleeds. - Diagnosis of pancreatic cancer (highest yield indication on USMLE). - Diagnosis of liver cancer and focal nodular hyperplasia. - Renal injury (ultra-HY; discussed in HY Renal PDF). - Blunt force trauma to abdomen in patient who is stable. - Diverticulitis. - Gallstone pancreatitis (USS → CT → ERCP).
Abdominal x-ray	- Used to look for bowel gas in suspected obstruction (e.g., sigmoid volvulus showing coffee bean sign). - Duodenal atresia (double-bubble sign). - Congenital diaphragmatic hernia (bowel gas in left hemithorax). - Necrotizing enterocolitis (pneumatosis intestinalis). - Toxic megacolon if patient is stable. - "X-rays of chest and abdomen" used for duodenal ulcer perforation to look for air under the diaphragm.
ERCP	- Endoscopic retrograde cholangiopancreatography; type of EGD that can also enter the biliary tree, remove stones there, and inject contrast if necessary. - Answer on USMLE for choledocholithiasis after ultrasound is performed (if gallstone pancreatitis, do USS → CT → ERCP). - Answer for diagnosis of cholangitis. - Answer for drainage of pancreatic pseudocyst (weird, but we can drain internally via ERCP). - Diagnosis of cholangiocarcinoma (vignette sounds like pancreatic cancer, but shows negative CT).
MRCP	- Magnetic resonance cholangiopancreatography. - Never seen this as correct answer on NBME, but I observe that students always pick it when they don't know what's going on, maybe because it sounds weird and specific. - Can visualize biliary tree much more safely than ERCP, but unlike ERCP, it isn't a form of treatment (ERCP is both diagnostic and therapeutic).
Endoscopic ultrasound	- Never seen this as correct answer, only wrong answer. - Can be done to diagnose pancreatic cancer if CT is negative or to drain pancreatic fluid collections in place of ERCP. But once again, never seen this as correct.
Paracentesis	- Refers to aspiration of peritoneal contents/fluid. Students often confuse this with pericardiocentesis, even though completely unrelated. - Used for spontaneous bacterial peritonitis. - As mentioned earlier, choose "white cell count and differential" before "gram stain and culture of the fluid."
Meckel scan	- Radiocontrast uptake scan that localizes to the diverticulum at terminal ileum.
Colonoscopy	- Diagnosis of colorectal cancer, IBD, pseudomembranous colitis, ischemic colitis, and hemorrhoids.

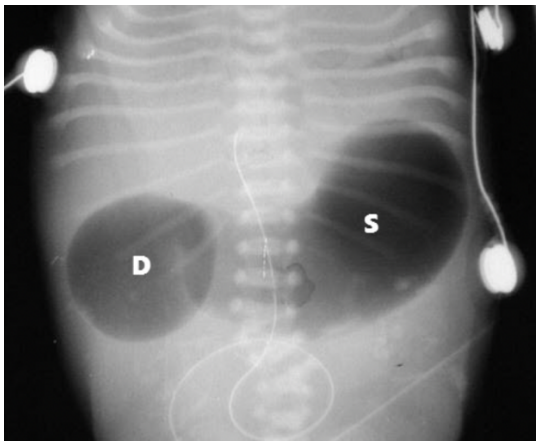
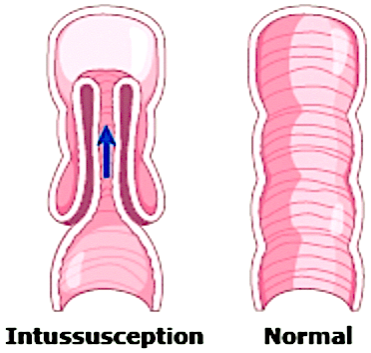
	- Do **not** do in diverticulitis (can cause perforation; do CT instead). - Do if patient has anal cancer prior to excision in order to first see if there's greater extent of cancer (might change management). Asked on 2CK form, where they say there is cancer at anal verge, and excision is wrong; answer is colonoscopy. - In general population, commence at age 45-50 (guidelines are evolving) and then do every 10 years. USMLE will not force you into a position where ohemgee is it 45 or 50. If they force you to choose 45 as the new guideline, it will be obvious the other answer choices are wrong. - If first-degree relative (parent or sibling) has colon cancer, commence at age 40 or 10 years prior to diagnosis in that family member, whichever is earlier. In other words, never later than 40. For example, if dad was diagnosed at 58, commence at age 40. If dad was diagnosed at 44, commence at age 34. Then do every 5 years. - If patient has IBD (UC or Crohn), commence 8 years after the diagnosis was made and then do every 2-3 years. 2CK NBME is real slick about this. They give mid-30s patient with IBD diagnosed in 20s + also has dad diagnosed with colon cancer in 50s; wrong answer = colonoscopy at age 40; correct answer is "colonoscopy now," since patient has IBD and should have had one done 8 years after his/her diagnosis. - Patients who have history of dysplastic polyps need repeat colonoscopies every 2-5 years, depending on size/morphology of polyp(s). - If patient has HNPCC (Lynch syndrome), start at 20-25, then do every 1-2 years. - For FAP, start at age 10-12 and do every 1-2 years. Then do prophylactic proctocolectomy at age 18 (on NBME). - For Peutz-Jeghers, start at age 8, then every 1-2 years.
Sigmoidoscopy	- "Sigmoidoscopy-guided insertion of rectal tube" is answer on NBME for treatment of sigmoid volvulus. - I've never seen sigmoidoscopy as a diagnostic tool. I make this point because there are alternative theoretic regimens to colon cancer screening – i.e., sigmoidoscopy + barium enemas, etc., but I've never seen these assessed.
Anoscopy	- Shows up on 2CK form as answer for diagnosis of hemorrhoids before banding is performed. This will help determine extent of hemorrhoids. - If the Q lists both anoscopy and colonoscopy as answers for the diagnosis of hemorrhoids, choose anoscopy if patient is under 45. If over 45, do colonoscopy.

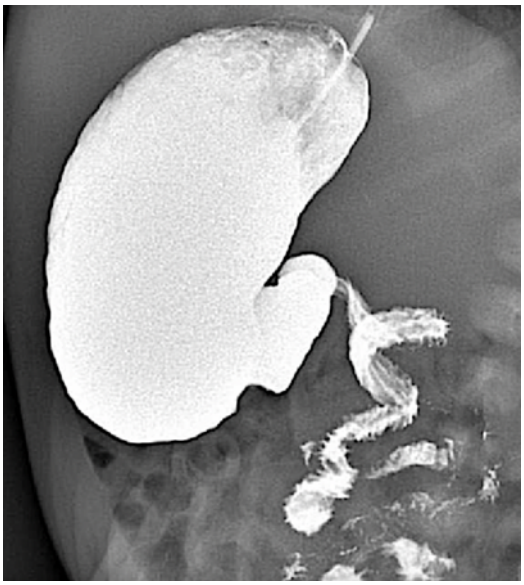
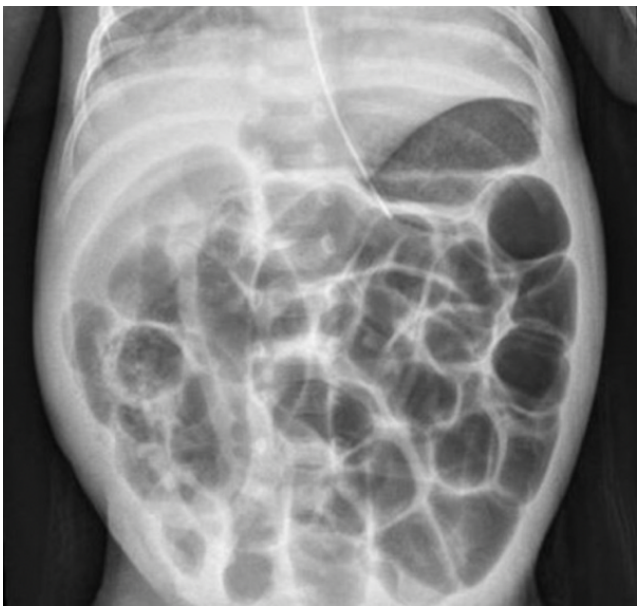
HY Polyp conditions for USMLE	
Lynch syndrome	- Hereditary non-polyposis colorectal cancer (HNPCC). - Mismatch repair genes *MSH2/6, MLH1, PMS2.* - Mutations cause "microsatellite instability." - In other words, the answer on USMLE for "mismatch repair" and "microsatellite instability." These are highly buzzy phrases. - Colonic polyps/cancer; also associated with gynecologic cancer. - Start colonoscopy at age 20-25, then do every 1-2 years.
Familial adenomatous polyposis	- Familial Adenomatous Polyposis (FAP); *APC* gene on chromosome 5; AD. - Hundreds to thousands of polyps on colonoscopy; 100% cancer risk. - Contrasts with Lynch syndrome because FAP will literally have "hundreds to thousands of polyps" whereas Lynch is only "some polyps." - Start colonoscopy at age 10-12 and do every 1-2 years. Then do prophylactic proctocolectomy at age 18 (on NBME). - FAP + soft tissue (e.g., lipoma) or bone tumors (e.g., of the skull) = Gardner syndrome. - FAP + CNS tumors = Turcot syndrome.
Peutz-Jeghers	- As talked about at the start of this PDF, this is combo of perioral melanosis and **hamartomatous** colonic polyps.

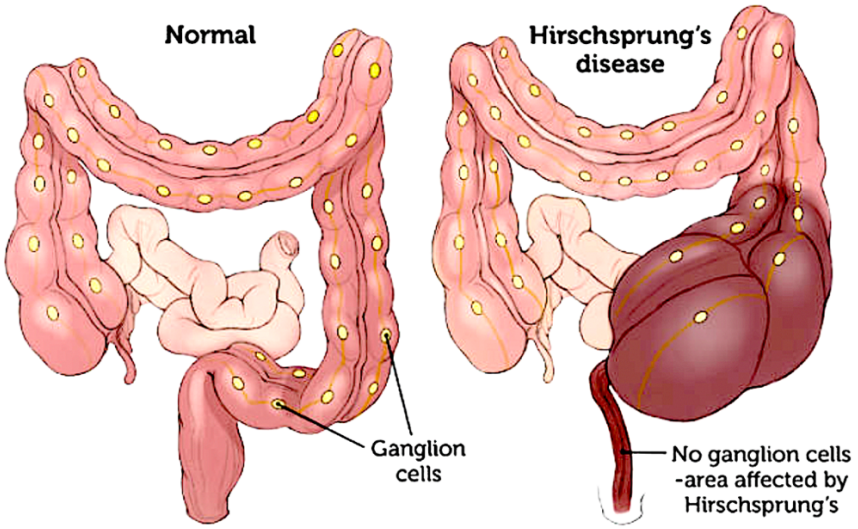
	- Start colonoscopy at age 8, then do every 1-2 years.
Juvenile polyposis	- Shows up on a 2CK NBME. - Q will tell you there's a teenager (i.e., juvenile, LOL!) with intermittent bleeding per rectum + colonoscopy shows scattered polyps + biopsy shows **"dilated, cystic, mucus-filled glands** with abundant lamina propria and inflammatory infiltrates" → answer = juvenile polyposis. - Only question I've seen on it, but it's on new 2CK NBME so I have to mention it.
Hyperplastic polyps	- All you need to know is these are **not pre-cancerous /** have **no dysplasia.**
Tubulovillous	- Dysplastic polyps that precede full-blown colorectal cancer are classically either **tubular or villous,** AND either **pedunculated or sessile.** - Villous is worse than tubular. Sessile is worse than pedunculated. - This means pedunculated tubular polyps are "best" and sessile, villous polyps are "worst." - Sessile means flat; pedunculated means "sticks out." - Polyps can have mixed characteristics and hence be tubulovillous. - What you need to know is: USMLE will give you a random 65-year-old with a polyp + show you a pic + ask you what it is → answer = "tubular polyp." Student freaks out and says, "Wait, we need to know polyp pics??" → No. The big picture concept is that older people who get colorectal cancer will have tubular, villous, or tubulovillous polyps as I just said. Wrong answers would be things like hamartomatous, juvenile, and hyperplastic. You can just eliminate to get there without even knowing the image. - 2CK wants you to know that patients with history of dysplastic polyps need repeat colonoscopies every 2-5 years, depending on size/morphology of polyp(s).

Colon cancer short points
- *KRAS* gene is the answer on USMLE for the first gene mutated in colonic polyps, prior to progression to overt colon cancer. - Colon cancer often develops as a result of *progressive* mutations, rather than one mutation straight-up. In other words, first *KRAS*, then *PTEN*, then *DCC*, then *TP53*. This is a HY path point for some reason. - If they tell you a colon cancer has metastasized and force you to choose a gene that's mutated, go with *TP53* for p53 protein. - If they tell you a polyp is seen and there is no evidence of invasion of the stalk, choose *KRAS*. This reflects earlier changes in the sequence as I just wrote above. - Can cause **colovesicular** fistulas, where a passageway develops between the GI tract and bladder, leading to UTIs and mixed enteric flora in the urine. This is also assessed for diverticulitis on 2CK and seems to be a new diagnosis USMLE likes. This could also be due to Crohn, in theory, but I haven't seen it on NBMEs yet. - Can increase the risk of *Strep bovis* endocarditis for whatever magical reason. Obscure, but assessed. - I already discussed colon cancer screening extensively above.

HY Peds GI diagnoses	
Pyloric stenosis	- Forceful/projectile **non**-bilious vomiting in neonate days to weeks old. Obstruction is above level of duodenum, so we won't see bile. - Students fixate on exact age of the kid for pyloric stenosis versus duodenal atresia. USMLE doesn't give a fuck and will give variable ages. - Hypertrophic pylorus; "olive-shape mass" in abdomen is buzzy but rarely seen in Qs. - **Ultrasound** done to diagnose; myotomy to treat. - USMLE wants you to know this is almost always a one-off / sporadic developmental defect, where the neonate will not have a broader syndrome. In contrast, duodenal atresia is usually Down syndrome.
Duodenal atresia	- Bilious vomiting in neonate. Bilious = obstruction at level of duodenum or lower. - Associated with **Down syndrome,** albeit not mandatory.

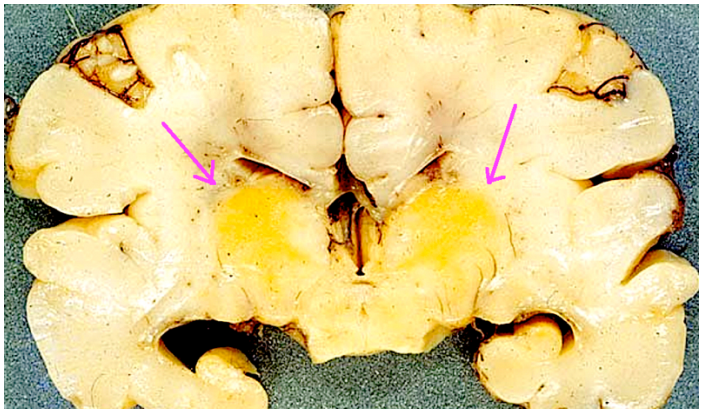
	- Will cause **double-bubble sign** on **abdominal x-ray.** 
Annular pancreas	- Another cause of duodenal obstruction with double-bubble sign, but much more rare than duodenal atresia. - Step 1 is pass/fail now, but you could be aware that this is caused by "abnormal migration of ventral pancreatic bud." This was the type of garbage we memorized back in the numerical Step 1 days. Mentioning it here because I think it still floats around on the Step.
Intussusception	- Intussusception is telescoping of the bowel into itself.  - Almost always under age 2. - Idiopathic, but can be triggered by viral infection or rotavirus vaccination, where mesenteric lymphadenopathy can act as a "lead point" for the intestinal telescoping. Underlying Meckel diverticulum is also a common lead point. - In ultra-rare scenarios, intussusception in elderly can occur due to colon cancer. - Classic presentation is intermittent/colicky abdominal pain, where the kid will have episodes of drawing the legs to the chest or squatting, with blood in the stool +/- bilious vomiting. - **An abdominal mass** (i.e., sausage-shaped mass) is a key feature, but not in all Qs. If they don't give the mass, which can cause confusion with midgut volvulus where there is no mass, they won't list the latter as a separate answer. - Diagnosis is first done with ultrasound, which may show a sausage mass or target sign (I've never seen NBME show the image, so they don't care). After the ultrasound, **enema is both diagnostic and therapeutic.** USMLE doesn't care what kind of enema is used; air-contrast enema is classic, but you will see all different types of enemas as correct answers. - "Air enema with ultrasound" is the answer on one 2CK form.
Midgut volvulus	- Failure of rotation of proximal bowel. - Almost always under age 1. - Presents similarly to intussusception (often mistaken for it), but there is **no mass.**

	- I've said intussusception Qs can sometimes omit mentioning a mass, but if they do this, they won't list midgut volvulus as a separate answer. - **Upper-GI series** shows corkscrew appearance.  - An upper-GI series is basically only ever done for midgut volvulus on USMLE. So if they give you a 10-line nonsense paragraph + tell you in last line an upper-GI series was performed, you know without even reading the Q that the answer is midgut volvulus. Or they'll just show you the above image, and then the answer is just "failure of rotation of proximal bowel." - If an abdominal x-ray is performed, it will show "dilated loops of small bowel with air fluid levels," since we have an obstruction. - Treatment is surgical.
Necrotizing enterocolitis	- Necrotic bowel occurring in premature neonates born <32 weeks' gestation. - Presents with pneumatosis intestinalis (air in bowel wall); resembles bubbles.  - USMLE will show you above image in kid born, e.g., at 26 weeks' gestation, and then answer is just necrotizing enterocolitis. Not hard.

	- It's to my observation necrotizing enterocolitis usually shows up as a *wrong* answer choice on NBMEs, where I see students pick it in, e.g., an adult, and I'm like, "Dude you see that in fucking hyper-preemies." And they're like Oh. So you need to remember this as specifically a premature neonate condition.
Hirschsprung	- Failure of migration of neural crest cells distally within the colon. - Causes an aganglionic distal segment that results in ↑ anal sphincter tone.  - Often associated with Down syndrome. - Presents usually as failure to pass stool at birth, or as chronic constipation in an infant, depending on the severity. But do not confuse with "meconium ileus," which is a term that refers to failure to pass stool at birth specifically due to cystic fibrosis. - Diagnose with anal manometry showing ↑ anal sphincter tone, followed by biopsy.
Cystic fibrosis	- Discussed in extensive detail in the HY Pulm PDF. - Just a reiteration that meconium ileus + **exocrine pancreatic insufficiency** are exceedingly HY. - Secretions within the pancreatic ducts are inspissated (meaning desiccated / dried up within a lumen), making them sticky. This means the enzymes can't make their way to the duodenum → fat-soluble vitamin malabsorption → NBME exams love vitamin E deficiency in CF in particular (presents as neuropathy). - D-xylose test is normal, since the intestinal lining/architecture is intact. Any malabsorption that occurs is due to mere paucity of enzymes in the small bowel. Recall that abnormal D-xylose test would be Celiac and Crohn on USMLE.
Biliary atresia	- As discussed earlier, ↑↑ direct bilirubin in a kid under the age of 6 weeks. - Caused by lack of development of the intrahepatic bile ductules and biliary tree. - Ultrasound will be done first, but USMLE wants **liver biopsy** to confirm the diagnosis. - Kasai procedure is done to treat, followed by liver transplant if unsuccessful.

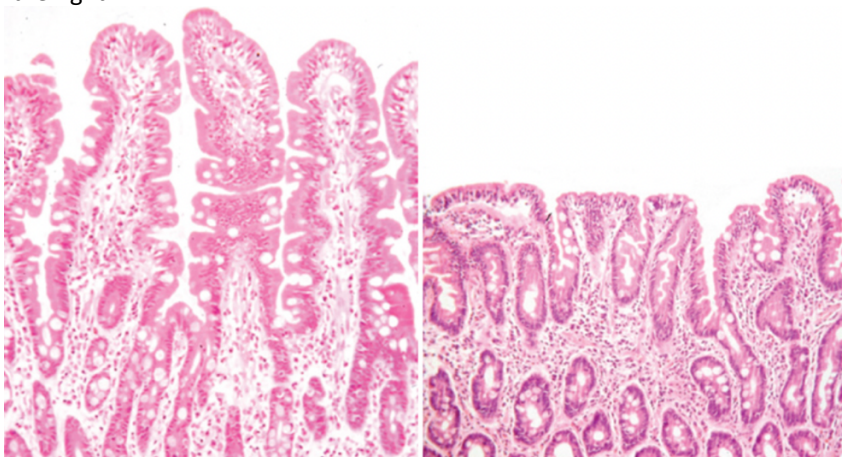
Jaundice in Peds (2CK only)
- Normal bilirubin should be ~1.0 mg/dL. Direct bilirubin should be ~0.1 mg/dL. - **Pathologic jaundice** in peds = if *any* of the following is/are present: 1) Any jaundice on the first day of life (first 24 hours of life), period = pathologic. 2) Jaundice present **after** one week if term, or after two weeks if preterm = pathologic. 3) Total bilirubin >15 mg/dL. 4) Direct bilirubin >10% of total bilirubin, *even if total bilirubin is <15 mg/dL.* 5) Rate of change of increase in bilirubin >0.5 mg/dL/hour.

- If none of the above 5 is met but the neonate has jaundice, the diagnosis is **physiologic jaundice.**
- If the kid has pathologic jaundice, we then want to explore the causes (all discussed below).
- Tx for pathologic jaundice = **phototherapy first, followed by exchange transfusion;** some literature makes a case for IVIG after phototherapy, but USMLE has exchange transfusion as correct, without listing IVIG.
- If ↑ bilirubin in the neonate accumulates in the CNS grey matter, this is called **kernicterus** and can cause irreversible neurologic damage. In the gross specimen below, bilirubin has deposited in the basal ganglia (pink arrows pointing to yellow areas). USMLE will show similar image, where the answer is just "putamen."



Breastmilk jaundice	- Due to β-glucuronidase in breast milk, which leads to de-conjugation of intestinal bilirubin + increased enterohepatic circulation. - Jaundice starts on day 3-5 and peaks at 2-3 weeks. - Tx = stop breastfeeding for ~48 hours (and do bottle feeding), which leads to a rapid ↓ in bilirubin; once breastfeeding is resumed, bilirubin might rise, but not back to pathologic levels.
Breastfeeding jaundice	- Insufficient feeding (e.g., failure of suckling, etc.) + decreased milk intake leads to reduced intestinal clearance of bilirubin + increased enterohepatic circulation. - Jaundice peaks at 3-5 days. - Tx = formula feeding (fluid + caloric supplementation). - What USMLE can do is tell you neonate has difficulty attaching to the breast during feeds + has jaundice, and you're like "Omg breastfeeding jaundice!" But then you'll see that 0/5 criteria are met for pathologic jaundice, and the answer to the Q is just "physiologic jaundice."
Hemolytic disease of the newborn (Rh type)	- Rh(-) mom experiences mixing of blood from Rh(+) fetus during first pregnancy (i.e., from traumatic birth, abortion, or instrumentation like amniocentesis). This causes mom to develop antibodies against Rh antigen. - **In a subsequent pregnancy,** these antibodies, which have now evolved to be IgG, will cross placenta and attack Rh(+) fetal RBCs that have Rh on their surface → hemolysis. - Can vary in severity between death *in utero* versus mild neonatal jaundice. - There is 2CK NBME Q where they say woman in 3rd trimester of first pregnancy has positive titers for anti-D (which means anti-Rh) + had Hx of bleeding in 1st trimester; they ask diagnosis → answer = Rh isoimmunization. In other words, there was mixing of blood during early pregnancy, and now she's developed antibodies against Rh. However, this won't cause Rh-type hemolytic disease of newborn in first pregnancy; the actual hemolytic disease will occur from 2nd pregnancy onward. - Another 2CK Q gives ↑ **fetal HR** in 3rd trimester in woman who's in 2nd pregnancy who has (+) "anti-D titers." They ask reason why fetal HR is ↑ → answer = "Rh isoimmunization" → destruction of fetal RBCs → ↓ oxygen delivery within fetal circulation → ↑ HR to compensate.
Hemolytic disease of the newborn (ABO type)	- Mothers with O blood type will have fractional IgG (instead of IgM) against A and B antigens → cross placenta → fetal hemolysis.

	- USMLE will give this to you as showing up in a **first pregnancy** in a mother who is O+**.** Can this occur in O(-) mothers in 2nd pregnancies onward? Yes. But USMLE does this so you can't accidentally get lucky with the Dx if you only know about Rh-type hemolytic disease of the newborn.

Autoimmune intestinal absorptive disorders	
Celiac disease	- Intolerance to gluten (i.e., to gliadin proteins found in wheat, oats, rye, and barley; but not rice). - Causes type-IV hypersensitivity response where T cells attack the small intestinal villi, resulting in **flattening of the villi** and malabsorption. The biopsy/image is pass-level. Below, we have normal villi on the left, and flattening of villi in Celiac on the right.  - Even though the disorder is T-cell-mediated destruction of the villi, patients still develop antibodies which are HY: anti-endomysial (aka anti-gliadin) and anti-tissue transglutaminase IgA. - Patients with concurrent IgA deficiency will have false (-) results for tissue transglutaminase IgA antibody. Since "autoimmune diseases go together," and "autoimmune and immunodeficiency syndromes go together" patients with IgA deficiency have 15x greater likelihood of developing Celiac. - Celiac presents as vague bloating/diarrhea in patients who might not be able to pinpoint what they're eating to cause the symptoms. - I'd say one of the highest yield points on USMLE regarding Celiac is that it can cause **iron deficiency anemia.** This is because iron absorption in the duodenum is impaired from the flattened villi. - For example, you might get a difficult/vague vignette where you're not sure if the diagnosis is Celiac or lactase deficiency, but you see that hemoglobin is low, and you're like "Boom. Celiac." - Associated with dermatitis herpetiformis, which presents as itchy, vesicular lesions on extensor areas. This causes "IgA deposition at dermal papillae."

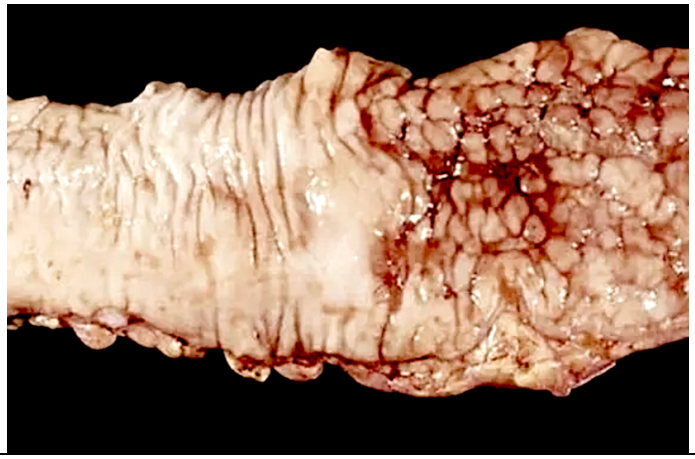
	- D-xylose test is abnormal because the intestinal lining/architecture is abnormal (i.e., we have flattening of villi). - Increased risk of gastrointestinal T-cell lymphoma in Celiac. This is called enteropathy-associated T-Cell lymphoma (EATL). - Diagnosis is made via antibody screening, followed by duodenal biopsy as confirmatory. - Treatment of Celiac is simply with gluten-free diet.
Lactose intolerance	- Deficiency of brush border disaccharidase (i.e., aka lactase deficiency). - Brush border is located at "tips of the villi." This is in contrast to stem cells of the GI tract, which are located at the "base of the crypts." USMLE asks the locations of these things. - Diarrhea/bloating in response to dairy. - In contrast to Celiac disease, there is **no iron deficiency anemia.** - ↑ incidence in Asians, but can also develop idiopathically starting in teenage years-onward. When a young adult has a new-onset diarrhea-related condition, it's usually lactose intolerance, as lactase production can ↓ with age. But once again, USMLE can trick you, so I'll be an asshole and reiterate that if you see ↓ Hb, choose Celiac instead. - Secondary lactose intolerance can occur following viral gastroenteritis (i.e, rotavirus or Norwalk virus) → sloughing of the tips of villi → vignette will say new-onset diarrhea/bloating following gastro illness 1-2 weeks ago. This is only transient and patient will recover. This is HY cause of lactose intolerance. - **Biopsy is normal.** This is in contrast to the flattened villi in Celiac. - Diagnosis is made using hydrogen breath test or detecting ↓ stool pH. - D-xylose test is normal, since the intestinal lining/architecture is intact. - Treatment is with avoidance of lactose-containing products, or with lactase pills.
Milk protein allergy	- HY for 2CK Peds but nonexistent on Step 1. - Biggest risk factor is not being exclusively breastfed for the first 6 months of life. - Will present as **blood in the stool** in a child who they say is, e.g., 4 months old, who was started on formula 3 weeks ago. - Treatment is switching to **hydrolyzed casein formula.** Switching to soy-based formula is wrong fucking answer. There is up to 50% crossover of allergy cases with kids who have milk- and soy-protein allergy. - Vignette can say kid was started on either a cow-milk or soy formula when symptoms started. It doesn't matter. Just choose hydrolyzed casein as answer.

Inflammatory bowel disease (IBD)
- Refers to both ulcerative colitis (UC) and Crohn. - Both can present with bloody, mucoid stools. - Both are associated with HLA-B27 → **PAIR** → **P**soriasis, **A**nkylosing spondylitis, **I**BD, **R**eactive arthritis. - For example, if a patient has psoriasis + bloody stools, you say, "The bloody stools are probably IBD." Or likewise, if patient with known IBD has lower back pain that's worse in the morning, you say, "that's probably ankylosing spondylitis." - Both UC and Crohn can be associated with other autoimmune diseases unrelated to HLA-B27, like vitiligo. - Both can cause anterior uveitis (non-specific finding seen in many autoimmune diseases). - Both have ↑ risk of colon cancer if the colon is involved. But if you're forced to choose, UC has > risk than Crohn because the colon is always involved in UC but not always in Crohn. - Both are treated with 5-ASA NSAID compounds **(mesalamine / sulfasalazine)** or steroids. If they ask first Tx, go with mesalamine or sulfasalazine, whichever they list (they won't list both) over steroids. Q on 2CK IM form has "prednisone therapy" as answer for Crohn, but 5-ASA isn't listed. - USMLE wants you to know anti-TNF-α agents (i.e., infliximab, adalimumab, etanercept) are used in IBD in patients who fail initial Tx with 5-ASAs and steroids.

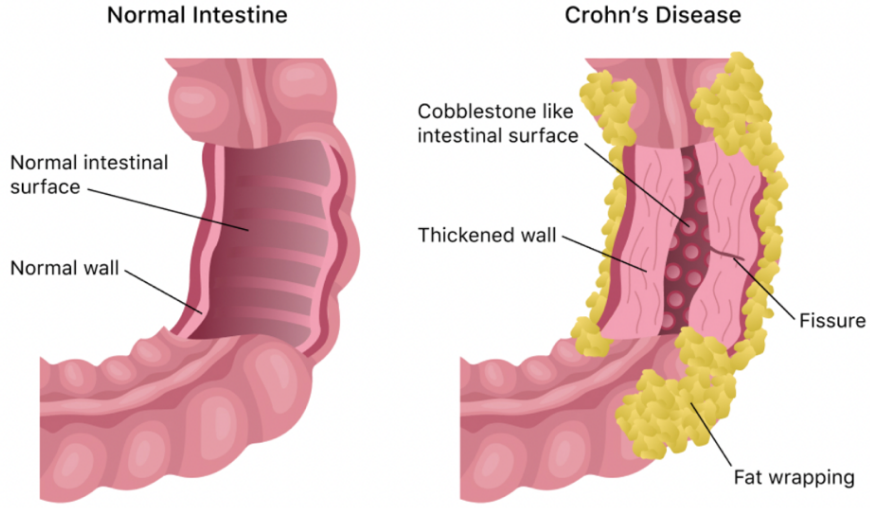
<table>
<tr><td colspan="2">- If they ask you for which cytokine can be anti-inflammatory for IBD, the answer is IL-10. Sounds weird, but just know IL-10 and TGF-β are mostly anti-inflammatory mediators. Don't worry about Th1 vs Th2 nonsense.</td></tr>
<tr><td>Ulcerative colitis</td><td>- Rectum-ascending.
- Not transmural – i.e., only affects mucosa and submucosa. This means fistulae are not seen. If they say fistulae to the skin overlying the anus or to any organ, this means transmural involvement (Crohn).
- Colonoscopy will show pseudopolyps and crypt abscesses. You don't need to know what these look like. You just need to know they = UC.
- Barium enema shows "lead-pipe appearance" due to loss of haustra. This detail is very important for UC.

Descending colon resembles a "lead pipe"

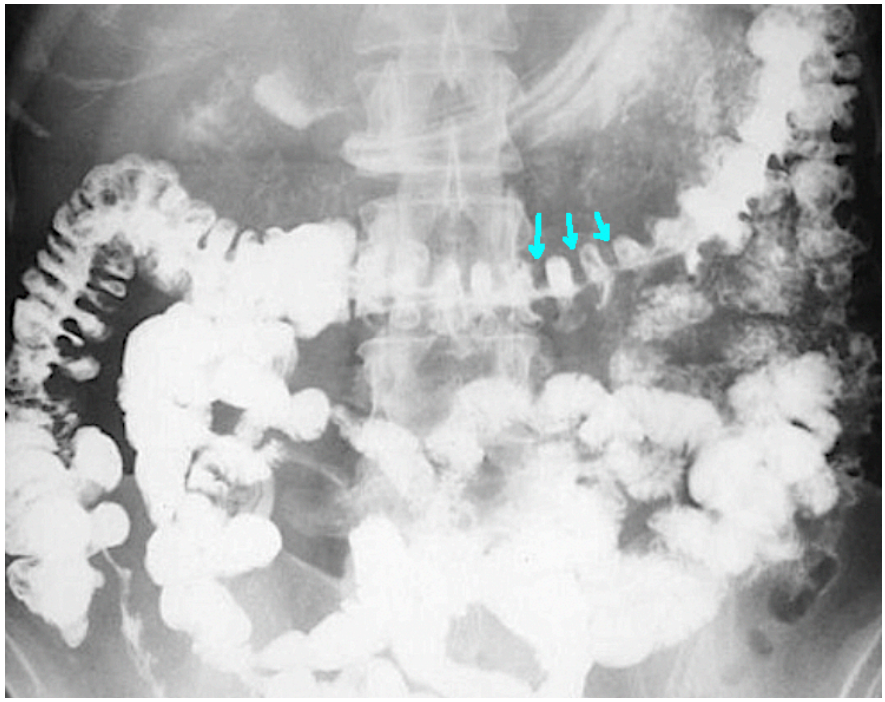
- Can cause primary sclerosing cholangitis, as discussed earlier. This can be associated with pANCA antibodies.
- Pyoderma gangrenosum is a necrotic skin lesion seen rarely in patients with UC. There is an NBME Q that describes this as an "ulcer with necrotic debris."
- Can cause toxic megacolon, which presents as SIRS and sometimes low BP in a patient with UC. They might say abdominal x-ray shows a 12-cm cecum (NR 3-8).
- If the patient has normal BP, NBME for 2CK wants steroids first for toxic megacolon. If the patient is unstable, go straight to laparotomy.
- Colectomy is sometimes performed in patients with severe UC, but very rarely in Crohn.</td></tr>
<tr><td>Crohn disease</td><td>- "Mouth to anus" – i.e., can occur anywhere in GI tract. USMLE loves giving mouth ulcers in Crohn. But terminal ileum is highest yield location.
- Transmural – i.e., can cause anal fistulae + to other organs.
- Colonoscopy shows skip lesions, where there is alternating diseased vs normal bowel segments, with cobblestone ulcers.

  </td></tr>
</table>

Skip lesion (left part normal + right part inflamed and cobblestoned)
- "Creeping fat" is buzzy term that can be seen on NBME, which refers to intestinal fat migration that wraps around the bowel.



- Barium enema shows "string sign," where inflamed segments are narrowed in comparison to normal bowel.



- Crohn disease can cause *intraluminal fibrotic strictures*, leading to small bowel obstruction. These are **not** the same as adhesions. There is a 2CK Q that lists both as answer choices. Strictures due to Crohn are internal / within the lumen; adhesions are external fibrotic bands due to prior surgery.
- Biopsy shows **non-caseating granulomas.** Very HY for USMLE you know that Crohn + sarcoidosis both have non-caseating granulomas.
- Sometimes associated with erythema nodosum. Not specific for Crohn in any regard, but tends to have ↑ association, whereas UC is pyoderma gangrenosum.
- Can cause anti-saccharomyces cerevisiae (yeast) antibodies. This is on a 2CK NBME, where they say (-) for these Abs, but (+) for pANCA, where answer is UC.
- Intestinal malabsorption can occur, resulting in B12 deficiency most commonly due to terminal ileum being classic inflammatory location.

	- Impaired fat absorption in small bowel can result in calcium oxalate urolithiasis. Malabsorption can cause higher fat content in small bowel lumen → chelation with calcium → less calcium available to bind oxalate → more oxalate is absorbed. USMLE will give sharp flank/groin pain in patient with Crohn, and then the answer is just "increased intestinal absorption of oxalate."

Irritable bowel syndrome (IBS)
- Don't confuse with IBD. IBS is psychosomatic (i.e., psych-related) condition. - Classic vignette will be a woman 20s-40s with stress factors who usually has alternating diarrhea / constipation; can also present as bloating or cramping. - Key detail is that symptoms are relieved with bowel motions. - NBME assesses "smooth muscle hypersensitivity" as the answer for the mechanism for IBS. - Treatment for diarrhea-predominant IBS is loperamide, which is an opioid that causes constipation. USMLE will give vignette of IBS with diarrhea, and then the answer is just "mu-opioid receptor agonist."

Other HY GI motility scenarios	
Diabetes	- Diabetic gastroparesis = ↓ peristalsis due to neuropathy of the GI tract. - Presents as GERD-like presentation in someone with Hx of advanced diabetes. - USMLE wants endoscopy first to rule-out physical obstruction. If endoscopy is negative, do gastric-emptying scintigraphy (aka gastric emptying scintigraphic assay) to confirm delayed gastric emptying. - Metoclopramide is used 1st for Tx. It is a prokinetic agent (i.e., ↑ peristalsis) and anti-emetic. It is a D2 antagonist but also an antagonist of serotonin 5HT3 and agonist of 5HT4 receptors. The effects on serotonin receptors ↑ gut peristalsis. - Erythromycin can also be used to agonize motilin receptors. - "2-week trial of PPI" is wrong answer. This is the answer for "regular GERD." If the patient has GERD-like presentation with advanced diabetes, however, the Dx is diabetic gastroparesis, not GERD. - Neuropathy to the hypogastric nerves (sympathetic) causes severe diarrhea. This is because the sympathetic nerves are "anti-peristalsis," so if we knock them out, we get too much peristalsis. - Neuropathy to the pelvic splanchnic nerves (parasympathetic) causes severe constipation. This is because the parasympathetic nerves are pro-peristalsis, so if we knock them out we get unopposed hypogastric. - In other words, diabetes + diarrhea → answer = hypogastric nerves are fucked up. - Diabetes + constipation → answer = pelvic splanchnic nerves are fucked up.
Thyroid	- Hypothyroidism can cause constipation. - Hyperthyroidism can cause diarrhea. - USMLE will give vignette of a thyroid condition with GI disturbance, and then the answer will just be "motility disorder." This is same answer for diabetes.
Supplements	- Iron and Aluminum can cause constipation ("Aluminimum amount of feces.") - Magnesium can cause diarrhea.
Drugs	- Verapamil and opioids cause constipation. - Macrolides, orlistat, and α-glucosidase inhibitors (e.g., acarbose) cause diarrhea.

Viral hepatitis
Two foundational points you need to know: 1) Hep A and E cause acute hepatitis only. Hep B, C, and D can cause chronic hepatitis. 2) HY point is that **hepatocellular damage from hepatitis is due to T-cell-mediated apoptosis, not direct viral cytopathicity.** Same goes for general hepatic inflammation. Choose T cell response, not direct viral effect.

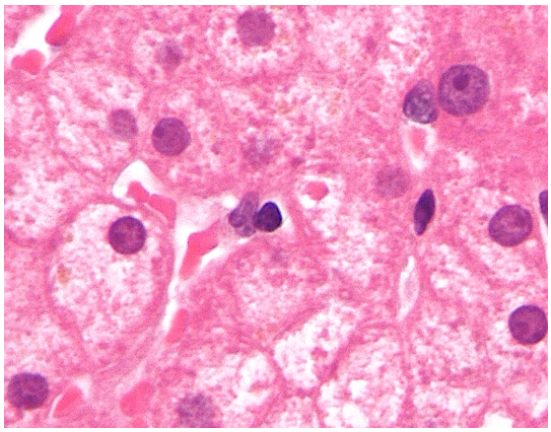
<table>
<tr><td colspan="2">- Hepatitis in general classically has ALT > AST, where #s can be in the hundreds to thousands, but I've seen plenty of variability on NBME Qs where this need not be the case, which is why I don't consider this a foundational point.</td></tr>
<tr><td>Hep A</td><td>- The answer for acute hepatitis in the United States most of the time. The Q might say the patient had recent travel to Mexico.
- Fecal-oral; only causes acute hepatitis.
- RNA virus; Picornaviridae (shared with polio, coxsackie, and echoviruses). Step 1 is P/F now, but that is some very mild-level stuff you could be aware of. If that makes you trip out, just ignore.
- IgM against HepA means acute infection.
- IgG against HepA means patient has cleared infection (because there is no chronic HepA).
- USMLE wants you to know HepA vaccine is indicated for IV drug users and MSM. There's a 2CK NBME Q where they just mention otherwise healthy MSM, and answer is Hep A vaccination.</td></tr>
<tr><td>Hep B</td><td>- Mandatory stuff for USMLE is the serology (I discuss below).
- Most common hepatitis infection worldwide. USMLE likes China for hepatitis B. Just a pattern I've noticed. Due to ↑↑ unvaccinated. In the USA, HepC is most common.
- Parenteral; can be acute or chronic.
- Transmitted vertically from mother to neonate, sex, via IV drugs, or blood exposure.
- Present in all body fluids, including breast milk.
- Virus is DNA, enveloped, circular (asked on USMLE, even though we have pass/fail exam). In contrast, Herpesviridae are DNA, enveloped, linear. I've discussed in my YouTube clips how memorizing viral structures is mostly a waste of time now that we have a P/F exam, but that this particular distinction between HepB = circular, and Herpesviridae = linear, is assessed.
- HepB produces a DNA-dependent DNA polymerase.
- Serology very HY:

- (+) Surface antigen = patient currently has HepB.
- (+) Surface antibody = patient is immune to HepB.
- (+) Core antibody = patient has HepB now or did in the past.
- (+) Core antibody IgM = patient has acute infection.
- (+) Core antibody IgG = patient has chronic HepB, OR cleared HepB.
- (+) Surface antibody / (-) Core antibody = Immune → Vaccinated against HepB.
- (+) Surface antibody / (+) Core antibody = Immune → History of HepB/cleared it/prior infection.
- (-) Surface antibody / (-) Core antibody = Not immune / Susceptible → need to vaccinate.
- (-) Surface antigen / (-) Surface antibody / (+) Core antibody IgM → window period (below).

- Vaccination against HepB is at birth, 2 months, and 6 months (no longer at 4 months).
- If mother's status is (+) or unknown for HepB, give HepB immunoglobulin (HBIG) to the neonate within 12 hours + give HepB vaccine as per normal. Do not wait to attain mother's status.
- You should know that a 2CK NBME Q gives mother's status as unknown when child is born → answer = "Give HepB vaccine now + only give IVIG if mother is positive." But new guidelines actually say to give HBIG to neonates within 12 hours of birth even if mom's status is unknown. I'm just letting you know this discrepancy because this answer is on the new NBME but doesn't follow current guidelines.
- If patient has Hx of completed HepB vaccination but has titers that show susceptibility, the answer is just "give more vaccine." Sometimes people's immunity wanes.

- Once a susceptible patient is exposed to HepB and the immune system attempts to clear it, sometimes Surface antigen will decline to the point that it is no longer detectible. But at the same time, the Surface antibody might not be high enough / at detectable levels yet. This is called the "window period," where both Surface antigen and antibody are negative, so it can appear as though the patient doesn't have an infection. However, Core antibody IgM will be (+). So the key point is that 1) you know the double-negative Surface antibody/antigen combo is seen in the window period, and 2) that Core antibody IgM is most reliable during the window period.

- HepB causes a "ground glass" hepatocyte appearance. You don't have to obsess over the image. Just memorize "ground glass" for HepB.</td></tr>
</table>

	 - USMLE really doesn't give a fuck about HepB pharm (i.e., entecavir, tenofovir). Waste of time. - You could be aware that interferon-α can be used for HepB.
Hep C	- **Hepatocellular damage is due to T cells / death is due to T-cell-mediated apoptosis,** not direct viral cytopathicity. I already mentioned this at top of chart, and this is applies to the other Heps as well, but I reinforce this as what I'd still say is the highest yield point for HepC on USMLE. - Parenteral; can be acute or chronic. - Transmitted almost exclusively from IV drugs/blood exposure. Not present in breastmilk and non-sanguineous body fluids (in contrast to HepB). - In contrast to HepB, HepC is not considered sexually transmitted. Large longitudinal study of couples with one HepC(+) partner showed sexual transmission almost nil (possibly due to menses exposure). If you're forced to choose for FM / behavioral science Qs, however, still inform that abstinence or barrier contraception minimizes risk. - RNA virus (Flaviviridae). - No vaccine due to ↑↑ antigenic variation (i.e., >7 genotypes and 80 subtypes of HepC exist). - IgM against HepC means acute infection. - IgG against HepC means usually means chronic HepC. - Histo shows "lobular necrosis," in contrast to the ground-glass appearance of HepB. USMLE doesn't give a fuck about you knowing the image. - Many drugs can be used to treat. USMLE doesn't care. What you could be aware of is pegylated interferon-α.
Hep D	- **Requires hepatitis B in order to infect,** which can be due to co-infection (happening at the same time) or superinfection (occurs later in someone who already has HepB). - If USMLE asks how to prevent HepD infection, answer = vaccination against hepatitis **B.** There is no vaccine against HepD. - Apparently HepB antigen forms the envelope for HepD (i.e., forms a circle around HepD). - HepD is RNA virus; aka delta virus.
Hep E	**- Causes fulminant hepatitis / ↑↑ risk of death in pregnant women.** - Same as with HepA, only causes acute hepatitis. - Seen more in Asia, e.g., Tibet. But if USMLE says Mexico + pregnant woman + fast death from hepatitis, you still have to use your head and know that's HepE over HepA.
Yellow fever	- Garbage diagnosis I don't think I've ever seen on USMLE material. Including it here as important negative. In theory: - Yellow fever is in *Flaviviridae* family (same as Zika, Japanese encephalitis, Dengue, West Nile). - Causes hepatitis and jaundice. - Can cause "councilman bodies" on biopsy with "mid-zone necrosis." These latter details date back decades in USMLE resources as "Oh em gee know this." Absolute nonsense. Waste of time.

Other HY hepatitis causes	
NASH	**- Answer on USMLE for liver condition in an overweight patient with normal labs.** - Non-alcoholic steatohepatitis. The new name is apparently metabolic dysfunction-associated steatohepatitis (MASH), but traditional NASH shows up a lot as answer choice on NBME material, so know both. - Underrated diagnosis in that students often have no idea about it but it shows up on both Steps (higher yield on 2CK IM). - Patient will have metabolic syndrome (i.e., ↑ BMI + ↑ lipids) + either mild transaminitis or completely normal labs. I'd say ~50% of Qs will give ALT and AST a very tiny bit elevated; the other ~50% of Qs will give you completely normal labs where you say WTF. - For example, they'll give you 40-year-old patient with BMI of 35 who comes in for routine health maintenance exam who has no complaints + lab studies are all completely normal; then the answer is just "fatty liver" or "non-alcoholic steatohepatitis." Student says, "Wait, but the labs are completely normal though." → Exactly. There's nothing wrong. But it's assumed most overweight people have some degree of fatty liver. - The condition starts out with completely normal labs, then progresses to very mild transaminitis. Over time, LFTs can worsen. Rarely it can cause cirrhosis and carcinoma. - Theoretical Tx is just hit the treadmill more and stop eating your dumb fast food.
Alcoholic	**- Classically AST > ALT.** - The ratio need not be 2:1. I've seen variations on NBME forms where there can be very mild transaminitis (i.e., both AST and ALT are <100, with AST only slightly higher than ALT). If the vignette is hyper-easy, they'll say something like AST is 400 and ALT is 200. Both should normally be <50 U/L. - Causes steatosis (fatty liver). - USMLE wants you to know **mallory hyaline** is seen on biopsy. You don't need to know the image, just this factoid.
Acetaminophen	- Patient will OD on acetaminophen and be asymptomatic for 24-28 hours, then will proceed to get ALT and AST in the tens of thousands + require liver transplant or death. - Breakdown of acetaminophen produces metabolite called NAPQI that depletes reduced glutathione. The latter normally mops up free radicals / prevents oxidation. - Acetaminophen overdose is most important for USMLE. ***N*-acetylcysteine is antidote.** - "*N*-acetylcysteine regenerates reduced glutathione." Memorize that. - Oxidized glutathione had disulfide bond (-S—S-); reduced glutathione has thiol (-SH). - Activated charcoal should apparently be administered to patients who present very acutely after massive ingestion, as this can ↓ acetaminophen absorption. But I've never seen this on NBME. It tends to be more Qbank that's gotchya-style this way. I've only ever seen *N*-acetylcysteine assessed across NBMEs.
Other drugs	- I'd say some important drugs that can cause transaminitis are: - Statins and fibrates, methotrexate, lithium, valproic acid, propylthiouracil. - **Mild transaminitis** is normal and expected with these agents; the answer is you do not need to decrease dose. **- For statins, myopathy is more common than toxic hepatitis** (offline Step 1 NBME 23).

Obgyn-related hepatic issues	
HELLP syndrome	- **HELLP** syndrome = **H**emolysis, **E**levated **L**iver enzymes, **L**ow **P**latelet count. - Can occur as part of severe preeclampsia (HTN + proteinuria after 20 weeks of pregnancy). - Vignette will be pregnant woman >20 weeks of pregnancy where you see the platelets and Hb are low and LFTs are up. Not complicated. - Highest yield point for USMLE is you get **schistocytes** on a blood smear. USMLE is obsessed with this. - Causes of schistocytes on USMLE = HELLP syndrome, HUS, TTP, DIC, and mechanical heart valves. If you're about to trip out, don't worry, I talk about these conditions in the HY Heme PDF.

Intrahepatic cholestasis of pregnancy **(2CK/3)**	- Answer on USMLE for woman in 3rd trimester who gets itchy palms and soles and ↑ serum bile acids. - Cause of 3rd trimester miscarriage. - Treat with ursodeoxycholic acid (ursodiol).
Acute fatty liver of pregnancy **(2CK/3)**	- Life-threatening disorder for both the mother and fetus presenting in 3rd trimester. - Due to long-chain 3-hydroxyacyl-coenzyme A dehydrogenase (LCHAD) deficiency in the fetus, leading to impaired mitochondrial fatty acid oxidation. Sounds highly pedantic and arcane, but it has been asked. - Presents with nausea, vomiting, abdominal pain, and jaundice that can progress to encephalopathy, coagulopathy, and multi-organ failure. - Labs can show impaired LFTs, RFTs, and coagulation tests. - Tx = immediate delivery of fetus.
Fitz-Hugh-Curtis Syndrome	- Pelvic inflammatory disease (chlamydia or gonorrhea) that has extended to the liver capsule. - USMLE wants you to know this causes **fibrin** deposition on the liver due to the inflammatory process. This is similar to post-surgical adhesions, where inflammation leads to fibrin deposition. - In other words, they'll show you some gross path specimen of a liver covered in yellow material + tell you a woman had severe pelvic inflammatory disease, and then the answer is just "fibrin." Sounds odd, but I'm telling you how it's tested.

Heavy metal liver diseases	
Hereditary hemochromatosis	- Autosomal recessive; *HFE* gene; chromosome 6. - Iron overload. - **Mechanism USMLE wants is "increased intestinal iron absorption."** - Main iron regulation is via shutting off intestinal absorption; this is impaired in hemochromatosis. - Body has poor ability to excrete iron; otherwise there are minor losses via skin shedding. Since women excrete naturally via menses, presentation occurs earlier in men (who clearly don't have this mechanism of excretion). - Can present as "bronze diabetes" → hyperpigmentation due to hemosiderin deposition in skin + ↑ fasting sugars (iron deposition in tail of pancreas). - Miscellaneous other findings can be seen like infertility (iron deposition in hypothalamus, anterior pituitary, or gonads), cardiomyopathy, or arthritis (pseudogout). - Hereditary hemochromatosis, primary hyperparathyroidism, and hypothyroidism are 3 most important causes of pseudogout. I used to only discuss the former two with students over the years, but the latter shows up on a new 2CK NBME exam where they mention chondrocalcinosis (calcium deposition in cartilage). - USMLE wants you to know there is ↑ risk of hepatocellular carcinoma. There is easy NBME Q of hemochromatosis where they ask what patient is at increased risk for, and the answer is simply "hepatocellular carcinoma." - Diagnose with **ferritin >300 ng/dL.** Transferrin saturation will also clearly be ↑. - It is exceedingly rare that ferritin is >300 in other conditions, however this can occur in lymphoma and leukemia, where ↑ ferritin is a poor prognostic marker in non-Hodgkin lymphoma. There is one NBME Q on 2CK where ferritin is 300 where it's not hemochromatosis, but USMLE won't play gotchya. - Treat with **serial phlebotomy, not chelators.** - Chelators such as deferoxamine or deferasirox are for secondary hemochromatosis due to transfusional siderosis (i.e., repeated blood transfusions that contain iron, for e.g., β-thalassemia major).
Wilson disease	- Autosomal recessive. - Copper overload.

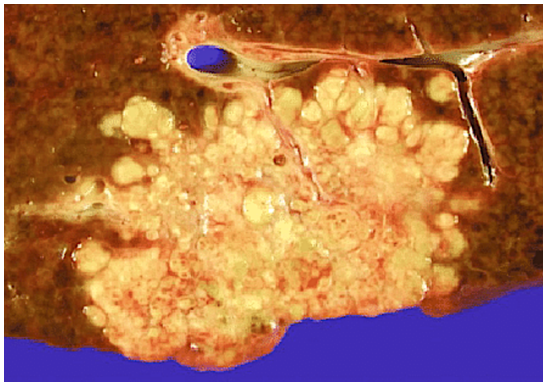
	- Inability to secrete copper into bile from the liver. Copper is normally excreted by the body via secretion into bile. - ↑ urinary copper + ↓ serum ceruloplasmin (copper-binding protein in the blood; in the case of copper overload, body tries to minimize amount carried in blood). - Buzzy / pass-level detail is Keiser-Fleischer rings, which is copper deposited in the cornea of the eye. Vignette can give you what sounds like Wilson disease, and then the answer is **"slit-lamp exam."** - Can cause ↑ LFTs with cirrhosis and Parkinsonism. - Copper deposits in basal ganglia, especially the putamen. - Parkinsonism in a young patient = Wilson until proven otherwise. - In old patient, Parkinsonism = Parkinson disease, normal pressure hydrocephalus, Lewy-body dementia, or progressive supranuclear palsy. - When a disease presents similarly to Parkinson disease but isn't, we call it a Parkinson-plus disorder. Wilson is an example of a Parkinson-plus disorder. - Treat with the copper chelator penicillamine.

Primary hepatocellular carcinoma (HCC) vs Metastases

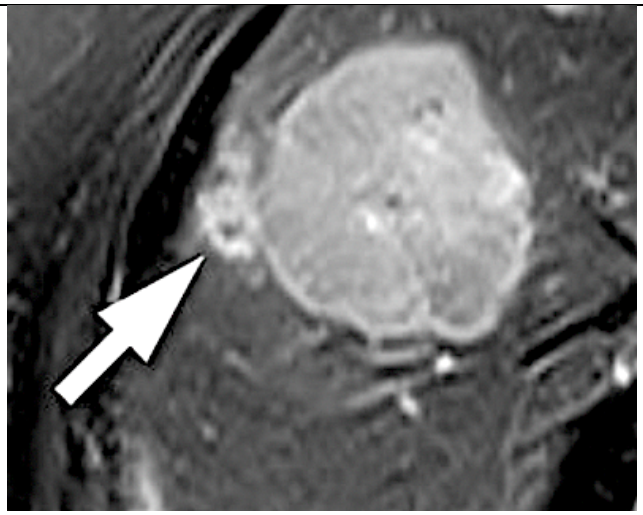
Three main points:
1) Primary HCC is going to be one large lesion, sometimes with tiny satellite lesions. In contrast, metastases to the liver will present as multiple lesions everywhere.
2) USMLE wants **colon cancer** as the most common type that goes to liver.
3) Do CT scan of abdomen as first step in diagnosis of suspected liver cancer. Biopsy is confirmatory.

- USMLE loves gross images of these conditions.
- HCC will present as one large solitary lesion, sometimes with mini surrounding satellite lesions.

Notice how the HCC is one solitary lesion.

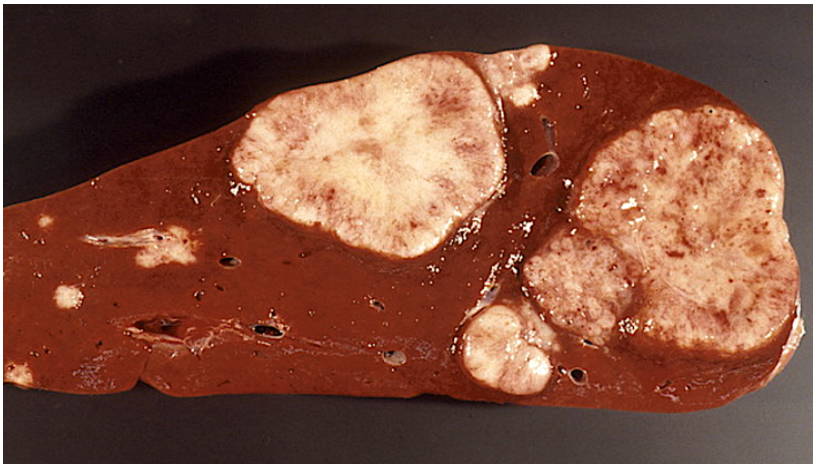


HCC can also grow as a "confluent multinodular" type.
But note how the cancer is still all relatively confined to one area.

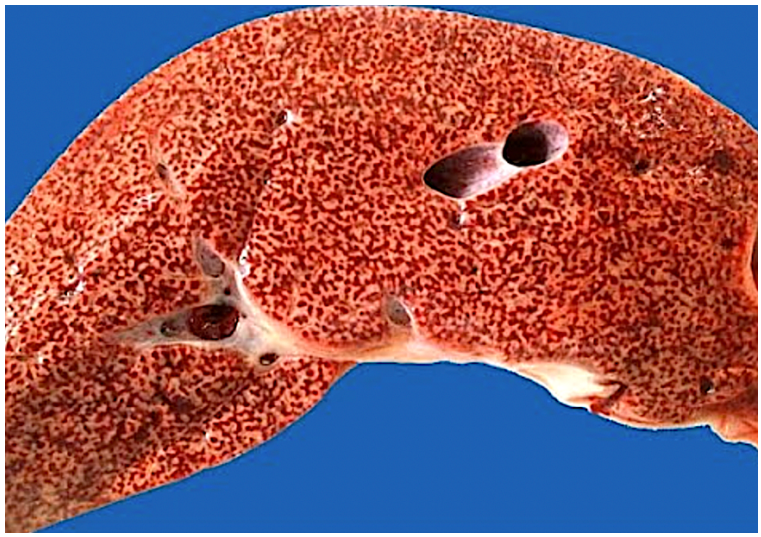


MRI shows satellite nodule of HCC that grows next to the larger, initial growth.

In contrast, these are hepatic metastases. Notice how the lesions are everywhere. USMLE will show image like this, and then the answer will just be “colorectal carcinoma.”



Another image of metastases. Not as obvious as the first image, but still you can see there are multiple lesions not confined to one location.

Liver in relation to heart/kidney	
Heart failure	- Right heart failure (either due to cor pulmonale or CHF) can result in venous congestion within the liver. This can cause a characteristic mottled appearance on gross specimen known as **"nutmeg liver."** 
Pulsatile liver	- Just know this is an examination finding seen in tricuscpid regurg. - As I talked about in the HY Cardio PDF, the highest yield cause of tricuspid regurg on USMLE is pulmonary hypertension / cor pulmonale.
Hepatorenal syndrome	- Just know that in chronic liver disease, sometimes the kidneys can also fail. - This is supposedly due to ↓ blood flow in patients with cirrhosis and ascites. - Biopsy shows no abnormalities. That is probably the only notable point. - Never seen this as correct answer on NBME. Only reason I'm including it is because I see it as a distractor on some 2CK Qs in particular, so you should at least know it exists and is likely always wrong.

Bowel ischemia	
Ischemic colitis	- Patient with cardiovascular disease (CVD) + blood in the stool. - Caused by bleeding at ischemic ulcers at watershed areas in the colon (i.e., splenic flexure and rectosigmoid junction). - Can occur randomly or due to inciting event like recent AAA surgery. - Diagnose with colonoscopy to visualize the ischemic ulcers.
Chronic mesenteric ischemia	- Patient with CVD + abdominal pain 1-2 hours after eating meals. - Caused by atherosclerosis of the SMA or IMA → consuming food ↑ oxygen demand of bowel → angina of bowel. - The timing of the pain in relation to meals sounds like duodenal ulcers, however instead of the vignette being a 29-year-old dude from Indonesia with *H. pylori*, it will be a 69-year-old dude with Hx of coronary artery bypass grafting, intermittent claudication, HTN, and diabetes. - Next best step is "mesenteric angiography" on USMLE.
Acute mesenteric ischemia	- Presents 3 different ways on USMLE: 1) Severe abdo pain in patient with AF → LA mural thrombus launches off to SMA or IMA. 2) Severe abdo pain in patient just cardioverted/defibrillated → can launch LA thrombus off to SMA or IMA. There's a Q on 2CK form where this is the case, where they don't mention AF in the stem. 3) Severe abdo pain in patient with Hx of chronic mesenteric ischemia (i.e., acute on chronic) → atheroma within SMA or IMA ruptures, effectively causing an "MI of the bowel."

	- USMLE wants "mesenteric angiography" as next best step. - Laparotomy is also answer on NBME form. - I've never seen medications or endarterectomy as answers.

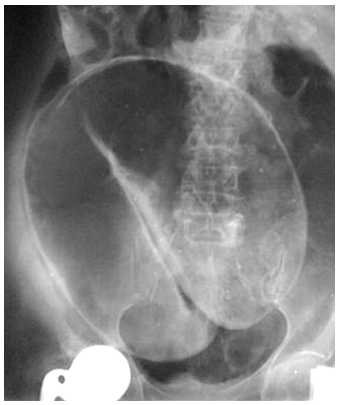
GI thromboses	
Mesenteric vein thrombosis	- The splenic vein and superior mesenteric vein (SMV) merge to form the portal. The inferior mesenteric vein (IMV) goes to the SMV. - Cirrhosis → ↑ portal venous pressure → ↑ SMV + IMV pressure → stasis. - In addition, any malignancy a patient might have → hypercoagulable state. - NBME Q gives patient with cirrhosis and lung cancer who has abdominal pain + dark, mottled small bowel on surgical examination → answer = "mesenteric venous thrombosis." Students ask why. I say, "Well there's ↑ mesenteric venous pressure from the cirrhosis + hypercoagulable state from the malignancy."
Splenic vein thrombosis	- Risk factors are same as above. - USMLE wants you to know splenic vein thrombosis can be a cause of esophageal varices in patients without cirrhosis (the usual risk factor for varices). This is due to collateral developments between the splenic vein and the left gastric vein. The left gastric vein drains the esophageal veins, so if there is high left gastric venous pressure, then there is high esophageal venous pressure. - Also, the short gastric veins feed into the splenic vein. So if we have ↑ splenic venous or portal venous pressure, then the short gastric veins also have ↑ pressure.
Budd-Chiari syndrome	- Aka hepatic vein thrombosis. - The hepatic vein drains the liver. So if we get thrombosis here, we get pressure back up to the liver. - Caused by polycythemia vera and pregnancy. - Triad = abdominal pain, ascites, and hepatomegaly.
Portal vein thrombosis	- Caused by portal venous stasis from cirrhosis (portal vein enters the liver).

Mandatory GI anatomy		
Structure	**Spinal level**	**HY points**
Celiac trunk	T12	- Supplies foregut (oral cavity until 1st part of duodenum). - Its 3 branches are: splenic artery, common hepatic artery, left gastric artery. - For whatever reason, the USMLE wants you to know the spleen is supplied by an artery of the foregut but is not itself derived from the foregut (i.e., it is derived from midgut). - The short gastric arteries come off the splenic artery and supply the superior greater curvature of the stomach.
SMA	L1	- Supplies the midgut (2nd part of duodenum until 2/3 distal transverse colon). - What the USMLE will do is give vignette of SMA thrombosis in peripheral vascular disease, causing acute mesenteric ischemia, and then ask which structure is involved → answer is anything that is midgut (e.g., **jejunum,** ascending colon, etc.)
IMA	L3	- Supplies the hindgut (distal 1/3 of transverse colon until the rectum). - Same as for SMA, they'll say there's acute mesenteric ischemia of IMA and then ask for involved structure (e.g., descending colon). - USMLE will also ask cancer lymphatic/venous drainage, which follows the same distribution as arterial supply. For instance, they'll say there's a cancer of the descending colon, and the answer is "inferior mesenteric lymph nodes" or "inferior mesenteric vein." Not complicated.

Renal/gonadal arteries	L2	- Gonadal arteries = testicular arteries in males and ovarian arteries in females.
Aortic bifurcation	L4	- Bifurcation of abdominal aorta into the common iliacs occurs as L4. - What USMLE will do is tell you a AAA repair is performed where a graft needs to be inserted between the aortic bifurcation and the renal arteries. Then they'll ask blood supply to which structure will get compromised. So you need to say, "Well the bifurcation is at L4, and the renal arteries are at L2, so anything that's IMA would get fucked up since IMA is L3." → answer = "descending colon."

Appendicitis
- Triad of RLQ pain, fever, and vomiting (Murphy's triad). - Pain starts at epigastrium (visceral pain) and migrates toward RLQ (parietal pain). USMLE will ask why there is movement of the pain → answer = "inflammation of the parietal peritoneum." In other words, organ pain (visceral pain) is felt somewhat non-specifically around the epigastrium. But once it becomes more superficial and involves the parietal peritoneum, it is felt more localized. - Pain at McBurney's point (1/3 of the way from the anterior superior iliac spine to the umbilicus). - (+) Rovsing sign is pain felt in RLQ upon palpation of LLQ. - USMLE wants you to know appendicitis pain can be **RUQ in pregnancy** due to displacement of bowel and a shift in appendiceal location. What they will do is tell you pregnant woman in 3rd trimester has RUQ pain, fever, and vomiting + negative abdo ultrasound (meaning not cholecystitis) → answer = "appendicitis." - "Inflammatory mass and fecalith in the cecum" is phrase used on 2CK NBME form to describe appendicitis. - Diagnosis is done via ultrasound first, and sometimes CT. However, I have not seen USMLE assess either of these, likely because their utility is debated. What I have seen is **laparoscopic removal in the stable patient.** - USMLE will give ethics/consent Q, where they say patient has operation performed for ovarian procedure but it is seen that she has an acutely inflamed appendix → answer = "remove due to necessity of medical emergency." - However, if the scenario is reversed and, while performing an appendectomy, the surgeon notices a suspicious lesion on, e.g., one of the ovaries, the answer is "do not biopsy." Consent must first be obtained for all non-emergencies.

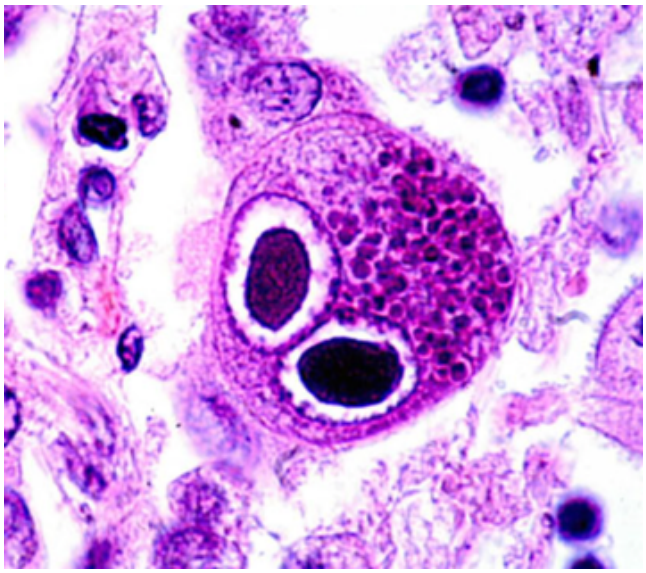
HY colorectal / proctocolonic diagnoses	
Angiodysplasia	- Tortuous, superficial vessels in the colonic wall that cause painless bleeding per rectum in elderly. - Classically associated with aortic stenosis (Heyde syndrome; possibly related to pressure backup to the colon). - NBME Q gives older guy arguing with his wife + gets chest pain and bleeding per rectum. Even though colonoscopy is used for diagnosis, they say in this Q that colonoscopy is negative, likely to tell you it's not cancer causing the bleeding.
Diverticulosis	- Mere presence of diverticula in the colon; >50% of people over age 60 in the US. - ↑ straining throughout life leads to herniation of mucosa + submucosa through the muscularis propria of the colonic wall. - Usually asymptomatic, but can bleed. Diverticular bleed is most common cause of painless bleeding per rectum in elderly, followed by colorectal cancer, followed by angiodysplasia. - Do not confuse with diverticulitis, which is when a diverticulum becomes inflamed.
Diverticulitis	- **LLQ pain + fever in patient over 55.** One of the highest yield diagnoses on USMLE. - Inflamed diverticulum, usually in the sigmoid colon. - Can cause **colovesical fistula** (on new 2CK NBME), where a passageway between the colon and bladder forms, leading to UTI and mixed flora in the urine. - **CT of the abdomen with contrast** is how to diagnose. - Do not scope acutely, as this can cause perforation.

	- After patient is treated with antibiotics, a colonoscopy should be scheduled weeks to months later to rule out malignancy. But once again, never scope acutely. - Perforated diverticulitis can require colectomy.
Sigmoid volvulus	- **Patient over 70** + 2-3 days of constipation + abdo pain. - Rotation around its mesentery causes "dilation of sigmoid colon" (answer on NBME as what is most likely to be seen in patient). - **Abdominal x-ray** is used to diagnose, which shows a **coffee bean sign,** which is one of the highest yield radiographic images on USMLE.  - Tx on NBME is "sigmoidoscopy-guided insertion of rectal tube."
Fecal impaction	- Patient over 70 + chronic constipation + **"hard stool palpated in the rectal vault."** - Can sometimes cause fecal incontinence and paradoxical overflow diarrhea leading to encopresis (i.e., word for shitting yo pants). - Idiopathic, but exacerbated by opioids. - Treatment is with enema and laxatives.
Hemorrhoids	- Bleeding per rectum that will be described as "blood on the toilet paper," or "blood that drips into the toilet bowl." - Hemorrhoids are technically normal vascular structures within the anal canal that facilitate/cushion the passage of stool. - Bleeding from internal hemorrhoids is painless. These are above the pectinate line in the anal canal. - Bleeding from external hemorrhoids is painful. These are below the pectinate line. - Pregnancy and cirrhosis are risk factors. - 2CK NBME wants rectal exam followed by **colonoscopy** for diagnosis. If colonoscopy is not listed, the answer will be **anoscopy.** - Hemorrhoids often self-resolve and can be managed conservatively with sitz bath, NSAIDs, and ↑ dietary fiber. However, if USMLE forces you to choose surgical management for more severe cases, the answer is **rubber band ligation.**
Anal fissure	- Young adult who has painful bowel movements +/- blood in the stool. - The key detail is **they refuse the rectal exam because the pain is so bad.** - For whatever reason, they can also say there's an associated skin tag. I've seen this more than once, where the student says, "What's with the skin tag?" No fucking idea. - If Step 1 forces you to choose a location for anal fissure, the answer is "posterior in the midline." Stupid, but it's asked. - Mechanism can be "increased anal sphincter tone." - Tx on 2CK Surg is **sitz bath.** - If the USMLE forces you to choose a med, topical nitrates or diltiazem is used.
Pilonidal cyst/abscess	- Answer on USMLE for cystic mass at the superior aspect of the gluteal cleft. - Contains hair; often caused by ingrown hairs; can be described as red/inflamed. - Tx on 2CK Surg = incision and drainage.
Perianal abscess	- Answer on USMLE for painful, erythematous mass near anal verge. - Increased risk in Crohn and diabetes.

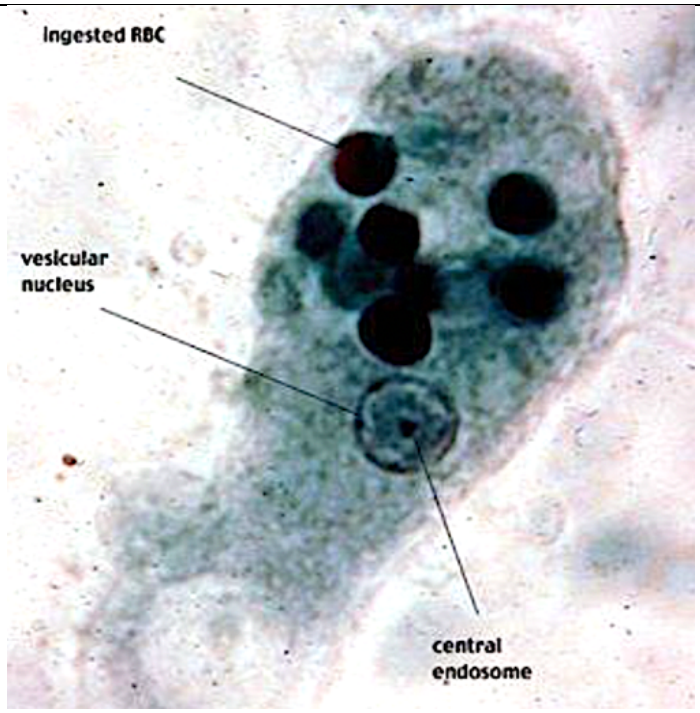
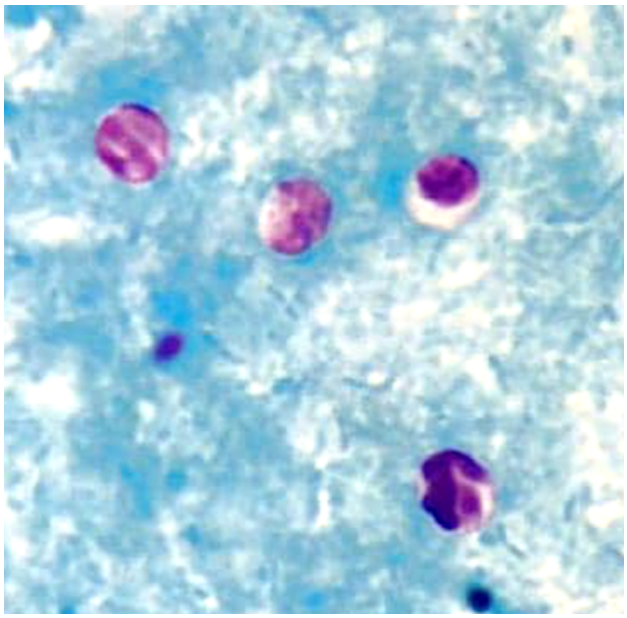
	- Tx on 2CK Surg = incision and drainage.
Anal malignancy	- 2CK Surg Q gives cancer at anal verge + asks for next best step in management → answer = colonoscopy; excision is wrong answer. Presumably the scoping is done to first investigate the extent of the disease, as that might alter management.

HY GI bacterial infections	
Gram-negative rods	
ETEC	- Enterotoxigenic *E. coli* causes traveler's diarrhea. - Will present as brown/green diarrhea in person who's gone to Mexico or Middle East classically. - Heat-labile toxin ADP ribosylates adenylyl cyclase → ↑ cAMP. - Heat-stable toxin ADP ribosylates guanylyl cyclase → ↑ cGMP.
EHEC	- Enterohemorrhagic *E. coli* causes bloody diarrhea 1-3 days after consumption of beef. - Produces shiga-like toxin, which can cause hemolytic uremic syndrome (HUS; triad of renal dysfunction, schistocytosis, and thrombocytopenia).
Shigella	- Bloody diarrhea 1-3 days after consumption of beef. - Also can cause HUS via shiga toxin. - Requires very few organisms to cause infection. - Main virulence is via its ability to invade, not the toxin itself. - Both shiga toxin of *Shigella* and shiga-like toxin of EHEC inhibit protein synthesis by cleaving the eukaryotic 60S ribosomal subunit. Step 1 is P/F now, although that's known to be asked, so you can decide yourself whether you want to memorize the **nonsense.**
Salmonella	- Food poisoning is caused by *Salmonella typhimurium* and *Salmonella enteritidis*. The nomenclature has changed with time, but those names are acceptable. - Bloody diarrhea 1-3 days after infection from consuming poultry, or following exposure to eggs or reptiles (i.e., turtles, lizards, etc.). New NBME Q mentions bloody diarrhea in someone with a pet lizard. - Requires more organisms than Shigella to cause infection. - *Salmonella typhi* causes typhoid fever, which will be rose spots on the abdomen in a patient who's "prostrated" (i.e., lying down in pain); can be either diarrhea or constipation. The reservoir for *S. typhi* is humans, not chickens/reptiles. - *Salmonella* can go latent in the gall bladder apparently.
Vibrio	- *Vibrio cholerae* (cholera) presents as "liters and liters" of rice-water stool in someone who went traveling to, e.g., Mexico. The way you can differentiate this from ETEC traveler's diarrhea is that cholera is notably **profusely high-volume.** - Acquired fecal-oral (i.e., fecal-contaminated food/water). - Both ETEC and cholera vignettes can tell you the patient has 8-12 stools daily, so it's not the # of stools that matters; it's the emphasis on volume. Cholera causes death via severe dehydration and electrolyte disturbance. - Vibrio toxin has same MOA as ETEC heat-labile toxin (i.e., ↑ cAMP). - Tx is **oral rehydration** on USMLE; if patient has low BP or altered mental status (i.e., confusion/coma), IV hydration is done. - *Vibrio parahemolyticus* doesn't cause profuse, watery diarrhea the same way cholera does. I've seen this organism asked once in an NBME vignette where the patient ate sushi (can be acquired from sushi and shellfish). - *Vibrio vulnificus* causes severe sepsis in half of patients. This is asked on offline NBME 19 where a dude went running on a beach and got sepsis, with no mention of consumption of food. But it's apparently acquired from shellfish.
Yersinia enterocolitica	- Causes bloody diarrhea + either **appendicitis-like (i.e., RLQ) pain** or **arthritis.** - The RLQ pain is from mesenteric adenitis or terminal ileitis. - Toxin has same MOA as ETEC heat-stabile toxin (i.e., ↑ cGMP).
Campylobacter jejuni	- Bloody diarrhea 1-3 days after consumption of **poultry.**

	- Can cause Guillain-Barre syndrome (ascending paralysis + ↓ tendon reflexes + albuminocytologic dissociation in the CSF → Tx with IVIG + plasmapheresis). - Grows best at high temperatures (42 degrees).
Gram-positive rods	
Bacillus cereus	- Watery diarrhea and/or vomiting in patient who's consumed reheated or fried rice. The process of heating/re-heating causes germination of spores. - Can also cause eye infections (weird, but on USMLE). They'll say patient who's had eye surgery + now has infection caused by gram(+) rod → will be only gram(+) rod listed.
Clostridium perfringens	- Watery/secretory diarrhea following consumption of **poultry.** - Causes gas gangrene (CO_2 gas) due to production of α-toxin/phospholipase; presents as black skin / **crepitus.** - Can also cause emphysematous cholecystitis (air in gall bladder wall).
Clostridium difficile	- Diarrhea (pseudomembranous colitis) ~7-10 days after commencing oral antibiotics. - Antibiotics kill off normal bowel flora, allowing *C. difficile* to overgrow. - *C. difficile* is not normal flora, however. It is acquired via **consumption of spores.** - Can be watery or bloody diarrhea on NBME. Can also cause LLQ cramping (not RLQ as with *Yersinia*). - There is NBME Q where they say 28-year-old with LLQ cramping and bloody diarrhea 7 days after starting oral antibiotics → answer = *C. diff*; wrong answer is *Yersinia.* - USMLE doesn't care which antibiotics cause it; can be any. - Diagnose with **stool AB toxin test; stool culture is wrong answer.** - Treat with **oral vancomycin.** - Vancomycin has poor oral bioavailability, so is usually given IV for things like endocarditis and meningitis. But for *C. diff* infection, *that's a good thing* because we want it to stay within the GI tract. - Other fancy Abx like fidaxomicin, rifaximin, etc., I've never seen on NBME. More just masturbation around dumb factoids.
Gram-positive cocci	
Staph aureus	- *S. aureus* pre-formed heat-stable toxin is acquired from two main sources on NBME: 1) various meats sitting under a heat lamp / out for long periods of time, e.g., at a buffet; 2) dairy products like creams, custards, potato salad. - Notably causes **vomiting 1-6 hours** after consumption. This is notable, as the symptoms occur rather quickly. Diarrhea, both watery and bloody, can occur, but is not mandatory. The main crux is the vomiting.
Other GI bacteria	
Tropheryma whipplei	- Causes Whipple disease, a GI malabsorptive syndrome where the patient can also get arthritis and renal and cardiac disease. - 100% of Qs will say "PAS-positive macrophages in the lamina propria." You don't need to know the image for this. But for whatever magical reason, the USMLE gives a fuck about this detail. Old-school pathology docs who reminisce maybe. - New highly pedantic Q on 2CK NBME 10 wants "ceftriaxone + daily TMP/SMX for one year" as the treatment. Organism isn't even HY. No idea why they care.
Tropical sprue	- Causes flattening of villi similar to Celiac, but considered infective/bacterial in origin. Literature says exact etiology not certain. - Answer on USMLE for patient living in tropical area with unknown malabsorptive disease, where you can easily eliminate the other answers. - Tx = tetracycline.

HY GI viral infections	
Rotavirus	- Most common cause of watery diarrhea in unvaccinated children < 5 years. - Vaccine normally given orally at 2, 4, and 6 months of age. - Double-stranded, segmented RNA (NBME asks it). - Wheel-shaped (also on NBME).
Norwalk virus	- Most common cause of watery diarrhea in adults and rotavirus-vaccinated children. - Cruise ships and business conferences are buzzy places to acquire (fecal-oral); basically any place with high density of people. - If the Q says a young child + family all have watery diarrhea, the answer is Norwalk, not Rota, since only the young child would get Rota, not the family also.
Herpes	- HSV1/2 causes herpes esophagitis → odynophagia + punched-out ulcers in esophagus.
CMV	- CMV esophagitis →odynophagia + linear (confluent) ulcers in esophagus. - CMV colitis → bleeding per rectum in AIDS patient with CD4 count <50. - CMV retinitis → blurry vision in HIV (or immunosuppressed) patient. - Most common organism transmitted via blood transfusions and organ transplants. - USMLE likes “intranuclear inclusions” or “intranuclear inclusion bodies” for CMV. This refers to the “owl eyes” that can be seen on histo. 

HY GI protozoal infections	
- A protozoan is a unicellular eukaryote. - **ECG** → ***E****ntamoeba histolytica,* ***C****ryptosporidium parvum,* and ***G****iardia lamblia* are all GI protozoa that are acquired via **cysts in water** (i.e., they are **water-borne**). If “water-borne” and “fecal-oral” are both listed as answers, USMLE wants “water-borne” as means of acquisition.	
Entamoeba histolytica	- Bloody diarrhea in person who went to Mexico. - Can cause “flask-shaped ulcers” in the small bowel and **liver abscess.** - Demonstrates “erythrophagocytosis,” where RBCs can be seen within it on LM.

	 - Treat with metronidazole. - Iodoquinol kills intraluminal parasite.
Cryptosporidium parvum	- Watery diarrhea in person who went to Mexico. - Appears as **acid-fast cysts** (same stain as TB).  - Self-limiting in immunocompetent persons → Tx = supportive care. - Chronic diarrhea in HIV → Tx = nitazoxanide.
Giardia lamblia	- Steatorrhea in person who went to Mexico. - Steatorrhea = bloating + extremely foul-smelling stool that floats. - The steatorrhea is due to *Giardia* causing **malabsorptive** diarrhea. - There is one NBME Q for *Giardia* where they say "foul-smelling *watery* diarrhea with bloating," which is audacious, since 9 times out of 10 it is not described as watery. But the "bloating" and "foul-smelling" are consistent with steatorrhea in this context. Just letting you know it exists.

	- Acquired via fresh water lakes / scuba diving. There is a *Giardia* NBME Q where they say a woman goes scuba diving in Mexico and then gets foul-smelling stools with bloating. Student says, "Wait but I thought *Giardia* was fresh water. If she went scuba diving in the ocean then how is that *Giardia*?" → People can go scuba diving in fresh water lakes too bro. Don't know what to tell you. - Flagellated protozoan. USMLE wants you to know the images for both the cyst as well as the flagellated trophozoite. Cyst form of *Giardia*. Not dramatic. Trophozoite form of *Giardia*. - Tx = metronidazole.

HY GI parasitic (helminth) infections	
- USMLE wants you to know **routes of acquisition** for the nematodes (i.e., ingested or through feet, etc.). - You do not need to obsess over exactly which anti-helminthic agent treats which organism. Waste of time. What USMLE wants you to know is the big-picture concept that -bendazoles (i.e., mebendazole) and pyrantel pamoate are the two agents that treat nematodes primarily. -Bendazoles microtubules (tubulin) and pyrantel pamoate causes helminth paralysis. - Praziquantel is used for cestodes and trematodes. It causes helminth paralysis. - Albendazole is the odd one out that is used for neurocysticercosis (*Taenia solium*; cestode). - USMLE doesn't give a fuck about diethylcarbamazine.	
Nematodes (roundworms)	
Ascaris lumbricoides	- Ingested. Giant roundworm; causes intestinal obstruction. (Ascariasis)
Enterobius vermicularis	- Ingested. Causes perianal itching / (+) tape test in children. (Enterobiasis)

Angiostrongylus	- Ingested by eating raw slugs/snails. Causes eosinophilic meningitis.
Trichinella spiralis	- Ingested by eating pork/**bear** meat. Causes triad of fever, periorbital edema, and myalgias. (Trichinosis)
Strongyloides stercoralis	- Acquired through soil/one's feet. Causes intestinal obstruction + pulmonary symptoms (apparently travels to lungs via blood, then travels up respiratory tree into esophagus, where it is swallowed).
Hookworms	- Acquired through soil/one's feet. "Hookworms" refers to *Ancylostoma duodenale* and *Necator americanus*. - Cause **microcytic anemia / ↓ hemoglobin** due to sucking blood in GI tract.
Cestodes (tapeworms)	
Diphyllobothrium latum	- Fish tapeworm. Nothing else you need to know.
Taenia solium	- Pork tapeworm. Causes cysticercosis (muscle pain/cysts) + neurocysticercosis ("Swiss cheese" appearance of brain or soap bubbles in ventricles).
Echinococcus granulosus	- Aka hydatid worm, or dog tapeworm; causes hydatid cyst disease. - Acquired from dogs (clearly). - Causes liver cysts. Only point you need to know is that you do not biopsy because this can cause anaphylaxis. Tx = surgically remove.
Trematodes (flukes)	
Schistosoma	- Penetrates skin when swimming; snail is host. - *Schistosoma mansoni/japonicum* can cause portal hypertension. Not HY but shows up once on NBME. Not to be confused with *Schistoma hematobium*, which causes squamous cell carcinoma of the bladder.
Paragonimus westermani	- Ingested from crab meat or crayfish. - Aka lung fluke. Most common cause of hemoptysis in the world.
Clonorchis sinensis	- Ingested from fish. Causes cholangiocarcinoma.

HY GI pharm	
PPIs	- Omeprazole. - Shut of proton pumps on parietal cells → ↓ acid secretion. - Irreversible and non-competitive; more efficacious than H2 blockers, which are reversible and competitive. - Choose PPIs over H2 blockers for Tx of most things, like GERD, ulcers, and *H. pylori.* - Diagnosis of GERD is 2-week trial of PPIs. - PPIs cannot be given with sucralfate or -azole antifungals (on NBME).
H2-blockers	- Cimetidine, ranitidine. - Not used often. PPIs more efficacious. - One 2CK Surg form has "2-week trial of H2 blocker" as correct for diagnosis of GERD, but PPIs aren't listed. As I said above though, if you're forced to choose between a PPI and H2 blocker, the PPI is correct basically always. - Cimetidine can cause gynecomastia (HY on USMLE) and inhibits P-450. - Ranitidine doesn't inhibit P-450. Sounds pedantic, but there's an NBME Q where they mention coma in someone taking diazepam + 2nd drug → answer = cimetidine as 2nd drug; ranitidine also listed but wrong (doesn't inhibit P-450).
Metoclopramide	- As discussed earlier, anti-emetic + prokinetic agent (means ↑ peristalsis). - D2 antagonist but also an antagonist of serotonin 5HT3 and agonist of 5HT4 receptors. The effects on serotonin receptors ↑ gut peristalsis. - HY use on USMLE is **diabetic gastroparesis** (i.e., sounds like GERD but patient has severe diabetes, where answer is metoclopramide, not PPI). - Because it's a D2-antagonist, adverse effects are similar to the anti-psychotics – i.e., prolong QT interval, hyperprolactinemia, anti-pyramidal effects (acute dystonia, parkinsonism, akathisia, tardive dyskinesia). - 2CK Psych Qs like metoclopramide as cause of parkinsonism, where the answer is "discontinue metoclopramide" as next best step before simply giving the patient Parkinson disease meds.

Ondansetron	- Powerful anti-emetic classically used for nausea/vomiting from chemoradiotherapy. - 5HT3 antagonist.
Erythromycin	- Motilin receptor agonist that can be used for diabetic gastroparesis. - Also a 50S ribosomal subunit inhibitor (antibiotic).
Misoprostol	- PGE_1 analogue. Used for **NSAID-induced ulcers** after PPIs. - NSAIDs ↓ prostaglandins. Therefore misoprostol is the replenished prostaglandin. - Prostaglandins ↓ acid production, ↑ mucous and bicarb production, and ↑ gastric mucosal blood flow.
Magnesium	- Antacid. Causes diarrhea.
Aluminum.	- Antacid. Causes constipation ("Aluminimum amount of feces.")
Calcium	- Antacid. Causes milk-alkali syndrome (↑ Ca^{2+} + ↑ $HCO3^-$+ renal dysfunction). - Bisphosphonates (e.g., alendronate) and tetracyclines (e.g., doxycycline) cannot be taken with calcium or iron supplements (i.e., divalent cations), since the latter chelate the drugs and ↓ absorption / oral bioavailability. USMLE loves this. - USMLE will give you, e.g., prostatitis treated with ciprofloxacin, + list all sorts of other meds the patient is on, as well as a calcium supplement, and then tell you the patient's condition isn't improving + ask why → answer = calcium carbonate.
Bismuth	- Can be used as part of *H. pylori* regimens (but not first-line).
Sucralfate	- Can be used to coat ulcers / form a barrier to protect against acid and ↓ pain. - Cannot take with PPIs. The latter ↓ ability of sucralfate to crosslink/work correctly.
5-ASA compounds	- Mesalamine, sulfasalazine; NSAIDs used for IBD, as discussed earlier.
TNF-α blockers	- Infliximab, adalimumab, and etanercept. - USMLE likes these for IBD *after* 5-ASA compounds and steroids are attempted. - Infliximab and adalimumab are monoclonal antibodies against TNF-α. - Etanercept is a recombinant receptor that mops up soluble TNF-α.
Octreotide	- Somatostatin analogue used for esophageal varices Tx after banding. - Acts by ↓ portal blood flow.
Propranolol	- Beta-blocker that can be used for esophageal varices prophylaxis (not Tx).
Lactulose	- Used to Tx hyperammonemia in cirrhosis. - Carbohydrate that is metabolized by gut bacteria into acidic substrates that trap ammonia as ammonium. The latter is then excreted (ions aren't absorbed as easily). - The answer on USMLE is "acidification, $NH4^+$" for what lactulose causes in the gut.
Neomycin	- Used to Tx hyperammonemia in cirrhosis. - Antibiotic that kills ammonia-producing bacteria.
Loperamide	- Mu-opioid receptor agonist used to treat diarrhea-predominant IBS. - Opioids cause constipation, so "this is a good thing" in this case.
Ezetimibe	- Blocks cholesterol absorption in the small bowel.
Cholestyramine	- Bile acid sequestrant. Causes reduced enterohepatic circulation of bile acids at terminal ileum → liver must now convert more cholesterol into bile acids in order to replenish them → liver pulls cholesterol out of the blood to accomplish this.
Orlistat	- Pancreatic lipase inhibitor used for obesity; can cause steatorrhea.
Nystatin	- Mouth wash used for oropharyngeal candidiasis. - Forms pores in fungal ergosterol membrane.
Fluconazole	- Used for candidal esophagitis. - Inhibits ergosterol synthesis (i.e., lanosterol → ergosterol) by inhibiting 14α-demethylase. USMLE can write this as "P-450-mediated demethylation reaction."
Acyclovir	- HSV esophagitis; HSV1/2 causes "punched out" ulcers. - DNA polymerase inhibitor; causes chain termination; resistance = altered viral thymidine kinase.
Ganciclovir	- CMV esophagitis and colitis; causes linear/confluent ulcers. Same MOA as acyclovir.
CAP	- Clarithromycin, Amoxicillin, PPI. - First-line Tx for *H. pylori*. - If patient still has (+) urease breath test after 4 weeks, assume antibiotics resistance and switch out the CA and add tetracycline, metronidazole, and bismuth (keep PPI).
Vancomycin	- Given orally to treat *C. difficile.*

Metronidazole	- "Anaerobes below the diaphragm." - Used for diverticulitis in combo with a fluoroquinolone. - Tx for *Giardia* + *Entamoeba.*
Imipenem	- Carbapenem antibiotic that has fantastic penetration for pancreatitis.
Mebendazole	- Used for most nematodes (i.e., *Ascariasis, Enterobiasis,* hookworms, etc.). - Inhibits microtubules. An NBME Q has "tubulin" as the answer for protein it inhibits.
Albendazole	- Same as mebendazole, but for whatever reason it's known as a preferred agent for neurocysticercosis, which is caused by a cestode (*Taenia solium*).
Pyrantel pamoate	- Agent used against nematodes. Shows up on an offline NBME as answer. - Causes paralysis.
Praziquantel	- Used for cestodes and trematodes. - Causes paralysis.

- 44M + fasting glucose of 112 mg/dL + dark skin on forearms + arthritis; Dx? → hereditary hemochromatosis → AR, chromosome 6, HFE gene, C282Y or H63D missense mutations account for 90% → **"Bronze diabetes"** → hyperpigmentation (from hemosiderin deposition) + diabetes due to iron deposition in tail of pancreas (normal fasting glucose is 72-99 mg/dL; impaired fasting glucose [pre-diabetic] is 100-125 mg/dL; diabetic is two fasting glucoses 126 or greater, or a single HbA1c >6.5%, or any random glucose >200 mg/dL) + third finding such as arthritis, cardiomyopathy, or infertility.
- 44M + fasting glucose of 130 mg/dL + hands are sore + x-ray of hands shows DIP involvement; what's the Dx for the type of arthritis? → **answer = pseudogout, not osteoarthritis.** Student says wtf? The two most common etiologies for pseudogout are hemochromatosis and primary hyperparathyroidism (pseudogout is calcium pyrophosphate deposition disease, and will present as either a monoarthritis of a large joint such as the knee, **or as an osteoarthritis-like presentation of the hands.**
- Tx of hereditary hemochromatosis → serial phlebotomy, **not** chelation therapy.
- 44M patient above + USMLE asks what's the mechanism for his disease → answer = "increased intestinal absorption of iron."
- 44M above + next best step in Dx? → check serum ferritin (>300 ug/L in men + post-menopausal women; or >200 in premenopausal women; USMLE will always say >300 so don't worry).
- Why do men get hemochromatosis younger + with worse Sx than women? → menstruation slows progression of disease.
- What is secondary hemochromatosis? → aka transfusional siderosis (amazing to remember if you want to sound sophisticated) → due to chronic blood transfusions → each transfusion of RBCs

contains iron → seen classically in beta-thalassemia major or any other patients receiving ongoing transfusions.

- Tx for secondary hemochromatosis (transfusional siderosis) → chelation therapy (e.g., deferoxamine), **not** serial phlebotomy.
- Three main conditions causing hepatocellular carcinoma? → HepB, HepC, hemochromatosis.
- Tumor marker of HCC? → AFP (same as yolk sac tumor, aka endodermal sinus tumor).
- Parkinsonism in young patient; Dx? → Wilson disease (AR, chromosome 13)
- Parkinsonism in older patient; Dx? → Parkinson disease
- Parkinsonism + axial dystonia; Dx? → progressive supranuclear palsy (this is on the USMLE!!)
- Parkinsonism + visual hallucinations + cognitive decline; Dx? → Lewy body dementia
- Parkinsonism + urinary incontinence + gait instability + cognitive dysfunction; Dx? → normal pressure hydrocephalus (NPH) = "wet, wobbly, wacky" + Parkinsonism.
- 23F + unilateral resting tremor + increased LFTs + hemolytic anemia; Dx? → Wilson disease
- 23F + unilateral resting tremor + increased LFTs + hemolytic anemia; next best step? → answer = do a slit-lamp exam.
- How is most copper normally excreted by the body? → through bile (hepatocyte transport pump).
- Important serum/blood findings in Wilson? → decreased ceruloplasmin; increased urinary copper.
- How do we normally excrete iron? → humans have poor elimination mechanism; losses are natural through skin (or menstruation in women).
- Where do we absorb iron and copper? → duodenum mainly.
- Where do we absorb bile acids and B12? → terminal ileum.
- Tx for Wilson disease? → penicillamine (copper chelator).
- "Opposite" of Wilson disease? → Menkes disease (can't absorb intestinal copper).
- What vitamin helps absorb iron? → vitamin C ferrireductase converts small bowel Fe3+ to Fe2+; only Fe2+ can be absorbed out of the lumen.
- What does cholecystokinin (CCK) do? → increases contraction of gall bladder, relaxes sphincter of Oddi, and increases exocrine pancreas secretion of lipases, proteases, and amylase.
- Which cells make CCK → answer = "enteroendocrine cells of the small intestine" → HY.

- Macronutrients entering the duodenum (i.e., fats, proteins, carbs); what hormone is notably secreted in response? → USMLE wants CCK for this one.
- Acid entering the duodenum; what hormone is notably secreted in response? → USMLE wants secretin → causes bicarb secretion from pancreas.
- What do gastric chief cells do? → secrete pepsinogen (inactive zymogen) → acid activates to pepsin.
- What do parietal cells secrete → acid + intrinsic factor.
- 22M + vitiligo + macrocytic anemia; Dx? → pernicious anemia causing B12 deficiency → "autoimmune diseases go together."
- Mechanism for pernicious anemia? → autoantibodies against parietal cells or intrinsic factor.
- Blood smear in pernicious anemia? → hypersegmented neutrophils + high MCV RBCs.
- Pt has B12 deficiency + atrophic gastritis; what is most likely to be increased in this pt? → **answer = gastrin** → need to assume pernicious anemia → atrophy of parietal cells due to Abs → decreased acid production → increased gastrin → USMLE likes this question!
- Which nerve must be severed to remove cancer at gastroesophageal junction → answer = vagus (just memorize it; it's on a retired NBME).
- HY structures passing through diaphragm? → "I Ate 10 Eggs At 12." → IVC T8; T10 Esophagus + thoracic duct; Aortic hiatus (aorta, azygous vein, thoracic duct) at T12.
- Drug-/pill-induced esophagitis → USMLE wants bisphosphonates (i.e., alendronate) + tetracyclines + clindamycin as causes.
- What part of the brain is damaged in Wilson? → USMLE wants putamen (they will show you a transverse head CT and expect you to pick out the letter labeling the putamen).
- Most active part of the bowel in terms of cell division? → answer = "base of the crypt." Memorize it. "Top of the villi" is the wrong answer.
- Where are Brunner glands → duodenum, proximal to the sphincter of Oddi.
- What do Brunner glands secrete? → alkaline mucous.
- What do Paneth cells secrete? → Defensins in the small intestine.
- Person who's vomiting; what's the biochemical disturbance? → hypokalemic hypochloremic metabolic alkalosis → low K, low Cl, high pH, high bicarb, low H, anion gap normal (even though it's alkalosis, not acidosis, the USMLE will still ask an arrow for the anion gap here).

- 2-week-old male + forceful non-bilious vomiting; Dx? → hypertrophic pyloric stenosis.
- Dx of pyloric stenosis? → abdominal ultrasound to show olive-shaped hypertrophied pylorus.
- Tx of pyloric stenosis? → myomectomy
- Who gets pyloric stenosis? → first-born males (weird, but it's on an old NBME) + neonates taking oral erythromycin for chlamydial ophthalmia neonatorum (erythromycin is a motilin-receptor agonist).
- 2-week-old male + bilious vomiting; Dx? → duodenal atresia, annular pancreas, congenital midgut volvulus, or Hirschsprung (correct, Hirschsprung can present with bilious vomiting).
- 2-week-old + Down syndrome + bilious vomiting + passed meconium ok; Dx? → duodenal atresia
- 2-week-old + Down syndrome + bilious vomiting + slow to pass meconium; Dx? → Hirschsprung
- How do you Dx duodenal atresia? → abdominal x-ray (AXR) showing double-bubble sign (very HY).
- How do you Dx Hirschsprung? → rectal manometry, followed by confirmatory rectal biopsy showing absence of ganglion cells.
- Mechanism for Hirschsprung? → failure of migration of neural crest cells distally to the rectum.
- Failure to pass meconium at birth. Most likely cause overall? → cystic fibrosis.
- 18-month-old + intermittent abdominal pain + crying + blood in stool; Dx? → intussusception.
- 18-month-old + intermittent squatting + crying + FOBT positive; Dx? → intussusception.
- 18-month-old + occasionally brings legs to chest + vomits + abdo mass; Dx? → intussusception.
- 18-month-old + occasionally brings legs to chest + vomits + no abdo mass Dx? → volvulus → this is congenital midgut volvulus.
- Presentation sounds like intussusception but no mass → answer = congenital midgut volvulus.
- Cause of intussusception? → >99% are in kids under age 2; caused by lymphoid hyperplasia due to viral infection (e.g., rotavirus) or recent vaccination; if in adult (usually elderly), it is caused by colorectal cancer.
- Dx and Tx of intussusception? → USMLE wants enema as the answer. Even though ultrasound can be done which shows a target sign, the USMLE always wants enema. And it can be any type. I've seen "air contrast enema", "air enema," "contrast enema" all as answers. I also had a student simply get "water-soluble contrast enema" on the exam, which means gastrografin. Barium would refer to regular contrast.

- For contrast swallows, when to do barium vs water-soluble (gastrografin)? → barium most of the time; if at risk of aspiration, must do barium because aspiration of gastrografin will cause pneumonitis. If patient has suspected esophageal perforation, do not do barium, as that will cause mediastinitis; must do gastrografin in this case.
- Level of celiac trunk and main branches + what's it supply? → T12; splenic artery, common hepatic artery, left gastric artery; supplies foregut (mouth to duodenum at ampulla of Vater).
- Level of SMA + what's it supply? → L1; supplies midgut (duodenum at ampulla of Vater until 2/3 distal transverse colon); so for instance, the right colic and middle colic arteries come off SMA.
- Level of IMA + what's it supply? → L3; supplies hindgut (2/3 distal transverse colon until the pectinate line 2/3 distal on the anal canal); left colic artery comes off IMA.
- Renal + gonadal (testicular in men; ovarian in women) arteries come off of L2 most often.
- Abdominal aorta bifurcates into common iliacs at L4.
- Abdominal aortic aneurysm occurs in males over 55 who are ever-smokers → a one-off abdo ultrasound is done in this patient group.
- Most common locations for atherosclerosis (in descending order) → abdominal aorta, coronary arteries, popliteal arteries, carotid arteries.
- USMLE favorite question → "Which of the following is supplied by an artery of the foregut but is not itself derived from the foregut?" → answer = spleen.
- Short gastric arteries come off which artery? → splenic.
- What's the main arterial supply to the pancreas? → Arteria pancreatica magna (greater pancreatic artery) → **a branch of the splenic.**
- 79M + Hx of atrial fibrillation + severe, acute, diffuse abdo pain; Dx? → acute mesenteric ischemia caused by mural thrombus embolizing to SMA or IMA.
- Above 79M; next best step in Mx? → mesenteric arteriography.
- Above 79M; Tx? → antibiotics (for necrotic bowel) then laparotomy (to remove necrotic bowel) → they will tell you in last line of vignette that IV Abx are administered and then ask for the next step, which is just laparotomy. It should be noted that the literature mentions various Txs like embolectomy, but the USMLE wants resection of nonviable bowel as the answer.

- 52F + short episode of ventricular fibrillation + defibrillated + now has severe abdo pain; Dx? → acute mesenteric ischemia due to ischemia caused by VF, not an embolus → antibiotics; CT if stable; if unstable go straight to laparotomy.
- 55F diabetic + Hx of intermittent claudication + Hx of abdo pain 1-2 hours after eating meals; Dx? → **chronic** mesenteric ischemia (CMI) caused by severe atherosclerosis of SMA or IMA (essentially angina of the bowel).
- 55F diabetic + Hx of CABG + Hx of abdo pain 1-2 hours after eating meals; next best step in Dx? → mesenteric arteriography (CMI).
- 55F diabetic + Hx renal artery stenosis + Hx of abdo pain 1-2 hours after eating meals; Tx? → angioplasty + stenting (CMI) to restore blood flow.
- Patient with CMI who has a 2-day Hx of severe abdo pain + fever; Dx? → acute mesenteric ischemia (acute on chronic due to a thrombosis; essentially akin to an "MI" of the bowel) → do mesenteric arteriography to Dx; Tx with Abx + laparotomy to remove necrotic bowel.
- What is pectinate line? → separates upper 2/3 of the anal canal (part of hindgut; endoderm-derived) from the lower 1/3 of anal canal (aka proctodeum, which is ectodermal).
- Lymphatic drainage above/below pectinate line? → above: internal iliac; below: superficial inguinal.
- Arterial supply above/below pectinate line? → above: superior rectal artery; below: middle/inferior rectal arteries.
- Venous drainage above/below pectinate line? → above: superior rectal vein; below: middle/inferior rectal veins.
- How does pectinate line relate to hemorrhoids? → above: internal hemorrhoids (painless); below: external hemorrhoids (painful).
- Tx for hemorrhoids? → conservative first, i.e., fiber + exercise; if they want intervention, banding (rubber band ligation) is the answer.
- How do you Dx congenital midgut volvulus? → upper-GI series (AXR + contrast follow-through of esophagus, stomach, and duodenum with barium or gastrografin).
- What does upper vs lower GI mean? → upper is above ligament of Treitz (suspensory ligament of duodenum; separates duodenum from jejunum; this ligament connects end of duodenum to the diaphragm) → below this ligament (jejunum and lower) is lower GI.

- Any significance to upper vs lower GI meaning? → upper-GI bleeds tend to cause melena (black, tar-like stools caused by blood exposed to acid); lower-GI bleeds tend to cause hematochezia (frank blood); large upper-GI bleeds can sometimes cause hematochezia.
- What are the mucous-producing cells in the stomach? → foveolar cells (aka surface mucous cells) secrete alkaline mucous; these are distinct from mucous neck cells, which secrete an acidic fluid containing mucin proteins.
- What are the mucous-producing cells in the intestines? → goblet cells
- "Wait, so there aren't goblet cells in the stomach?" → correct.
- Barrett esophagus; what are the changes in mucosa (from what to what)? → metaplasia of non-keratinized stratified squamous → intestinal simple columnar epithelium (intestinal means "has goblet cells").
- What does Barrett look like on endoscopy? → bright red mucosa (UWorld has a Q where they show the endoscopy).
- Barrett will go to what kind of cancer? → adenocarcinoma of distal third of esophagus
- Biggest risk factor for Barrett? → GERD (often in obese patients due to lower LES tone)
- Tx of Barrett? → PPIs are standard to decrease GERD; they're more efficacious than H2-blockers.
- Acute gastritis (inflammation of lining) main causes? → NSAIDs, glucocorticoids, H. pylori.
- Leading cause of chronic gastritis? → H. pylori.
- What is atrophic gastritis? → a type of chronic gastritis in which ongoing inflammation of glandular cells causes them to be replaced with fibrotic tissue.
- What is type A vs B atrophic gastritis? → Type A = non-antral (mainly fundus); caused by autoimmune attack against parietal cells (pernicious anemia), resulting in B12 deficiency due to insufficient intrinsic factor production. Type B = antral; caused by H. pylori.
- How do you Dx H. pylori? → urea breath test or stool antigen test.
- How do you Tx H. pylori? → CAP = Clarithromycin, Amoxicillin, Proton pump inhibitor. If patient has positive urea breath test four weeks after Tx, assume resistance of Abx, so switch out the clarithromycin and amoxicillin and give metronidazole + tetracycline + bismuth instead (with the PPI).
- Urease, oxidase, catalase; H. pylori is positive for which ones? → all three (asked in a USMLE Q, where they had different + and – combos, and the answer was all three +).

- What is peptic ulcer? → gastric ulcer and/or duodenal ulcer.
- Most common cause of gastric ulcers → H. pylori, then NSAIDs, then smoking.
- What about alcohol? → EtOH doesn't cause ulcers; it just prevents their healing.
- Weird causes of gastric ulcers? → Cushing ulcers (head trauma → increased ACh outflow → increased M3 receptor agonism on parietal cells → increased acid secretion); Curling ulcers (sloughing of intestinal mucosa due to acute fluid losses typically seen with burns; think "Curling irons are hot.")
- Most common cause of duodenal ulcers → caused almost exclusively by H. pylori.
- What does USMLE care about relating to sucralfate? → can coat the base of ulcers + protect them; activated by acidic pH, so can be inactivated in patients on antacids.
- Mechanism via which H. pylori causes ulcers? → secretion of proteinaceous products that damage intestinal mucosa (on USMLE Free 120 I believe).
- Gastrin levels in H. pylori? → elevated; H. pylori causes increased acid secretion
- Cancer caused by H. pylori? → MALT lymphoma (mucosa-associated lymphoid tissue lymphoma); H. pylori is considered a pre-MALT lymphoma condition, where eradication causes remission of 80% of low-grade stage 1E MALT lymphomas (confined to mucosa + submucosa).
- Key points about Whipple disease? → caused by bacterium *Tropheryma Whipplei* → causes PAS-positive macrophages in the lamina propria of the small bowel (USMLE is obsessed with this detail); can cause arthritis and diarrhea. Tx = Abx (e.g., doxycycline).
- What is Tropical sprue? → malabsorptive disease characterized by flattening of intestinal villi (similar to Celiac histo); etiology obscure/manifold but bacterial infection is accepted as one cause; Tx = Abx (e.g., doxycycline).
- Celiac disease important points? → gluten intolerance; fluten found in wheat, oats, rye, and barley, so therefore will get Sx after eating, e.g., pasta (too easy for the vignette to mention though); causes flattening of intestinal villi on biopsy (image HY); patients often present with **iron deficiency anemia (HY way to differentiate from lactose intolerance);** Dx with Abs: anti-endomysial IgA (anti-gliadin IgA), anti-tissue transglutaminase IgA → after you get positive Abs in Dx of Celiac, USMLE wants **duodenal biopsy** to confirm ("no further studies indicated" is the wrong answer) → Tx = dietary changes to avoid gluten.

- "I've heard IgA deficiency relates somehow to Celiac. Can you explain." → Remember that "autoimmune diseases go together," so increased risk of one means increased risk of another; the HLA associations are not super-strict → if patient has Celiac, he or she is 10-15x more likely to have IgA deficiency (which one comes first is up for debate) → because these patients are IgA deficient, they will have false-negative results on antibody screening (since Abs are IgA).
- Weird factoid about Celiac? → increased risk of T cell lymphoma; HLA-DQ2/8 positive.
- What immunoglobulin is produced at Peyer patches (GALT; gut-associated lymphoid tissue)? → IgA. Peyer patches contain large number of IgA-secreting B cells. IgA is a dimer connected by a J-chain.
- USMLE wants you to know **colipase deficiency** is a reason why a patient with chronic pancreatitis might not be able to digest triglycerides → yes, weird and random, but I don't know what to tell you. It's on the NBME.
- What's Zollinger-Ellison syndrome? → gastrin-secreting tumor → USMLE likes "recurrent" duodenal ulcers, or the presence of any single jejunal or ileal ulcer → frequently seen as part of MEN1 (pancreas, pituitary, parathyroid).
- Dx of ZES? → answer = check serum gastrin levels; if USMLE mentions secretin-stimulation test, it's only because they'll say in the vignette that "gastrin is not suppressed with secretin stimulation" as a way to tell you the Dx is ZES → secretin should normally lower gastrin levels, but they remain elevated in ZES.
- Tx of ZES? → PPIs, but definitive is surgical resection of tumor.
- What is Menetrier disease? → don't confuse with Meniere disease; Menetrier is atrophy of parietal cells (causing achlorhydria) and hypertrophy of foveolar cells (surface mucous cells) to the extent that the inner lining of stomach resembles brain gyri; can be caused by CMV infection; Meniere disease, in contrast, is a tinnitus/vertigo syndrome caused by defective endolymphatic drainage from the inner ear.
- Types of stomach cancer (apart from MALT lymphoma) → intestinal vs diffuse type; USMLE doesn't ask about intestinal (it's characterized by irregular tubular histology); diffuse = linitis plastica, which is "leather bottle" appearance of the stomach → cells contain mucin and are called signet ring cells → often associated with Virchow node (pronounced ver-cough), which is a palpable left supraclavicular

lymph node → this positive node is called Troisier sign of malignancy → if gastric cancer metastasizes hematogenously to ovaries, the mets are called Krukenberg tumors → biopsy shows signet ring cells containing mucin; you'll know it's not mucinous cystadenocarcinoma (MC) of ovary because 1, MC isn't the bilateral type (serous cystadenocarcinoma is), 2, MC is associated with pseudomyxoma peritonei (peritoneal infiltration by mucous from tumor), and 3, MC has a "locular," or "loculated" appearance.

- High ALP + high direct bilirubin + high amylase or lipase → gallstone pancreatitis = choledocholithiasis.
- High ALP + high direct bilirubin + high amylase or lipase + remote Hx of cholecystectomy → sphincter of Oddi dysfunction (can't be a stone cuz the gallbladder was removed ages ago).
- High ALP + high direct bilirubin + normal amylase or lipase in someone with recent cholecystectomy → choledocholithiasis (retained stone in cystic duct that descended, but not distal to pancreatic duct entry point).
- Dx and Tx of choledocholithiasis → ERCP.
- High ALP + high direct bilirubin + normal amylase or lipase in someone with remote cholecystectomy → pancreatic cancer.
- Dx of pancreatic cancer → CT abdo with contrast.
- High ALP + high direct bilirubin + normal amylase or lipase in someone with remote cholecystectomy + CT is negative → cholangiocarcinoma.
- High ALP + high direct bilirubin + normal amylase or lipase + diffuse pruritis + high cholesterol → primary biliary cirrhosis (PBC).
- High ALP + high direct bilirubin + normal amylase or lipase + autoimmune disease (in pt or family) → PBC.
- Dx of PBC → anti-mitochondrial Abx next best step; liver biopsy is confirmatory.
- Recent cholecystectomy + fever + abdo pain → post-op bile leak.
- High ALP + high direct bilirubin + normal amylase or lipase + CT shows cystic lesion in bile duct → choledochal cyst → do simple excision of cyst (cholangiocarcinoma not cystic + CT can be negative).

- Imaging to view liver or pancreas → CT with contrast.
- Imaging to view gallbladder → abdominal ultrasound.
- Imaging to view gallbladder in suspected cholecystitis only if USS negative → HIDA scan.
- Imaging to view bile ducts → ERCP or MRCP (choose ERCP > MRCP if both listed).
- 22M + stressed studying for exams + yellow eyes + has had a few similar episodes in the past + is otherwise healthy; Dx? → answer = Gilbert syndrome (proncouned jeel-BEAR).
- Mechanism for Gilbert? → decreased expression of glucuronosyltransferase (bilirubin conjugating enzyme) in the liver → decreased ability to take up unconjugated bilirubin at the liver → jaundice.
- Tx for Gilbert? → reassurance; benign condition.
- Criteria for pathologic jaundice? → student says "I really need to know that?" Absolutely. HY on peds shelves and 2CK. If any one or more of the following is positive, the etiology of the kid's jaundice is considered pathologic:
    - Any jaundice present in first 24 hours of life, period.
    - Any jaundice present after one week if term or two weeks if preterm.
    - Total bilirubin >15 mg/dL.
    - Direct bilirubin >10% of total bilirubin, *even if total is under 15 mg/dL.*
    - (The one everyone forgets) Rate of change of increase of total bilirubin >0.5 mg/dL/hour.
- How do those pathologic jaundice guidelines relate to actual USMLE Qs though? → if pathologic, USMLE wants phototherapy as the Tx; if that's insufficient, do exchange transfusion → in addition, even in adults, if you see a Q where someone's direct bilirubin is >10% of total, that'll be a huge clue to you that an obstructive pathology is present.
- What are normal bilirubin levels? 0.1 mg/dL direct; 1.0 mg/dL total (yes, the lab values will be there for you on the exam, but do you want to wear training wheels forever? You must know these for 2CK and Step 3).
- 8-day-old neonate + jaundice + direct bilirubin 14 mg/dL + total bilirubin 15 mg/dL; Dx? → answer = biliary atresia → super HY Dx.
- 8-day-old neonate + jaundice + direct bilirubin 14 mg/dL + total bilirubin 15 mg/dL; next best step in diagnosis? → answer = straight-up liver biopsy → if positive, must do liver transplant.

- 8-day old neonate + jaundice + direct bilirubin 1 mg/dL + total bilirubin 20 mg/dL; Dx? → Crigler-Najjar syndrome.
- Mechanism of Crigler-Najjar? → absence (type I) or deficiency (type II) of glucuronyosltransferase.
- Tx of Crigler-Najjar? → phenobarbital is helpful in type II (why the USMLE occasionally asks this I don't know why); Type I does not respond to phenobarbital; plasmapheresis + phototherapy are temporizing measures; liver transplant is curative.
- What are Dubin-Johnson vs Rotor syndrome? → never fucking asked on the USMLE but I mention them here otherwise some students would probably spasm out → high direct bilirubin due to decreased ability to secrete it into bile → DJ has black liver; Rotor does not.
- Over-arching HY point about hepatitis infections for USMLE → they want you to know that hepatocellular damage is due to "cytotoxic T lymphocytes," not "viral cytopathic effects."
- Hepatitis A HY points → fecal-oral; acute; shortest incubation period (2-6 weeks); vignette will mention person getting acute hepatitis in US or Mexico → can be asymptomatic, but jaundice, fever, anorexia common; self-limiting → there's a Q on one of the 2CK NBMEs where a patient gets HepA followed by all cell lines (RBCs, WBCs, platelets) down, and the diagnosis was simply viral-induced aplastic anemia, but this was slightly unusual as we classically associate Parvo B19 with viral-induced aplastic anemia.
- Hepatitis B HY points → parenteral transmission; in all body fluids and can be transmitted through breastmilk, sex, and IV drug use → most common transmission is vertical at birth (through birth canal); vignettes associate HepB with China; 30% of patients with polyarteritis nodosa are HepB seropositive; HepB can also cause membranous glomerulonephritis; at birth, give HepB vaccine, followed by a second dose at 2 months, and a third dose at 6 months (apparently some vaccine schedules are saying it's no longer given at 4 months); if mom is positive for HepB, give neonate immunoglobulin + vaccine; if mom's status is unknown, give neonate vaccine + only give immunoglobulin if mom's results come back positive; liver shows ground-glass appearance on biopsy → I believe a UWSA2 question for Step 1 gave an IV drug user + they showed a liver with a ground-glass appearance, and the answer was HepB; everyone selects HepC because they say, "oh wow IV drug user," but it was HepB; HepC has a nodular appearance of the liver; both HepB and C and cause hepatocellular carcinoma.

- USMLE wants you to know HepB is “DNA, enveloped, circular” and has a polymerase enzyme with dual function: **DNA-dependent DNA polymerase** + RNA-dependent DNA polymerase (reverse transcriptase).
- Tx for HepB? → a variety of meds; the USMLE isn’t really fussed and won’t ask you; this is more Qbank where they may show up; but some drugs are **interferon-alpha,** entecavir, tenofovir, telbivudine, and lamivudine.
- If you get a Q where they tell you someone was vaccinated for HepB but their surface Ab is still negative, the next best step in Mx = give HepB vaccination; some patients don’t seroconvert immediately, or their surface antibody titer wanes.
- USMLE is obsessed with HepB serology:

MEHLMANMEDICAL.COM						
Hepatitis B serology interpretation						
	HBsAg	HBsAb	HBcAb	HBcAb IgM	HBcAb IgG	HBeAg
Acute infection (low infectivity)	+	-	+	+	-	-
Acute infection (high infectivity)	+	-	+	+	-	+
Chronic infection (low infectivity)	+	-	+	-	+	-
Chronic infection (high infectivity)	+	-	+	-	+	+
Immune (previous infection) (<6 months)	-	+	+	+	-	-
Immune (previous infection (>6 months)	-	+	+	-	+	-
Immune (vaccinated)	-	+	-	-	-	-
Window period	-	-	+	+	-	-
Susceptible (not immune)	-	-	-	-	-	-

- HepC important info → longest incubation period (2-26 weeks); can become chronic; parenteral transmission; transmitted by blood only; **not sexually transmitted** (if you spend an hour and do a comprehensive literature review yourself, you'll learn that the transmission rate among heterosexual couples where one partner is infected is exceedingly low, i.e., on the order of 1 in 190,000 sexual contacts); if transmission occurs sexually, it is due to blood exposure; HepB, for instance, is in sexual secretions; HepC is not.
- Appearance of liver in HepC? → nodular; in contrast, HepB → ground-glass.
- Dx of HepC → ELISA + serum HCV RNA levels.

- Tx of HepC → USMLE likely won't ask you; but **pegylated-interferon-alpha** has the greatest chance of showing up on the exam; other drugs are daclatasvir and sofosbuvir.
- HepD important points → called a "subviral satellite" because it depends on HepB to cause infection (i.e., without HepB, HepD exposure won't cause infection) → what's the best way to prevent it? → answer = simply vaccinate against HepB → three types of HepB proteins form an envelope around the HepD ssRNA → infection with HepD can occur in someone with preexisting HepB infection (superinfection) or at the same time as HepB infection (coinfection); coinfection with HepB+D carries the highest mortality rate of any of the hepatitis viruses (~20%).
- HepE important points → fecal-oral transmission (enteral); acute disease only; classically when someone travels to India or Tibet → **high mortality in pregnant women is highest yield point.**
- Patient takes Abx for several days + has watery diarrhea; Dx? → C. difficile (pseudomembranous colitis).
- Patient takes Abx for several days + has crampy LLQ pain + bloody diarrhea; Dx? → C. difficile → this is on a 2CK NBME → Yersinia enterocolitica was also listed and was wrong; this is a good distractor because Y. enterocolitica causes pseudoappendicitis due to ileitis / mesenteric adenitis, but is RLQ pain, not LLQ.
- Which Abx cause C. diff overgrowth? → clindamycin, cephalosporins, ampicillin are highest yield.
- Mechanism for C. diff infection → "Ingestion of spores" is correct over "bacterial overgrowth" on the USMLE → yes, disruption of normal flora leads to C. diff overgrowth, but ingestion of spores is correct answer on NBME exam.
- Dx of C. difficile? → answer = stool AB toxin test, **not** stool culture (exceedingly HY).
- Tx of C. difficile? → **guidelines as of Feb 2018 say oral vancomycin first-line, not metronidazole** → apparently UW is updated on this too now → note that vanc is given orally → apart from C. diff, it's always given IV because it has terrible oral bioavailability, but in the case of C. diff, where we want the drug confined to the lumen of the colon, that makes sense.
- Mechanism of colonic necrosis in C. diff colonic necrosis? → answer = "cytoskeletal disruption."
- Patient is treated with vanc for C. diff but gets recurrence weeks later; why? → answer = "regermination of spores."

- C. diff + fever of 104F + tachy + diffuse abdominal pain; next best step in Mx? → AXR → look for toxic megacolon → Tx w/ NPO (nothing by mouth), NG decompression + rectal tube (decompression) + Abx (vancomycin or fidaxomicin) + steroids (if UC) + correct any electrolyte imbalances (sometimes low K) → if patient doesn't improve with conservative therapy, must do surgery (subtotal colectomy + ileostomy); do **not** do a colonoscopy on a patient with toxic megacolon as this will cause perforation.
- Damage to which nerves can cause constipation? → answer = pelvic splanchnic (because these are parasympathetic, which drive bowel function) → in contrast, hypogastric nerves = sympathetic (internal anal sphincter); pudendal nerve = somatic (external anal sphincter).
- Hepatocellular carcinoma + peanut farmer from China; cause? → aflatoxin → you don't have to like this factoid, but it's on the NBME.
- Vinyl chloride exposure + liver pathology; what's the Dx? → answer = hepatic angiosarcoma.
- 17M + fever + tonsillar exudates + cervical lymphadenopathy + cough + hepatomegaly; Dx? → EBV mononucleosis.
- Alcoholic + liver biopsy shows what? → answer = Mallory hyaline → damaged intermediate filaments.
- 45M + pan-acinar emphysema + increased LFTs; Dx? → alpha-1-antitrypsin deficiency → inability to break down elastase in the lungs, but enzyme is synthesized in liver; also causes cirrhosis.
- 45M + cirrhosis + fluid wave + fever + abdo pain; Dx? → spontaneous bacterial peritonitis (SBP)
- 69F diabetic + undergoing peritoneal dialysis + fever + abdo pain; Dx? → SBP
- 8M + viral infection + pedal/periorbital edema + fluid wave + fever + abdo pain; Dx? → SBP → minimal change disease (nephrotic syndrome) causing ascites.
- SBP Dx → paracentesis (peritoneal aspiration; don't confuse with periocardiocentesis) + do gram-stain + look for >250 WBCs per HPF.
- Tx for SBP → ceftriaxone.
- Cause of spider angiomata, palmar erythema, and gynecomastia? → answer = failure of the liver to process estrogen → answer also = hyperestrogenism.
- Clubbing causes? → pulmonary disease like CF and COBD; cardiac R→L shunts; **GI disease;** familial, etc. Bottom line is → just be aware GI disease can cause clubbing (i.e., IBD, Celiac, primary biliary cirrhosis).

- Woman 20s-50s + high cholesterol + diffuse pruritis + sister has rheumatoid arthritis; Dx? → primary biliary cirrhosis → USMLE likes "autoimmune diseases go together" in patient (or family).
- How to Dx PBC? → anti-mitochondrial Abs first; if positive, liver biopsy is confirmatory.
- Neonate born 32 weeks gestation + fever + abdominal distension + feeding intolerance + hematochezia; Dx? → necrotizing enterocolitis → bowel infection in premature neonates usually <32 weeks gestation.
- Dx of NE? → abdominal x-ray (AXR) visualizing **pneumatosis intestinalis** (air in the bowel wall), **air in the portal vein,** or free air under the diaphragm.
- Tx of NE? → NPO (nil per os; nothing by mouth), NG decompression, broad-spectrum Abx; if necrotic bowel, must do surgery.
- 49M + Hx of abdo pain after meal; now presents with sepsis + diffuse, acute abdo pain; Dx? → ruptured viscus (duodenum).
- Above 49M; next best step in Dx? → answer = "x-ray of chest + abdomen" to look for air under the diaphragm (confirms diagnosis); USMLE will never give you choice A) CXR; B) AXR, etc.; they'll either give you CXR alone, AXR alone, or both; one of the 2CK surgery NBMEs has both as the answer.
- 28F + pregnant + severe, acute abdo pain + jaundice + hepatomegaly + ascites + encephalopathy; Dx? → Budd-Chiari syndrome → hepatic vein thrombosis → rare, but associated with pregnancy and malignancy.
- Bad type of colonic polyp/adenoma? → villous + sessile characteristics are more sinister than tubular + pedunculated.
- Colon cancer progression? → step-wise → APC → KRAS → PTEN → p53. That is highly simplified, but the point is that USMLE wants you to be aware CRC occurs as a result of many mutations in sequence.
- 18F + CRC in family + has hundreds or thousands of polyps on colonoscopy; Dx? → FAP (familial adenomatous polyposis) → AD, chromosome 5, APC gene.
- FAP + soft tissue or bone tumors; Dx? → Gardner syndrome
- FAP + CNS tumors; Dx? → Turcot syndrome
- 18F + FAP; Tx? → answer = total proctocolectomy (answer on 2CK NBME; sounds overkill right off the bat, but it's the answer) → 100% chance of colon cancer.
- 20M + FAP + skull tumor; Dx? → Gardner, not Turcot → skull is bone, not CNS. Oh wow, craziness.

- 20M + FHx of CRC + has ten polyps seen on colonoscopy; Dx? → HNPCC (Lynch syndrome), not FAP → FAP has hundreds or thousands of polyps on colonoscopy; HNPCC has "some polyps."
- Mechanism of HNPCC? → answer = "microsatellite instability," or "defect in mismatch repair genes" → mutations in MLH1, MSH2, PMS1.
- 22F + has 30 polyps on colonoscopy + mom died of endometrial cancer; Dx? → HNPCC → Lynch syndrome associated with gyn tumors such as ovarian + endometrial, as well as other organ system tumors such as pancreas, stomach, and small bowel. Testicular + prostate very rare, but gyn common.
- 26F + they show you pic of spoon-shaped nail + tell you her lips have been cracked + they ask you what other symptom she might have; answer = dysphagia. Dx = **Plummer-Vinson syndrome** → triad of iron deficiency anemia (causes koilonychia; spoon-shaped nails) + angular cheilosis (cracked corners of mouth) + esophageal webs (dysphagia).
- USMLE wants the arrow combination (up, down, unchanged) for LES tone and peristalsis in CREST syndrome; answer = down arrow for both (this is on retired NBME 13 I think for Step 1).
- What does CREST stand for? → Calcinosis, Raynaud phenomenon, Esophageal dysmotility, Sclerodactyly, Telangiectasias
- 42M + dysphagia to solids + liquids + no other Hx; Dx? → achalasia → inability to swallow solids + liquids together implies neurogenic etiology.
- 42M + EtOH Hx + dysphagia to solids that progresses to include liquids; Dx? → esophageal cancer (SCC) → dysphagia to solids **that progresses to** solids + liquids = cancer until proven otherwise.
- HY points about esophageal cancer? → adenocarcinoma is distal 1/3 and is caused by GERD (obesity → low LES tone → GERD → Barrett → adenocarcinoma); SCC is upper 2/3 of esophagus and is caused by smoking + alcohol; can also be caused by webs, burns, chemicals, and achalasia (difficult, because achalasia is LES so your thought is, "how can that cause SCC of upper 2/3?" → probably the dysphagia causes increased esophageal irritation, which then becomes the risk factor for SCC).
- Mechanism for achalasia? → loss of NO-secreting neurons in myenteric plexus of LES → increased LES tone → bird's beak appearance on contrast swallow + increased tone on esophageal manometry; cause is often idiopathic, but Chagas disease (Trypansoma cruzi) is a known infective cause (rare).

- How do you Dx achalasia? → **USMLE wants barium (or gastrografin) swallow, then manometry, then confirmatory biopsy, in that order.** There is a Q on an NBME for 2CK where both barium and manometry were listed, and the answer was barium swallow, not manometry.
- So when is manometry the answer for achalasia? → when they show you a pic of the bird's beak from the barium swallow already performed, so clearly the next best step is manometry. The USMLE will sometimes show a graph of a manometry that's been performed, and you'll simply see that the pressure is high at the LES → hence Dx = achalasia. The confirmatory / most accurate test is biopsy of LES showing loss of neurons.
- Tx for achalasia → surgery first-line (endoscopic myotomy is superior to pneumatic dilatation); if patient has high surgical risk, can use botulinum toxin as first-line therapy → if fails, dCCB or nitrates.
- 42M + overweight + halitosis + gurgling sound when drinking fluids + occasionally regurgitates undigested food; next best step in Mx? → barium (or gastrografin) swallow; Dx = Zenker.
- Above 42M + Hx of GERD → go straight to endoscopy as the answer (cancer, not Zenker).
- Location + mechanism of Zenker? → false diverticulum just superior to the cricopharyngeus on the posterior pharyngeal wall → USMLE answers are "increased oropharyngeal pressure" and "cricopharyngeal muscle spasm." They want you to know it is **not** a congenital weakness. Dysphagia is a risk factor because this increases oropharyngeal pressure. I've noticed Zenker vignettes often mention the patient is overweight, although the association is non-specific.
- What is Whipple triad? → means insulinoma → 1) hypoglycemia, 2) Sx of hypoglycemia (diaphoresis, tachycardia, tremulousness), 3) relief of Sx with meals (or they'll say it gets worse between meals).
- If patient has Whipple triad, what's next best step in Mx? → check serum C-peptide levels → now this is where I get you a point: if C-peptide is high, the wrong answer is CT abdo to look for insulinoma because **the Dx is not automatically insulinoma.** Now you're probably like, "Really? Wait, why? I'm not following." → if C-peptide high, answer = **first check serum hypoglycemic levels** → meaning, some patients can surreptitiously take sulfonylureas (i.e., glyburide, etc.), which are insulin secretagogues, **so they're C-peptide levels will be high.** Only after the serum hypoglycemic screen is negative do you consider CT abdo.
- 32M + high glucose levels + body rash; Dx? → glucagonoma → rash is called necrolytic migratory erythema and is seen in glucagonoma.

- 32M + watery diarrhea + hypokalemia + achlorhydria; Dx? → VIPoma, aka WDHA syndrome (Watery Diarrhea, Achlorhydria, Hypokalemia).
- 32M + increased bowel motions + facial flushing; Dx? → VIPoma (another presentation I saw that was harder to Dx).
- 32M + steatorrhea + pancreatic tumor; Dx? → somatostatinoma.
- Biggest risk factor pancreatic cancer? → smoking
- Biggest risk factor for gall bladder cancer → smoking
- 52F + 2 kids + BMI 28 + recurrent colicky epigastric pain; next best step in Dx? → ultrasound (cholelithiasis).
- Above 52F + USS shows calcification in the gall bladder wall; next best step in Mx? → cholecystectomy → porcelain gallbladder carries 1/3 risk of cancer → must do surgical removal.
- Why increased risk of cholesterol stones in pregnancy? → estrogen upregulates HMG-CoA reductase + progesterone slows biliary peristalsis → **both of these hormonal effects are exceedingly HY.**
- Brown pigment stones → bacterial infection → bacteria deconjugate bilirubin, making it less water soluble → stone precipitation.
- Black pigment stones → hemolytic anemia / increased RBC turnover syndromes (e.g., sickle cell).
- What are pigment stones composed of? → calcium bilirubinate.
- Random vignette of sickle cell anemia; Q asks you which of the following is the patient at increased risk of + all answers seem obscure → answer = cholelithiasis → black pigment stones.
- USMLE Q will ask you whether high vs low cholesterol, bile acids, and phosphatidylcholine = good or bad for cholesterol stone formation? → high cholesterol = bad; high bile acids = good; high phosphatidylcholine = good.
- 45F + recurrent duodenal ulcers + Hx of renal calculi + serum gastrin levels elevated + she is started on PPI; next best step in Mx? → **check serum calcium levels** → MEN1 (pancreatic, parathyroid, pituitary).
- Travel + watery (or brown-green) diarrhea; Dx? → Travelers diarrhea → ETEC HL/HS toxins.
- MOA of ETEC HL toxin? → HL toxin ADP ribosylates adenylyl cyclase → **increases cAMP** → increases Cl secretion into small bowel lumen → Na follows Cl → water follows Na → secretory diarrhea.

- Which organism has same MOA as ETEC HL toxin? → Cholera toxin → difference is cholera is described as "liters and liters" of high-volume stool ("rice-water stool" is buzzwordy and rarely seen); if vignette wants to describe rice-water, they'll say "specks of mucous" in high-volume watery stool.
- MOA of ETEC HS toxin? → ADP ribosylates guanylyl cyclase → **increases cGMP** → decreases Cl reabsorption from lumen → more Na stays in lumen → more water stays with Na → watery diarrhea.
- Which organism has same MOA as ETEC HS toxin? → Yersinia enterocolitica toxin → difference is Y. enterocolitica causes bloody, not watery, diarrhea, and Y. enterocolitica also causes pseudoappendicitis in children (due to terminal ileitis / mesenteric adenitis) and arthritis in adults.
- Reheated fried rice + watery diarrhea; organism + mechanism? → Bacillus cereus → activation of spores.
- Kid + bloody diarrhea + petechiae + red urine; Dx? → hemolytic uremic syndrome (HUS) caused by EHEC O157:H7 or Shigella → HUS = triad of thrombocytopenia + schistocytosis (microangiopathic hemolytic anemia) + renal insufficiency; toxin will inhibit ADAMTS13 in afferent arterioles + cause endothelial damage → platelet consumption + clumping (thrombocytopenia) → RBCs shear as they pass by the platelet clumps (schistocytosis).
- How does HUS contrast with TTP? → TTP is caused by a mutation that results in defective ADAMTS13, or antibodies against ADAMTS13, resulting in the inability to cleave vWF multimers → platelet clumping → similar progression as HUS. One of the points of contrast is that TTP is not toxin-induced, and TTP also tends to be a pentad of the HUS findings + fever + neurologic signs.
- Bloody diarrhea + chicken / turtles → Salmonella enteritidis / typhimurium.
- Chicken + bloody diarrhea + most common cause overall in the US → Campylobacter jejuni
- Bloody diarrhea + ascending paralysis → Guillain-Barre syndrome caused by Campylobacter jejuni (many vignettes for GBS **will not** mention recent infection; GBS can also be caused by Shigella, Salmonella, and Yersinia).
- Homeless shelter + diarrhea or constipation + rose spots on abdomen + prostrated (lying down in pain); Dx? → typhoid fever → Salmonella typhi → don't confuse with food poisoning Salmonella strains (enteritidis + typhimurium) → the reservoir for typhoid is **humans;** it is not spread by chickens or turtles.

- Shigella vs Salmonella points → Salmonella produces H2S gas, is motile, and requires many organisms to cause infection; Shigella does not produce H2S gas, is non-motile, **and very few organisms cause infection;** they are both oxidase-negative.
- Incubation period for infective diarrhea? → gram-negative rods are 1-3 days (E.coli, Salmonella, Shigella, Yersinia).
- Tx for food poisoning diarrhea → don't treat majority of time → answer = "Abx increase duration of fecal excretion of organism."
- Creams + custards + mayo + potato salad + vomiting 1-6 hours after meal; Dx? → S. aureus heat-stable preformed toxin.
- Immunocompromised + watery diarrhea; Dx? → Cryptosporidum parvum.
- Tx for C. parvum? → supportive is correct over nitazoxanide.
- Fresh water lake / scuba diving + floaty stools; Dx? → Giardia causing steatorrhea.
- Who notably gets Giardia on USMLE? → IgA deficiency.
- Tx for Giardia → metronidazole.
- Travel + bloody diarrhea + epigastric/RUQ pain; Dx? → Entamoeba histolytica + liver abscess.
- Tx for the liver abscess → drain.
- Intestinal obstruction + rough-surfaced eggs in stool; Dx? → Ascariasis (Ascaris lumbricoides)
- Tx for most nematodes (roundworms) → mebendazole or pyrantel pamoate.
- Kid + perianal itching; Dx? → Enterobius vermicularis (pinworm).
- Walking on soil / beach + microcytic anemia → hookworm (Ancylostoma duodenale / Necator Americanus) → USMLE will ask how you acquired hookworm, and the answer is "through your feet."
- Strongyloides stercoralis; how do you acquire? → through skin ("through your feet").
- Pregnant woman + ring-enhancing lesions on CT + cats is not an answer; how did she acquire? → answer = pork → can also get Toxo from pork.
- Travel to Mexico + cystic lesion seen in lateral ventricle on CT and/or "swiss cheese" appearance of brain; Dx? → cysticercosis Taenia solium → pork tapeworm (cestode).
- Travel to Mexico + muscle cysts → taenia solium.
- Pork or bear meat + fever + myalgia + periorbital edema; Dx? → Trichinella spiralis (trichinosis) → pork nematode (in contrast pork cestode [tapeworm] is Taenia solium).

- Fish + high MCV; Dx? → Diphyllobothrium latum (fish tapeworm) causing B12 deficiency.
- Beef + tapeworm; Dx? → Taenia saginata.
- Crabmeat + hemoptysis; Dx? → Paragonimus westermani.
- Worm + portal hypertension; Dx? → Schistosoma mansoni / japonicum
- Worm + cholangiocarcinoma (bile duct cancer); Dx? → Clonorchis sinensis.
- Dogs or travel + liver cysts + diarrhea; Dx? → Hydatid cyst disease → Echinococcus granulosis.
- Neonate + regurgitating milk while feeding; next best step in Mx? → answer = insertion of NG tube → Dx = tracheoesophageal fistula → USMLE wants "endoderm" as the answer if they ask embryo → most common variant is proximal esophagus ends in blind pouch + distal esophagus connects to trachea.
- Rupture of gastric ulcer → left shoulder pain; which structure is irritated? → answer = diaphragm.
- Alcoholic liver disease → AST/ALT classically ~2/1 and in the low-hundreds (e.g., 250/125), but I've seen plenty of Qs where this is not the case (e.g., ALT is normal and AST slightly elevated); in chronic disease, enzymes can be completely normal; in contrast, if you see ALT and AST in the thousands, e.g., both are 1200 and ALT is equal to or greater than AST, think viral hepatitis.
- Acetaminophen toxicity → metabolite called NAPQI causes necrosis; give activated charcoal acutely (apparently there's a UW Q where this is the answer); give N-acetylcysteine otherwise to "regenerate reduced glutathione" → reduced-G has a thiol group (-SH) group.
- Other notable hepatotoxic drugs? → RIP from the TB RIPE drugs (i.e., rifampin, INH, and pyrazinamide are hepatotoxic); statins + fibrates; antifungal -azoles + terbinafine; methotrexate; halothane; acetaminophen; those are highest yield probably.
- 45F + BMI 29 + BP 140/90 + high TGAs + low HDL + elevated fasting glucose + slightly elevated AST and ALT; Dx? → NASH (non-alcoholic steatohepatitis) → caused by metabolic syndrome → Tx is lifestyle modification to eradicate the metabolic syndrome.
- Above 45F + completely normal liver enzymes; Dx? → NASH → I've seen plenty of Qs where all labs values they show you are completely normal + the only thing you're left with is, "well she's fat / has metabolic syndrome," and you eliminate the others to just say, "well, this is NASH."
- Histo of liver showing you large-ish circular lesion + tiny circles immediately next to it; Dx? → primary hepatocellular carcinoma → smaller lesions are satellite lesions, not mets.

- Histo of liver showing you cancer everywhere; Dx? → **colon cancer;** bc cancer is everywhere it's mets.
- Grehlin stimulates appetite (greatest when haven't eaten for a while).
- Leptin causes satiety (feeling of fullness); greatest **just after** a meal has started → sounds obvious, but they show this in graph form, and the wrong answer is immediately when you start eating.
- Liver disease + ascites; why the ascites? → **increased hydrostatic pressure** (portal HTN).
- Liver disease + peripheral edema; why the edema? → **decreased oncotic pressure** (decreased albumin production by the liver).
- Tx for chronic pancreatitis → give patient pancreatic enzymes → "pancrelipase" is an answer on another NBME Q, which is a mixture of pancreatic lipases, proteases, and amylases.
- Cause of chronic pancreatitis → multiple bouts of acute pancreatitis.
- Major causes of pancreatitis → alcohol + gallstones are major causes; hypertriglyceridemia + hypercalcemia are other known causes; may also be caused by ERCP (endoscopic retrograde cholangiopancreatography), mumps, drugs (e.g., sulfa). Absolutely do not say "scorpion sting" without saying the other causes first or you'll get upbraided on your surg rotation.
- Gallstone pancreatitis = choledocholithiasis = stone in ampulla of Vater = high ALP + high direct bilirubin + high lipase or amylase → Dx + Tx = ERCP.
- First Tx for pancreatitis in general → USMLE wants triad of NPO (nil per os; nothing by mouth) + fluids + NG tube → if USMLE asks for imaging, do CT with contrast to look for fluid collections + degree of necrosis → **drain fluid collections percutaneously;** if pseudoabscess forms and doesn't regress, answer = **internal drainage via ERCP or EUS (endoscopic ultrasound)**; if frank pus (fat enzymatic necrotic pancreas) >30%, must do necrosectomy (excision of necrotic pancreas) or aggressive percutaneous drainage.
- Acute pancreatitis + dyspnea + low O2 sats + bilateral infiltrates = ARDS.
- 72M + fatigue + low Hct → answer = do colonoscopy; most common cause of per rectum blood loss in elderly → diverticular bleed + colorectal cancer + angiodysplasia
- What is angiodysplasia → tortuous, superficial vessels on colonic wall that rupture + bleed → painless bleeding in elderly.

- Question on 2CK NBME mentions elderly guy with 2/6 mid-systolic who gets per rectum bleeding after argument with wife → answer = angiodysplasia → Heyde syndrome = aortic stenosis + angiodysplasia → mechanism might be due to increased pressure within colonic vasculature.
- Difference between diverticulosis vs diverticulitis → diverticulosis is simply the presence of diverticula → 50% of the US population over age 60 has them → most commonly in sigmoid colon due to law of Laplace (decreased diameter of sigmoid means greater pressure on the wall → greater propensity for outpouching); diverticular bleed is most common cause of per rectum blood loss in elderly → they need not be inflamed; in contrast, diverticulitis is when there is inflammation of diverticula.
- 69M + LLQ pain + fever = diverticulitis → Dx with CT with contrast of abdomen → Tx w/ Abx (metronidazole, PLUS fluoroquinolone or Augmentin; USMLE won't ask you the exact Abx, but you should be aware that metro covers anaerobes below the diaphragm) → **never** do a colonoscopy on someone with suspected diverticulitis, as you may cause perforation. However, after the diverticulitis is fully treated + cleared, patient will need a follow-up colonoscopy to rule out malignancy.
- 12M + pic showing you perioral melanosis (sophisticated way of saying hyperpigmentation around the lips); Dx? → Peutz-Jeghers syndrome → they'll sometimes just show you the pic and then ask what kind of colonic polyps you'll see → answer = hamartomatous polyps
- They show you pic of PJS hamartomatous polyp; high cancer risk from this lesion? → answer = non-cancerous.
- Patient with PJS; need special CRC screening? → yes, not bc of the hamartomatous lesions, but patients with PJS have increased risk of many types of cancer; for 2CK: do colonoscopy at age 8; if polyps present, repeat every three years; if not present, repeat at age 18 and then every three years.
- What is IBS? → Irritable bowel syndrome → classically constipation +/- diarrhea +/- other GI Sx like cramping pain or GERD-like Sx that are relieved with defecation → there are many ways to Tx IBS, such as starting with psych screen, but if the USMLE asks about meds, they like **lubiprostone,** which is used for constipation-predominant IBS (PGE1 analogue that causes increased Cl secretion in bowel → Na follows Cl → water follows Na → softens stool).
- What is IBD? → IBD = inflammatory bowel disease = Crohn and ulcerative colitis (UC) collectively.
- HLA association with IBD? → HLA-B27 → "PAIR" → Psoriasis, Ankylosing spondylitis, IBD, Reactive arthritis.

- Crohn GI findings? → mouth to anus; classically terminal ileum; frequently intermittent bloody diarrhea if colonic involvement; skip lesions causing "string sign" on contrast studies; "cobblestone mucosa"; transmural inflammation with non-caseating granulomas; perianal fistulae; B12 + fat-soluble vitamin malabsorption.
- 25M + oral ulcer + bloody diarrhea; Dx? → Crohn
- Extra-intestinal manifestations of Crohn? → classically erythema nodosum (red shins; not a rash; this is panniculitis, which is inflammation of subcutaneous fat); anterior uveitis (red eyes); oxalate nephro-/ ureterolithiasis (decreased fat absorption → more binding of fat to calcium [saponification] → therefore less calcium binds to oxalate → more oxalate absorbed → oxalate stones).
- Any weird factoid about Crohn? → sometimes patients (+) for anti-Saccharomyces cerevisiae Abs
- How to Dx Crohn? → USMLE wants colonoscopy
- Tx for Crohn? → USMLE wants NSAIDs (either sulfasalazine or mesalamine [5-ASA] will be listed) before steroids → use immunosuppressants late (i.e., infliximab, azathioprine, etc.).
- Does Crohn share anything with UC? → Yes, bear in mind in real life, there is overlap between the two diseases, so don't pigeonhole things; think of these disease-associations as propensities rather than as intransigent rules.
- 28M + lower back pain worse in morning and gets better throughout the day + mouth ulcer; Dx? → Crohn disease (oral involvement only) + sacroiliitis (back pain Sx of ankylosing spondylitis).
- 20F + bloody diarrhea + sore joints + eczematoid plaque on forehead → IBD + psoriatic arthritis
- UC GI findings? → rectum-ascending (meaning, starts at rectum and ascends; does not involve anus; Crohn of course is mouth to anus); bloody diarrhea; **not** transmural (mucosa + submucosa involved only; unlike Crohn); **no** granulomas (unlike Crohn); **lead pipe** appearance of colon on contrast studies (unlike "string sign" of Crohn); crypt abscesses (just memorize) → lead pipe means **loss of haustra** (so the colon looks smooth from the outside; this is really HY!) → USMLE might also there are "microabscesses within the crypts" on colonoscopy (meaning crypt abscesses in UC).
- Extra-intestinal manifestations of UC? → primary sclerosing cholangitis (beaded appearance of common bile duct; can be p-ANCA positive); pyoderma gangrenosum (crater on the forearm with necrotic debris); like Crohn, is associated with anterior uveitis + HLA-B27 associations.
- How to Dx UC? → USMLE wants colonoscopy.

- Tx for UC? → same as Crohn for USMLE purposes, but just be aware in severe cases colectomy is performed.
- 65M + intermittent bloody diarrhea + now has fever of 104F + abdominal pain + high leukocytes; Dx? → answer = toxic megacolon → **Dx with AXR, not colonoscopy!** → if you scope, patient will perforate and die → AXR will show dilated bowel (e.g., one NBME Q says "12-cm cecum"); in general, know that AXR is done when you're looking for gas.
- Where do most colonic ischemic ulcers occur? → watershed areas → splenic flexure (watershed of SMA and IMA) + sigmoidal-rectal junction (watershed of IMA and hypogastric artery).
- Ondansetron → 5HT3 (serotonin) receptor antagonist → anti-emetic classically for those with malignancy / undergoing chemotherapy → for Step 1, USMLE is content with you knowing MOA for ondansetron + that it acts at the chemoreceptor trigger zone (CTZ) of the caudal medulla → you need to be able to identify this on sagittal MRI (i.e., they'll show you letters at different locations and you need to choose caudal medulla for where ondansetron acts; I'd post an image here but I have zero interest in copyright infringement → Googling "ctz medulla mri" is more than sufficient; for 2CK be aware that ondansetron is used for hyperemesis gravidarum during pregnancy (metoclopramide also used, however ondansetron decreases vomiting significantly more than metoclopramide; nausea reduction is same).
- Metoclopramide → D2 receptor antagonist used as a prokinetic agent / anti-emetic → don't fuck up this MOA; students all the time remember that it acts at D2 receptors, but not whether it's an antagonist or agonist → easy way to remember is based on knowing its side-effects (similar to antipsychotics!) → hyperprolactinemia + tardive dyskinesia. **USMLE Step 1 also wants you to know that it prolongs the QT interval on ECG.** I had a student come out of the exam saying they were asked which drug doesn't prolong QT, and they had listed agents such as metoclopramide, azithromycin, risperidone, amitriptyline, which all CAN prolong QT.
- Really HY point is that metoclopramide is first-line pharmacologic agent in those with **diabetic gastroparesis.** USMLE will slam people on how this contrasts with GERD:
- 55M + BMI of 33 + vignette doesn't mention diabetes + 3 months burning in throat → Dx = GERD → Tx? = **trial of PPIs** (i.e., **trial of omeprazole)** for two weeks → relief of Sx is consistent with GERD as the correct Dx.

- 55M + BMI of 33 + poorly controlled diabetes (type I or II) + 3 months of burning in throat → Dx = **diabetic gastroparesis, NOT GERD** (woahhh crazy) → first pharm **Tx = metoclopramide, not PPIs.** If metoclopramide not listed, choose erythromycin (motilin receptor agonist).
- Regarding gastroparesis, the USMLE vignette will make an explicit point about bad diabetic disease, i.e., peripheral edema (renal insufficiency due to decreased oncotic pressure from albuminuria) +/- cataracts (osmotic damage from intracellular sorbitol) +/- urinary retention (neurogenic bladder due to osmotic denervation leading to hypocontractile detrusor) +/- they simply say HbA1C of 12% (diabetes is 6.5% or greater; prediabetic is 6-6.49). There will be no question as to whether they want gastroparesis vs simple GERD.
- If the vignette (more 2CK here) doesn't ask straight-up which drug you choose and they ask for next best step in Mx for suspected gastroparesis → first do endoscopy to rule out physical obstruction → if negative, then do gastric-emptying scintigraphy (scintigraphic gastric-emptying assay) → if delayed gastric emptying, first Tx = smaller meals; if insufficient, then do metoclopramide, then add erythromycin.
- Ursodeoxycholic acid (Ursodiol) → naturally occurring bile acid given to patients with cholesterol cholelithiasis as an alternative to cholecystectomy. On USMLE, Dx of cholelithiasis = abdominal ultrasound, followed by Tx = cholecystectomy. Ursodiol can be given to select patients but is not considered the universal next best step.
- USMLE also wants you to know ursodiol is given to pregnant women with **intrahepatic cholestasis of pregnancy (ICP)** → classically itchiness of palms + soles in 3rd trimester of primigravid women → answer = yes, there is increased risk of fetal demise → Dx by checking serum bile acids (high in ICP) → mechanism = estrogen + progesterone may impair bile secretory transporters, resulting in release into blood → mechanism of ursodiol in the Tx is unclear, but its use is first-line and HY for 2CK + Step 3 obgyn. I mention it here because it's otherwise a HY drug for gastro as per above.
- Octreotide → somatostatin analogue → in general, think of somatostain as a hormone that shuts of secretion of other hormones (e.g., GH, VIP); it also decreases portal blood flow → used for Tx of esophageal varices AFTER banding (endoscopic ligation) is performed; in other words, on the USMLE, choose banding for varices before octreotide; propranolol is mere prophylaxis (also decreases portal blood flow).

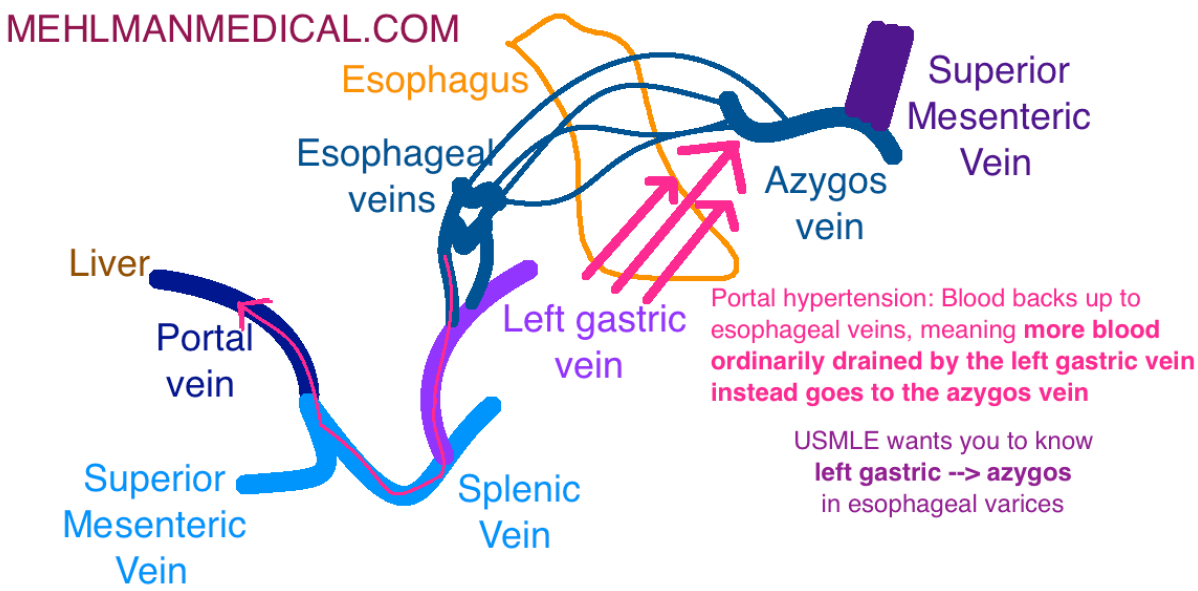
- Octreotide also can be used for carcinoid tumors by causing decreased serotonin release from tumor → carcinoid tumors are usually small-bowel or appendiceal (can also be bronchial) → small blue cells + S-100 positive + neuroendocrine origin → Dx with urinary 5-HIAA (5-hydroxyindole acetic acid) → symptomatic only if metastasizes to liver → diaphoresis + flushing + tachycardia + **tricuspid regurg + clonus** (considered UMN finding).
- Cyproheptadine (serotonin receptor antagonist) can also be used for carcinoid, but is classically used for serotonin syndrome instead → classically drug interactions like switching to an MOAi from an SSRI without not enough time passing; can also be caused by taking St John Wort if on SSRI, or by tramadol alone → serotonin syndrome does not cause tricuspid valve lesions because it's too acute.
- Octreotide + somatostatinomas cause steatorrhea due to inhibition of pancreatic lipase secretion.
- Aluminum (antacid) → causes constipation → "Aluminimum amount of feces."
- Magnesium (antacid) → causes diarrhea → this actually showed up as a case on 2CS (correct, CS).
- Calcium carbonate (antacid) → can cause rebound gastric acid hypersecretion + milk alkali syndrome → hypercalcemia + metabolic alkalosis.
- Orlistat → pancreatic lipase inhibitor used in some patients for weight loss → could theoretically cause fat-soluble vitamin malabsorption due to decreased intestinal fat absorption → for USMLE simply know the MOA and that the drug exists.
- Loperamide → mu-opiod receptor agonist used in the Tx of diarrhea → NBME exam asks this drug as an arrow question → **addictive potential LOW (DOWN arrow)** → because it can be used in the Tx of diarrhea, it can also therefore theoretically cause constipation (not rocket science).
- Lactulose → used to Tx hepatic encephalopathy by decreasing serum NH3 → undigestible carbohydrate that gut bacteria convert to acidic end-product → intraluminal NH3 (absorbable) produced by bacteria is converted to NH4+ (not absorbable) → USMLE Q will ask you whether the drug makes gut conditions more or less acidic, as well as whether it's NH3 or NH4+ that's not absorbed (they give you different combos) → answer = "acidic; decreased NH4+ absorption."
- Neomycin → used to Tx hepatic encephalopathy by killing NH3-producing bacteria in the gut; USMLE will give you a big, rambling paragraph on hepatic encephalopathy and simply tell you this drug is given then ask for MOA → answer = "kills intraluminal gut bacteria."

- Proton pump inhibitors (e.g., omeprazole) are more efficacious than H2-blockers. PPIs are irreversible and non-competiive; H2-blockers are reversible and competitive.
- Three mechanisms for stomach acid secretion are 1, ACh binding directly to M3 receptors on parietal cells (Vagus activity), 2, gastrin binding directly to gastrin receptors on parietal cells, and 3, gastrin causes enterochromaffin-like cells secrete histamine, which then binds to H2 receptors on parietal cells → these three effects are **synergistic** → USMLE, in contrast, wants "permissive" for the effects of cortisol on catecholamines (cortisol upregulates alpha-1 receptors so NE + E can bind and do their job), and "additive" for the effects of anti-platelet agents used together.
- What is Dumping syndrome? → caused by gastric bypass surgery, diabetes, or malfunctioning pyloric sphincter, in which stomach contents following a meal enter the duodenum too quickly; there are two types: early vs late → both show up in vignettes (without people even realizing they're seeing a Dumping syndrome Q).
- Early Dumping syndrome → 10-30 minutes after a meal → rapid entry of hyperosmolar gastric contents into duodenum → osmotic expansion of small bowel lumen → diarrhea + bloating → on USMLE, answer = "rapid emptying of hyperosomlar chyme into small bowel."
- Late Dumping syndrome → 1-2 hours after meal → rapid absorption of carbohydrates through small bowel wall → hyperglycemia → pancreas secretes lots of insulin → **rebound hypoglycemia** → USMLE merely wants you to identify this in a vignette as Dumping syndrome. They might say Hx of gastric bypass + now there's a meal + patient gets diarrhea +/- hypoglycemia → Dx simply = Dumping syndrome.
- What is Blind loop syndrome? → disturbance of normal floral balance in the small bowel due to disruption of peristalsis (i.e., surgery / post-surgical ileus), but may also be caused by conditions like IBD and scleroderma → leads to steatorrhea + B12 def + fat-soluble vitamin deficiencies → USMLE merely wants you to be able to make the diagnosis from a vignette → Tx is with antibiotics (doxycycline or fidaxomicin).
- Important points about intestinal transporters? → apical = side of intestinal lumen; basolateral = side of blood; SGLT-1 are GLUT5 are apical transporters that take in monosaccharides from small bowel lumen; GLUT5 takes in fructose; SGLT1 takes in glucose + galactose (think 5 for fructose being a

pentose, so SGLT1 is for the hexoses, glucose + galactose). Once the monosaccharides are in the enterocyte (small bowel cell), GLUT2 on the basolateral membrane takes them into the blood.

- 32M + exquisitely painful anal verge + refuses rectal exam; Dx? → anal fissure.
- Where do anal fissures occurs? → posterior in the midline, below the pectinate line.
- Tx for anal fissure? → Sitz bath
- Acanthocytes on blood smear; Dx? → abetalipoproteinemia or liver disease → USMLE loves heat stroke as cause of acanthocytes → 82F found unconscious on summer day + body temperature of 107F + blood smear shows acanthocytes; Dx = liver failure (heat stroke) → heat stroke = end-organ damage due to hyperthermia; heat exhaustion is hyperthermia + mental status change + fatigue + **no** end-organ signs.
- 8M + bloody stool + perfectly healthy otherwise; Dx? → Meckel diverticulum; student says, "huh, I thought that was age 2." I agree with you. But there's an NBME Q where the kid was 8, and the answer was Meckel → bleeding due to "heterotopic gastric / pancreatic tissue."
- How to Dx Meckel diverticulum? → Meckel scan (Tc99 uptake scan that localizes to diverticulum).
- Tx for Meckel → if asymptomatic, can leave alone; if symptomatic, surgical removal.
- Meckel diverticulum (true or false diverticulum?) → true → contains all layers of bowel → mucosa + submucosa + muscularis propria + adventitia; in contrast, false (Zenker) is just mucosa + submucosa.
- 16F + fever + high leukocytes + RLQ pain that migrated from epigastrium; Dx? → appendicitis (easy, but so HY how can I not at least mention it classically) → USMLE wants you to know that migration is because, initially, epigastric pain = visceral pain; RLQ pain = inflammation of parietal peritoneum. Must do a pregnancy test if female + adnexal ultrasound to look for gyn causes, i.e., ruptured cyst, etc. If male, go straight to laparoscopy. If rule out gyn cause in female, do laparoscopic removal. Ultrasound + CT can be done, but false-negatives have led to rupture + death, so they don't change management if clinical suspicion is high, which is why pt goes straight to laparoscopy if under high suspicion for appendicitis → if during surgery the appendix is normal, **answer = still remove it.**
- Mallory-Weiss tear vs esophageal varices HY point → MWT usually caused by vomiting/retching in alcoholic + **presents with a little bit** of blood in the vomitus; varices present with **LOTS of blood** → about half of patients with ruptured varix die.

- Mallory-Weiss tear vs Boerhaave → MWT is not transmural; Boerhaave is transmural → causes subcutaneous emphysema (crepitus due to air under the skin).
- Tx of varix? → banding + octreotide.
- Prophylaxis for varix? → propranolol.



YouTube

@mehlmanmedical

Instagram

@mehlman_medical

MEHLMANMEDICAL
HY GASTRO

All material is copyrighted and the property of mehlmanmedical.

Copyright © mehlmanmedical